

MODULE CATALOG

ELECTRONICS & TELECOMMUNICATIONS ENGINEERING

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GENERAL SUBJECTS

1. Marxist philosophy - BAA00101

Module designation	Marxist philosophy
Code, if applicable	BAA00101
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	The course equips students with the basic contents of the worldview and the Marxist-Leninist philosophical methodology. Helping students apply knowledge about the worldview, Marxist-Leninist philosophy, and philosophy creatively in cognitive and practical activities, to solve problems that the social life of a country or of the time being set.
Content	Marxist-Leninist philosophy is a course in the Marxist-Leninist knowledge block and Ho Chi Minh Thought. This module equips students with basic, general, and systematic knowledge of the problems of Marxist-Leninist philosophy. From there, learners have a basis and reasonable research and learning methods and apply them to the process of evaluating life phenomena.

2. Marxist-Leninist Political Economy - BAA00102

Module designation	Marxist-Leninist Political Economy
Code, if applicable	BAA00102
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	Firstly, equip students with basic and core knowledge of Marxist-Leninist political economy in the context of economic development of the country and the world today. Ensure the basic, systematic, scientific, update new knowledge, associate with practice, creativity, skills, thinking, learner quality, connectivity to overcome duplication, enhance integration and reduce the load, reduce content that is no longer relevant or scholastic content for students of non-theoretical colleges and universities. Second, on that basis, forming thinking and analytical skills, assessing and identifying the nature of economic benefit relations in the country's socio-economic development, contributing to helping students build appropriate social responsibility in the job position and life after graduation. Third, contribute to building the stance and ideology of Marxism-Leninism towards students.
Content	The program content consists of six chapters: in which chapter one discusses the objects, research methods, and functions of the Marxist-Leninist Political Economy. Chapters 2 to 6 present the core content of the Marxist-Leninist political economy according to the subject's objectives. Specifically, issues such as goods, markets and the role of actors in the market economy; Producing surplus value in a market economy; Competition and monopoly in the market economy; Socialist-oriented market economy and economic interest relations in Vietnam; Industrialization, modernization, and international economic integration in Vietnam.

3. Calculus 1B - MTH00003

Module designation	Calculus 1B
Code, if applicable	MTH00003
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	Introduction to Calculus, with two major contents of differential and integral calculus. - Knowledge: visual, quantitative, conceptual understanding of essential definitions, theorems, and properties in calculus - Skills: understanding of concepts, ability of using calculus in practical problems, ability to solve calculus problems, ability to use computer computation softwares - Attitude: diligence
Content	The course provides basic knowledge of calculus for non-majors, including IT, physics, electronics and telecommunications, material science, oceanology, meteorology and hydrology, ... , helping students to acquire necessary background for professional study. Content includes: real numbers, sequences and series of numbers, continuity, convergence, derivative, Riemannian integral of functions of one real variable.

4. Calculus 2B - MTH00004

Module designation	Calculus 2B
Module level, if applicable	General Education
Code, if applicable	MTH00004
Semester(s) in which the module is taught	2nd semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	Introduction to Calculus, with two major contents of differential and integral calculus of functions of several variables. - Knowledge: visual, quantitative, conceptual understanding of essential definitions, theorems, and properties in calculus - Skills: understanding of concepts, ability of using calculus in practical problems, ability to solve calculus problems, ability to use computer computation softwares - Attitude: diligence, ask questions
Content	The course provides basic knowledge of calculus for non-majors, including IT, physics, electronics and telecommunications, material science, oceanology, meteorology and hydrology, ... , helping students to acquire necessary background for professional study. Content includes: The set of $\mathbb{R}^n$, functions of several real variables, continuity, partial derivatives, extrema, multiple integrals, line integrals, Green theorem, surface integrals, Stokes and Gauss–Ostrogradski theorem, differential equations.

5. Linear Algebra - MTH00030

Module designation	Linear Algebra
Code, if applicable	MTH00030
Semester(s) in which the module is taught	2nd semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	Introduction to higher mathematics. - Knowledge: solid grasp of knowledge on matrices on number fields and applications to solving systems of linear equations; determinants and applications; vector spaces and linear maps. - Skills: computation on matrices; solving systems of linear equations; computing coordinates of vectors in a linear basis; change of coordinates following change of bases; presentation of linear operators by matrices; computing images and kernels of linear operators; using MAPLE computation software. - Attitude: diligence, participating in discussions
Content	The course leads first year students to higher mathematics. Aside from fundamental knowledge for all students, the course lays foundation for later study for all majors.

6. General physics 1 (Mechanics - Thermodynamics) - PHY00001

Module designation	General physics 1 (Mechanics - Thermodynamics)
Code, if applicable	PHY00001
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	3 Credits
Module objectives/intended learning outcomes	This course covers the principles of kinematics, dynamics, statics, work, energy, linear momentum, gravitation, and thermodynamics. Students who complete this module could be achieved the following: - <i>Knowledge: Be able to understand and apply laws of mechanics to explain physical phenomena and solve problems; Be able to understand and apply mechanisms of heat transfer, equations of state, the first and the second law of thermodynamics.</i> - <i>Skills: Be able to work at individual level and group work.</i> - <i>Competences: Ability to apply mechanics and thermodynamics knowledge to analyze physical situations.</i> - <i>Attitude: Honest</i>
Content	This module includes the following topics: 1. Physics and measurement 2. Kinematics of particles 3. Force and Newton's laws 4. Conservation laws in classical mechanics 5. Kinetics of rigid bodies 6. The ideal gas 7. The first law of thermodynamics 8. The first law of thermodynamics

7. Modern physics - PHY00004

Module designation	Modern physics
Code, if applicable	PHY00004
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	3 Credits
Module objectives/intended learning outcomes	This module provides students with fundamental knowledge of quantum optics, atomic and nuclear physics. Students who complete this module could be achieved the following: - <i>Knowledge: Be able to understand quantum optics, atoms and nuclei in the discovery and study of matter. Be able to apply quantum optics, atoms and nuclei in science activities.</i> - <i>Skills: Be able to work in individual, group work, and problem solving.</i> - <i>Competences: Ability to apply modern physics knowledge to analyze new physical situations.</i> - <i>Attitude: Honesty and diligence</i>
Content	This module includes the following topics: 1. Wave-particle duality of light 2. Waves of matter 3. Fundamentals of quantum mechanics 4. Fundamentals of atomic physics 5. Fundamentals of nuclear physics

8. Earthscience - GEO00002

Module designation	Earth science
Code, if applicable	GEO00002
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	Earth Science introduces general knowledge about the Earth related to the rights of the Earth, inside and outside the Earth including position and operation of the earth in space; composition and structure of the atmosphere, wind, and weather, climate, climate change; distribution of fresh water in hydrosphere, sea and ocean behavior, El Nino-La Nina phenomena: Geosphere: composition and structure of the earth's crust, weathering, erosion-accumulation, landslides, the internal structure of the earth, earthquakes, volcanoes, plate tectonic activities; learn about the history of the earth through the record of fossil remains. Knowledge of Earth science is a necessary basis for understanding the natural environment of the Earth.
Content	Presentation on deformations of the earth's crust and earthquake, volcanic activities and membrane tectonic mechanism Learn about Earth's history through fossil and stratigraphic records. Apply this knowledge to explain some issues in the main profession Skilled in group discussion, presentation and criticism Attitude, diligence: enthusiasm, honesty in learning; Serious and honest in checking.

9. Environmental studies - ENV00001

Module designation	Environmental studies
Code, if applicable	ENV00001
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	This is a compulsory subject in the general knowledge block in the training program for students of all disciplines. The focus of this module is to provide basic knowledge of Environmental Science: concepts, classification of resources, environment, basic problems and core of the environment. Students are also introduced to measures to protect the environment, conserve resources towards the goal of sustainable development.
Content	Chapter 1: Overview of the Environment 1. General concepts of environment 2. Basic composition of the environment (volumes) Chapter 3: Natural Resources 1. Definition 2. Classification 3. General issues of natural disaster Chapter 4: Human Impact on the environment 4.1 History of human impact on the environment 4.2 Human impact on environmental components Chapter 5: Environmental issues and sustainable development 5.1 Population and environment Chapter 5: Environmental issues and sustainable development 5.2 Environmental pollution Chapter 5: Environmental issues and sustainable development 5.3 Climate change Chapter 5: Environmental issues and sustainable development 5.4 Sustainable development

	Chapter 6: Environmental management and Environmental Education
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10. General physics 1 (Mechanics - Thermodynamics) - PHY00001

Module designation	General physics 1 (Mechanics - Thermodynamics)
Code, if applicable	PHY00001
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	3 Credits
Module objectives/intended learning outcomes	This course covers the principles of kinematics, dynamics, statics, work, energy, linear momentum, gravitation, and thermodynamics. Students who complete this module could be achieved the following: - <i>Knowledge: Be able to understand and apply laws of mechanics to explain physical phenomena and solve problems; Be able to understand and apply mechanisms of heat transfer, equations of state, the first and the second law of thermodynamics.</i> - <i>Skills: Be able to work at individual level and group work.</i> - <i>Competences: Ability to apply mechanics and thermodynamics knowledge to analyze physical situations.</i> - <i>Attitude: Honest</i>
Content	This module includes the following topics: 9. Physics and measurement 10. Kinematics of particles 11. Force and Newton's laws 12. Conservation laws in classical mechanics 13. Kinetics of rigid bodies 14. The ideal gas 15. The first law of thermodynamics 16. The first law of thermodynamics

11. History of Vietnamese Communist Party - BAA00002

Module designation	History of Vietnamese Communist Party
Code, if applicable	BAA00002
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	In terms of content: providing systematic and basic knowledge about the birth of the Communist Party of Vietnam (1920-1930), the Party's leadership over the Vietnamese revolution during the period of political struggle. government authority (1930-1945), in two resistance wars against French colonialism and American imperialism (1945-1975), in the cause of national construction and defense during the country's transition to socialism. association, conducting the renovation work (1975-2018). Ideologically: Through historical events and experiences on the leadership of the Party to build a sense of respect for objective truths, raise pride and confidence in the Party's leadership cause. About skills: Equipping with scientific thinking methods on history, skills in choosing research materials, studying subjects and the ability to apply historical awareness to practical work, criticizing misconceptions on the history of the Party.
Content	The course provides systematic and fundamental knowledge about the birth of the Communist Party of Vietnam (1920-1930), the Party's leadership for the Vietnamese revolution during the period of struggle for power (1930-1945), during the two resistance wars against the French colonialists and the American imperialists (1945-1975), in the cause of national construction and defense during the period of the whole country's transition to socialism, conducted doi moi (1975-2018). Through historical events and experiences in the leadership of the Party to build a sense of respect for objective truth, heighten pride and confidence in the Party's leadership cause. Equip with scientific thinking methods on history, skills in choosing research materials, studying subjects, and the ability to apply historical awareness to practical work, and criticize misconceptions about the history of the Party.

12. HoChiMinh's Ideology - BAA00003

Module designation	HoChiMinh's Ideology
Code, if applicable	BAA00003
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Relation to curriculum	General
Credit points	2
Module objectives/intended learning outcomes	About knowledge: Equip students with basic knowledge about the concept, origin, the process of formation and development of Ho Chi Minh thought; the basic contents of Ho Chi Minh's thought; the application of the Communist Party of Vietnam in the national-democratic revolution and the socialist revolution, in the current national renewal process. About skills: Helping students to think, analyze, evaluate, and creatively apply Ho Chi Minh's Thoughts to solve problems in real life, study and work. About attitudes: Helping students improve their political bravery, patriotism, loyalty to the goal, the ideal of national independence associated with socialism; aware of the role and value of Ho Chi Minh's thought for the Vietnamese Party and nation; realize their responsibility in studying and training to contribute to the construction and defense of the country.
Content	Description of course content: the subject equips students with basic knowledge about objects, research methods, and learning meanings of Ho Chi Minh's ideology; on the basis, of the process of formation and development of Ho Chi Minh thought; on national independence and socialism; on the Communist Party and the State of Vietnam; on great national and international solidarity; about culture, ethics, people.

13. General law - BAA00004

Module designation	General law
Code, if applicable	BAA00004
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	General Objective: By the end of the course, students will be able to understand the basic legal concepts and terms related to the country's legal system and state apparatus; apply legal provisions to solve some simple case studies; help students form and develop some skills such as looking up legal documents, analyzing legal regulations, and working in groups, thereby improving their sense of survival, learning and working following the Constitution and regulations. The law, the right behavior orientation in life. Specific objectives/course output standards: • Knowledge: Present basic legal concepts and terms related to the state apparatus and the Vietnamese legal system; Solve some exercise cases based on the provisions of a law book in the legal system of Vietnam; • Skills: Analyzing legal regulations; Lookup legal documents; Working group • Attitude, diligence: Raise awareness of living, studying, and working following the Constitution and the law.
Content	The module provides knowledge about the structure of the State apparatus as well as the functions, authority, and legal status of agencies in the State apparatus of the Socialist Republic of Vietnam in terms of economic management; Legal nature, and structure of the system of legal documents. From an overview of the system of legal branches in our State's legal system, a course is devoted to studying the basic contents of administrative law, civil law, and criminal law as branches of law. the main law (original branches of law) of the legal system, so that learners can easily access themselves to other branches of law arising from these major branches of law.

14. Probability and Statistics - MTH00040

Module name:	Probability and Statistics
Code, if applicable	MTH00040

Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	The course provides basic knowledge of the theory of probability and mathematical statistics. The theory of probability studies random phenomena, while the theory of mathematical statistics proposes general models and statistical decisions. - Knowledge: the course provides the most basic knowledge and notions of probability and statistics to be background knowledge for later courses. - Skills: employs probability and statistics to solve some real-world problems related to analysis and presentation of data. - Attitude: the course helps students acquire initial knowledge of probability and statistics, and recognition of the role of probability and statistics in science and in life, from which an enthusiasm for science can be formed, then a serious and proactive attitude in study.
Content	Basic notions on probability, random variables, common probability distributions, hypothesis checking, regression.

15. General Economic - BAA00005

Module designation	General Economic
Code, if applicable	BAA00005
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	Knowledge: Grasp the basic content of Microeconomics - a part of economics. Specifically : Understand the theory of economic choice, the influence of the law of scarcity, and economic models on economic choice. Understand the theory of supply and demand. Understand the theory of consumer behavior. Understand the theory of producer behavior. Understand the theory of competition and monopoly. Understand the theory of factor markets. Understand the theory of the role of government. Understand the analysis of the influence of factors on the balance of the market, in terms of skills Having the ability to apply the knowledge learned to study the nature of economic phenomena, the laws, and trends of the phenomena, and the laws of the market economy. Ability to apply the knowledge learned in the study of macroeconomics, development economics, and several other economic subjects. Forming and developing (one step) capacity to collect information, skills to synthesize and systematize issues in an overall relationship; skills to compare, analyze, comment, and evaluate micro-economic issues. Develop reasoning and public speaking skills. about attitude Trying to be righteous in recognizing and evaluating the lines, policies, and laws of the State of Vietnam in the development of the market economy with the state's regulation. Other Objectives: Through presentations and problem-solving. Forming and developing collaboration and teamwork skills; Develop skills of creative thinking, discovery, and discovery; Cultivate and develop assessment and self-assessment capacity; Develop public speaking and commenting skills.

Content	The course presents some basic problems of economics; principles of economics, supply and demand patterns and market equilibrium; theory of consumer behavior and business behavior; types of markets; aggregate supply, aggregate demand, and measure national output.
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16. Psychology - BAA00006

Module designation	Psychology
Code, if applicable	BAA00006
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	About knowledge - Understand the system of basic concepts of psychological science and research methods in psychology. - Understand the origin, formation and development of psychology and consciousness. - Understand the nature of human psychological processes: perception; emotion - affection; act. - Identify human psychological states. - Understand the psychological attributes that make up the personality structure. Understand the factors affecting the formation and development of personality. About skills Developing the capacity to study documents: Analyze, synthesize, compare, and generalize. Form and develop the ability to identify psychological phenomena, and apply learned knowledge to solve practical problems. Consulting and consulting skills. About attitude: Cultivate a passion for learning and studying subjects. Forming a sense of initiative and positivity in self-study. Form the right motivation in learning. Raise a sense of responsibility for group activities. Other goals: Forming personality qualities in accordance with the requirements of the integration period. Forming communication and behavioral skills in the community. Forming a modern and scientific way of living and working. Forming and developing the ability to think creatively, independently and critically. Skill formation: Reasoning skills; Public speaking skills; Form and develop teamwork skills.
Content	The course of general psychology helps learners to acquire basic knowledge about the nature and characteristics of psychological phenomena and basic psychological laws of

	humans (perception, emotion, will, etc.) actions and personalities...). On that basis, it helps learners to apply knowledge in practice to identify and distinguish basic psychological phenomena in humans.
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17. Innovative methodology - BAA00007

Module designation	Innovative methodology
Module level, if applicable	General Education
Code, if applicable	BAA00007
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	3
Module objectives/intended learning outcomes	
Content	After completing the course, students can apply concepts and subject knowledge to the following specific benefits: - Provide a system of ways of looking at things - Increase observation, curiosity, creativity - Analyze and logically explain existing creative solutions - Increase the agility of absorbing and assessing the value of information - See the unified similarity between seemingly very different systems - Overcoming psychological inertia - Helps to discover available reserves in the system, especially free and easy to use heavenly reserves - Give and choose an appropriate approach to solve the problem - Play out ideas for improving a given system - Forecasting the development trend of a given system in the future - Help detect, place and select problems to be solved - Used to practice developing creative imagination - Used to improve yourself, build your style, think and work scientifically and creatively - Contributing to building system-dialectical thinking

18. Science socialism - BAA00103

Module designation	Science socialism
Code, if applicable	BAA00103
Semester(s) in which the module is taught	3 rd semester
Language	Vietnamese
Credit points	2
Module objectives/intended learning outcomes	The subject equips students with the basic contents of scientific socialism (one of the three components constituting Marxism-Leninism). Helping students apply basic knowledge of scientific socialism creatively in cognitive and practical activities, solving problems that the social life of a country, of the times being set.
Content	The subject equips students with the basic contents of scientific socialism (one of the three components constituting Marxism-Leninism). Helping students apply basic knowledge of scientific socialism creatively in cognitive and practical activities, solving problems that the social life of a country, of the times being set.

19. English 1 - BAA00011

Module designation	English 1
Code, if applicable	BAA00011
Semester(s) in which the module is taught	1st semester
Language	Vietnamese
Credit points	4
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: • Learners will be able to understand and use vocabulary in various topics such as leisure activities, important life events, emotion, attitude, physical appearance description, travel plans, presenting dreams, countries, people, and languages. • Learners can understand and use grammar structures at the pre-intermediate level such as basic tenses and other related matters. • Learners will be able to choose the answer that best describes the given picture, choose the correct response to the questions, and understand dialogues and short monologues. • Learners will be able to pronounce single words, word clusters and sentences, describe a given picture, and build basic communications in daily life. • Learners will be able to comprehend 300-500 word passage of familiar topics, and gain more knowledge of different cultures around the world. • Learners can write essays about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English major students at the University of Science - Vietnam National University - Hochiminh City, using the first eight modules in the book New Cutting Edge (Pre-intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics such as leisure activities, important life events, feelings and emotions, attitudes, physical appearance descriptions, travel plans, presenting dreams, countries, people, and languages. Students need to complete various tasks, including presentations, debates, role-plays, assignments, tests and so on.

20. English 2 - BAA00012

Module designation	English 2
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Code, if applicable	BAA00012
Semester(s) in which the module is taught	2nd semester
Language	Vietnamese
Credit points	4
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: - Learners will be able to understand and use vocabulary in various topics such as everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, finance- related issues. - Learners can understand and use grammar structures in pre-intermediate level such as basic tenses and more complex grammatical structures including conditional sentences, passive, and verb patterns. - Learners will be able to choose the correct response for the questions, and understand dialogues and short monologues. - Learners will be able to pronounce words, generate short conversations, discuss real-life familiar topics, understand and quickly respond to generated questions, and improve basic communication skills in daily life. - Learners will be able to comprehend 500-700 word passages of familiar topics, and gain more knowledge of different cultures around the world. - Learners can write appropriate responses to written requests or complaints in business and social contexts, applying theories into real life practice.
Content	This course is designed for non-English major students at the University of Science - Vietnam National University - Ho Chi Minh City, using seven modules (modules 09-15) in the book New Cutting Edge (Pre-intermediate). These modules

	cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics such as everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, and finance-related issues. Students need to complete various tasks, including presentations, debates, role-plays, assignments, tests and so on.
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21. English 3 - BAA00013

Module designation	English 3
Code, if applicable	BAA00013
Semester(s) in which the module is taught	3rd semester
Language	Vietnamese
Credit points	4
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their basic knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners will be able to understand and use vocabulary in various topics such as leisure activities, important life events, emotion, attitude, physical appearance description, travel plans, dreams, countries, people, and languages. Learners can understand and use new language in a natural, communicative way. Learners will be able to present their opinions about some social and cultural issues and understand dialogues and talks. Learners will be able to comprehend 500-700 word passages of familiar topics, and gain more knowledge of different cultures around the world. Learners can write paragraphs about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English majors at the University of Science - Vietnam National University - Hochiminh City, using the first six modules in the book New Cutting Edge (Intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics, namely leisure activities, important life events, feelings and emotions, attitudes, physical appearance descriptions, travel plans, presenting dreams, countries, people, and languages. Students need to complete various tasks, including presentations, debates, role-plays, doing homework, tests and so on.

22. English 4 - BAA00014

Module designation	English 4
Code, if applicable	BAA00014
Semester(s) in which the module is taught	4th semester
Language	Vietnamese
Credit points	4
Module objectives/intended learning outcomes	Upon completing this course, learners will enhance their intermediate knowledge of general English of vocabulary and grammar in four skills: Listening, Speaking, Reading and Writing. Here are the objectives in detail: Learners can understand and use the language needed in more complex real-life situations in a natural, communicative way. Learners will be able to express their own ideas in interviews, mini-talks, problem-solving and story-telling. Learners will be able to comprehend 700-1000 word passages of up-to-date topics of international interest, and learn more about the world and other cultures. Learners can write essays about familiar topics related to daily life, learning activities, entertainment, events, etc.
Content	This course is designed for non-English majors at the University of Science - Vietnam National University - Ho Chi Minh City, using six modules (modules 07-12) in the book <i>New Cutting Edge</i> (Intermediate). These modules cover vocabulary, grammar, reading, listening, speaking and writing in a wide range of topics namely <i>everyday items, important life events, holiday plans, health problems, hobbies and interests, personalities, and finance-related issues</i>. Students need to complete various tasks, including presentations, debates, role-plays, doing homework, tests and so on.

MODULES IN CHARGE OF THE FACULTY

1. Basic Electronics- ETC00001

Module name	Basic Electronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC00001
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	1st semester
Person responsible for the module	MSc. Tran Xuan Tan
Lecturer	MSc. Tran Xuan Tan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 08:30 – 11:00 Monday 12:30 – 15:00
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	None
Module objectives/intended learning outcomes	- Express basic concepts of electronics. - Express technical documentations in English that are used in this course. - Apply electrical circuit laws and theorems to calculate values in basic electrical circuits. - Analyze basic electronic circuits. - Apply discrete components, ICs, electronic design software to design basic electronic circuits.
Content	Chapter 1: Overview + Basic concepts + Electrical circuit laws and theorems Chapter 2: Semiconductors, p-n junction and diodes Chapter 3: Bipolar junction transistors – BJTs - Structure, Characteristics - Biasing - Small-signal model - Power amplifiers and voltage regulators - Differential amplifiers, OpAmp Chapter 4: Field effect transistors – FETs - Structure, Characteristics - Biasing - Small-signal model

	- TTL and CMOS logics.
Study and examination requirements and forms of examination	Reinforcement exercises: Reinforcement exercises in each lesson: 30 Attendance: attendance all the lectures: 30 Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40
Media employed	Text Books
Reading list	- Bài giảng điện tử: Môn học Điện tử, Nguyễn Thành Long, Đại học Quốc gia Thành phố Hồ Chí Minh, Năm 2006 - Microelectronics Circuit Analysis and Design 3rd Edition, Donald A. Neamen, McGraw Hill, 2007

Written/updated by
Tran Xuan Tan

2. Introduction to Engineering-ETC00003

Module name	Introduction to Engineering
Module level, if applicable	Undergraduate
Code, if applicable	ETC00003
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	PhD. Le Duc Hung
Lecturer	PhD. Le Duc Hung
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	<i>None</i>
Module objectives/intended learning outcomes	- Have basic knowledge of Electronics and Telecommunications - Form ideas, analyze, and implement a basic system on a project - Experiment, investigate and discover knowledge - Experience-based project - Identify and formulate problems, Problem solving and decision making - Project management - Teamwork skills - Critical thinking - Creativity thinking - Technical Communication Skills - Communication and Presentation skills - Professional ethics and behave professionally
Content	Chapter 1: Introduction to Technology Chapter 2: The fields of Electronics - Telecommunications Teamwork skills Chapter 3: Basics of Microcontroller Effective learning Critical thinking skills Chapter 4: Basics of Printed Circuit Board Basics of Electronic - Telecommunications (P1) Problem solving and decision-making skills

	Chapter 5: Communication and presentation skills Technical communication, writing technical documents Chapter 6: Basics of Electronic - Telecommunications (P2) Creativity thinking Chapter 7: Basics of Electronic - Telecommunications (P3) Professional ethics Project management skills Chapter 8: Seminar
Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 10 - Labwork: Lab reports during the course: 20 - Seminar: Teamwork presentation: 20 - Final exam (60 minutes): Multiple-choice test: 50
Media employed	Text Books
Reading list	1. Introduction to Engineering – Pham Ngoc Tuan, Ho Thi Thu Nga, Do Thi Ngoc Khanh, Tran Dai Nguyen, Nguyen Van Tuong, Nguyen Minh Ha – Vietnam National University Ho Chi Minh City Press, 2014.

Written/updated by
Le Duc Hung

3. Programming Technique of Electronics – Telecommunications Branch - ETC00005

Module name	Programming Technique of Electronics – Telecommunications Branch
Module level, if applicable	Undergraduate
Code, if applicable	ETC00005
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	1st semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	MSc. Cao Tran Bao Thuong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 90 hrs Assignments and independent learning: 180 hrs Total Working hours: 270 hrs
Credit points	4
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	<i>None</i>
Module objectives/intended learning outcomes	- Describe the principles of object oriented programming - Understand basic program constructs in C/C++ - Ability to implement software in C/C++ language - Ability to develop applications with C/C++ language.
Content	Chapter 1: An Introduction to Programming Chapter 2: Variables and Constants Chapter 3: The Selection Structure Chapter 4: The Repetition Structure Chapter 5: Functions Chapter 6: Arrays Chapter 7: Strings Chapter 8: Files Chapter 9: Struct
Study and examination requirements and forms of examination	- Assignments: 1 problems during the course: 20 - Mid-term exam (45 minutes): Multiple choice exam or Written exam: 30 - Final exam (60 minutes): Written exam: 50
Media employed	Text Books
Reading list	Nhập môn lập trình , Trần Đan Thu, Nguyễn Thanh Phương, Đinh Bá Tiến, Trần Minh Triết, nxb KH và KT, 2011.

4. Digital Electronics- ETC00002

Module name	Digital Electronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC00002
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	2nd semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	PhD. Bui Trong Tu
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Electronics.
Module objectives/intended learning outcomes	- Calculate number conversion between different number systems. - Derive digital circuits from Boolean equations and the truth table. - Use Karnaugh maps to optimize combinational logic, including incompletely specified logic. - Analyze characteristic tables and equations and timing diagram of latches and flipflops. - Create state diagram from sequential circuits and design synchronous and asynchronous sequential circuits from state diagrams.
Content	Chapter 1: Number Representation, Arithmetic and Logic Circuits: Radix Number Conversion (Binary, Octal and Hex) Binary Arithmetic (Unsigned and Signed) Chapter 2: Algebra and Logic Gates: Truth Tables Algebraic Reduction/Expansion to Standard Forms Chapter 3: Logic Minimization: Boolean algebra

	Karnaugh maps Chapter 4: Common Combinational Components: Adder, Subtractor, Multiplexer, Demultiplexer, Encoder, Decoder, Converter, Comparator Chapter 5: Storage Components and Counters: Latches and Flip-Flops Registers and Memory Counters Other Examples
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30 - Mid-term exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30 - Final exam: Written exam, a four-page cheatsheet per student is allowed in the exam room: 40
Media employed	Text and Reference Books
Reading list	1. Nguyễn Hữu Phương, Mạch số, NXB Thống kê, 2000. 2. Thomas L. Floyd, Digital Fundamental, Prentice Hall. 3. Tokheim Roger L., Digital electronics, Glencoe/Mcgraw-Hill.

Written/updated by
PhD. Bui Trong Tu

5. Basic Electronics Lab- ETC00081

Module name	Basic Electronics Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC00081
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	1st semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	MSc. Nguyen Thi Thien Trang MSc. Bui An Dong Mr. Phan Viet Dung
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Electronics
Module objectives/intended learning outcomes	- Understand the fundamentals of electronics, diode circuits and models, BJT and FET transistor circuits and models, analog ICs and logic ICs - Know how to use electronic devices and analyze basic circuits - Perform small-signal analysis or equivalent circuit for electronic circuits - Design electronic circuits with specific description - Know how to design, debug, solder, and build electronic circuits - Demonstrate the systematization of knowledge
Content	Chapter 1: Electronic circuit Fundamentals Introduce of passive and active electronic devices using in lab, bread boards, power sources, VOM, function generators, oscilloscopes. Chapter 2: PN junction – Rectifiers with filter Half-wave rectifier with and without filter part, full-wave rectifier with two diodes or bridge network. Chapter 3: Regulators Regulators using Zener diode and transistor BJT. Chapter 4: Small-signal amplifiers using BJT Chapter 5: Analog ICs and applications

	Inverting amplifiers, non inverting amplifiers. Chapter 6: Digital ICs and applications NOR gate, EXOR gate, multiplexer and demultiplexer. Chapter 7: Circuit implementation using simulation tools – Proteus Introduction, basic guidance, simulation with diode applications Chapter 8: Circuit implementation using simulation tools – Proteus (cont) Simulation with small-signal amplifier using BJT and FET Chapter 9: Circuit implementation using simulation tools – Proteus (cont) Simulation with analog and logic ICs Chapter 10: Review week
Study and examination requirements and forms of examination	Daily quizzes: 6 quizzes during the course:15 Assignment: 3 reports during the course: 25 Final exam: 60
Media employed	Text Books
Reading list	1. Fundamentals of electric circuits, Charles Alexander and Matthew Sadiku, McGraw-Hill, 2009 2. Electronic devices and circuit theory, Robert L. Boylestad and Louis Nashelsky, Pearson, 2013.

Written/updated by
 Nguyen Thi Thien Trang

6. Analog Electronics- ETC00004

Module name	Analog Electronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC00004
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	MSc. Nguyen Thi Hong Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Wednesday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Electronics, Digital Electronics.
Module objectives/intended learning outcomes	- Understand the basic concepts related to analog electronics – Op Amp - Explain the working of analog circuits, particularly of Op Amp based circuits - Understand and explain the datasheet of analog ICs and apply to design of specific circuits - Describe the working of Active filters, Oscillators and Linear-Digital Ics in communication - Assemble related documents, imitate and calibrate analog circuit designs, combine theory and tools - Demonstrate the systematization of knowledge
Content	Chapter 1: Op-Amp Fundamentals Analysis of the Differential Amplifier circuits, Op Amp basics, Op Amp specification - DC Offset Parameters & Frequency Parameters, Op Amp unit specification – IC 741 (datasheet, electrical and operating characteristics, equivalent and detail circuits). Chapter 2: Op-Amp Applications Summing amplifier, Difference amplifier, Integrator, Differentiator, and applications. Instrumentation amplifier and applications, V-to-I and I-to-V converters. Chapter 3: Active Filter Design

	All types of filter responses, First-order and Second-order active filters of Low-pass filter, High-pass filter, Band-pass filter and Band-stop filter, sensitivity analysis. Chapter 4: Feedback and Oscillator circuits Introduction to Feedback circuits, Phase-Shift Oscillator, Wien Bridge Oscillator, Tuned Oscillator Circuit, Crystal Oscillator. Chapter 5: Linear-Digital ICs Introduction to V-to-F and F-to-V converters, ADC and DAC, Phase-Locked Loop (operation and application).
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20 - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30 - Final exam (60 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 50
Media employed	Text and Reference Books
Reading list	1. Bộ Khuếch Đại Xử Lý & IC Tuyến Tính, translated by Trinh Trung Thanh, Nguyen Huu Ngoc and Dang Van Su, NXB Khoa Học Và Kỹ Thuật, 1994.

Written/updated by
 Nguyen Thi Hong Ha

7. Analog and Digital Electronics Lab- ETC00082

Module name	Analog and Digital Electronics Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC00082
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	3rd semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	MSc. Nguyen Thi Thien Trang MSc. Bui An Dong Mr. Phan Viet Dung
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Electronics, Digital Electronics, Analog Electronics
Module objectives/intended learning outcomes	- Understand the basic concepts related to analog electronics and digital electronics - Explain and analysis the working principle of analog and digital circuits - Implement analog and digital circuits for various parameter measurements - Know how to design, debug, solder, and build analog and digital electronic circuits - Demonstrate the systematization of knowledge
Content	Chater 1: Logic gates and oscillators using NAND gates and IC 555 Chater 2: Flip Flop Counters using IC 7476, 7474, 7490 and 74193. Chater 3: Shift registers, Ring counter and Johnson counter – Twisted Ring counter Chater 4: Encoder, decoder, multiplexer and demultiplexer Chater 5: Priority encoder, BCD decoder Chater 6: Applications of OP AMP Chater 7: Oscillators Phase-shift oscillators, Wien bridge oscillators

	Chater 8: Active filters – High-pass, low-pass, bandpass, bandstop filters Chater 9: Trigger Schmitt, Exact rectifier Chater 10: DAC (Digital to Analog Converter)
Study and examination requirements and forms of examination	- Daily quizzes: 8 quizzes during the course: 20 - Assignment: 2 reports during the course: 20 - Final exam: 60
Media employed	Text Books
Reading list	1. Electronic devices and circuit theory, Robert L. Boylestad and Louis Nashelsky, Pearson, 2013. 2. Electronic Circuit Analysis and Design 2nd edition, Donald A. Neamen, McGraw-Hill, 2001.

Written/updated by
 Nguyen Thi Thien Trang

8. Physics of Electronic Devices- ETC10001

Module name	Physics of Electronic Devices
Module level, if applicable	Undergraduate
Code, if applicable	ETC10001
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	MSc. Nguyen Thi Thien Trang
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Electronics, Digital Electronics.
Module objectives/intended learning outcomes	- Understand the semiconductor theory, the structure and operating principles of semiconductor devices - Understand and analyze the working principles of semiconductor device applications, both in DC and AC analysis - Analyze electrical properties and design simple circuits to given design specification - Demonstrate the systematization of knowledge
Content	Chapter 1: Semiconductor theory - Intrinsic and Extrinsic Properties of Semiconductors Chapter 2: PN Junction – Diodes and applications Chapter 3: Bipolar Junction Transistor Chapter 4: Field Effect Transistor Chapter 5: Thyristor Chapter 6: Opto-Electronics and applications
Study and examination requirements and forms of examination	- Daily quizzes: 6 quizzes during the course: 10 - Assignments: 3 problems during the course: 20 - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30 - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40
Media employed	Text Books

Reading list	1. Semiconductor Physics and Devices - Basic Principles, Donald A. Neamen, McGraw-Hill, 2002. 2. Electronic devices and circuit theory, Robert L. Boylestad and Louis Nashelsky, Pearson, 2013.
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Written/updated by
 Nguyen Thi Thien Trang

9. Signals, Systems, and Circuit Analysis- ETC10002

Module name	Signals, Systems, and Circuit Analysis
Module level, if applicable	Undergraduate
Code, if applicable	ETC10002
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	MSc. Tran Xuan Tan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 08:30 – 11:00 Monday 12:30 – 15:00
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Maths, Physics, Basic Electronics.
Module objectives/intended learning outcomes	- Express basic concepts of signals, systems, Laplace transform, response of system in time domain, response of system in frequency domain. - Express technical documentations in English that used in this course. - Apply electrical circuit laws and theorems to calculate values in electrical circuits - Analyze electronic circuits - Appraise electrical circuits: stable of system, response of system in time domain and frequency domain.
Content	Chapter 1: * Signals - Overview of signals - Mathematical formulas * Laplace transform * Systems - Overview of systems - Classification - Description of system in time domain - Description of system in frequency domain - Discrete-Time systems Chapter 2: Electrical circuit laws and theorems

	- Ohm's Law, Kirchhoff's Law, Superposition, Thévenin's theorem, Norton's theorem, Miller's theorem - Base analysis: Node-voltage analysis, Mesh-current analysis Chapter 3: Response of system in time domain - Models of passive components R,C,L using Laplace transform - Transitional problem in RC and RL circuits: Impulse Response, Transfer Function, response of circuits that are applied DC and AC input signals Chapter 4: Response of system in frequency domain - Poles and Zeros - Stable of system - Stability criterions - Frequency response, Bode plot - Filter Circuits Chapter 5: Two-port networks - z-parameters, y-parameters, h-parameters, A-parameters - Interconnection of two-port networks - Feedback
Study and examination requirements and forms of examination	- Reinforcement exercises in each lesson: 30 - Attendance: 30 - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40
Media employed	Text Books
Reading list	1. Phương Xuân Nhân, Hồ Anh Tuý, Lý thuyết mạch (tập 1), NXB KHKT 2008. 2. Phương Xuân Nhân, Hồ Anh Tuý, Lý thuyết mạch (tập 2), NXB KHKT 2007.

Written/updated by
Tran Xuan Tan

10. Measurement and Instruments- ETC10003

Module name	Measurement and Instruments
Module level, if applicable	Undergraduate
Code, if applicable	ETC10003
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Prof. PhD. Dinh Sy Hien
Lecturer	Prof. PhD. Dinh Sy Hien
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hours/week Office hours: Thursday: 7:30 AM – 11:30
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	None
Module objectives/intended learning outcomes	- Understand the basic concepts and characteristics of measurement instruments. - Analyze, classify the working principles, structures of various types of measurement instruments along with their applications. - Demonstrate, imitate and analyze some applications of measurement instruments using LabVIEW software development tool.
Content	Chapter 1: Basic Concepts of Measurements Introduction 1.1 Configuration of measurement systems 1.2 Classification of transducers 1.3 Characteristics of measurement instruments Chapter 2: Strain Gauge Introduction 2.1 Types of measurement devices of distortion 2.2 Working principle of measurement devices of resistive distortion 2.3 Types of measurement devices of electric distortion 2.4 Electric circuits of distortive measurements Chapter 3: Temperature Measurements Introduction 3.1 Temperature scale 3.2 Mechanic temperature sensors 3.3 Electric temperature sensors 3.4 Thermistor

	3.5 Thermocouple
	3.6 Semiconductor sensors
	3.7 Quartz thermometer
	3.8 Temperature measurements using irradiation methods
	3.9 High temperature thermometers
	Chapter 4: Optical Sensors
	4.1 Phototransistors
	4.2 Solar cells
	4.3 Opto-couplers
	Chapter 5: Displacement
	5.1 Potentiometers
	5.2 Inductive sensors
	5.3 Capacitive sensors
	5.4 Hall effect sensors
	5.5 Level measurements
	Chapter 6: Operational Amplifiers
	6.1 Block-diagram of operational amplifiers
	6.2 Feedback principle
	6.3 Frequency response
	6.4 Buffer amplifier
	6.5 Noninverting amplifier
	6.6 Inverting amplifier
	6.7 Summing amplifier
	6.8 Subtract amplifier
	6.9 Differentiator and integrator
	6.10 Logarithmic amplifier and anti-logarithmic amplifier
	Chapter 7: Oscillators
	7.1 Characteristics of oscillators
	7.2 Condition of oscillation
	7.3 Circuits of RC oscillators
	7.4 Circuits of LC oscillators
	7.5 Quartz oscillator
	7.6 Functional generators
	Chapter 8: Filters
	8.1 Active and passive filters
	8.2 Types of filters
	Chapter 9: Data Conversion
	9.1 Digital to analog converters
	9.2 Analog to digital converters
	Chapter 10: Electronic Testing Instruments
	10.1 Oscilloscopes
	10.2 Digital Multimeters
	10.3 Electronic counters
	10.4 Signal spectrum analyzers
	10.5 Functional generators
	Chapter 11: Microprocessor Based Measurement Instruments
	11.1 Overview of microprocessors

	11.2 Microprocessor based measurement instruments 11.3 LabVIEW software development tool 11.4 PC based measurement instruments
Study and examination requirements and forms of examination	- Assignment 30 - Midterm exam 30 - Final exam 40
Media employed	Text Books
Reading list	1. Dinh Sy Hien, Sensors and measurement instrumentation, VNU HCMC publishing, 2004 (Vietnamese Language), recommended textbook. 2. Robert B. Northrop, Introduction to instrumentation and measurements, Second Edition, Taylor & Francis Group, 2005.

Written/updated by Dinh Sy Hien

11. Measurement Lab - ETC10004

Module name	Measurement Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10004
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	MSc. Le Trung Khanh
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Measurement and Instruments, Basic Electronics Lab, Analog and Digital Electronics Lab, Introduction to Engineering, Programming Technique of Electronics – Telecommunications Branch
Module objectives/intended learning outcomes	- Identify and analyze the characteristics and working principles of various kinds of sensors - Manipulate of popular measurement instruments - Design schematic of some popular sensors - Program measurement instruments
Content	Chapter 1: Voltage and current measurements Chapter 2: Design DC voltage supply with dedicated IC Chapter 3: Sensor: photoresistor Chapter 4: Sensor: phototransistor Chapter 5: Current transformer and AC current measurement Chapter 6: Sensor: semiconductor temperature sensor Chapter 7: Sensor: thermocouple Chapter 8: Sensor: Hall effect sensor Chapter 9: Current measurement using Hall effect sensor Chapter 10: Review
Study and examination requirements and forms of examination	- Assignment: 30 - Final exam: 70

Media employed	Text Books
Reading list	- Thực hành Đo, Lê Trung Khanh

Written/updated by Trung-Khanh

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12. Computer structure- ETC10005

Module name	Computer structure
Module level, if applicable	Undergraduate
Code, if applicable	ETC10005
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Tuesday 08:00 AM – 11:00 AM Friday 08:00 AM – 11:00 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Digital electronics, Programming Technique
Module objectives/intended learning outcomes	- Understand data representation and operations - Identify the computer Organization - Analyze and design of ALU - Simulate and build the instruction set - Assemble the language programming
Content	Chapter 1: Basic components of a computer systems Chapter 2: Instructions: Language of the computer Chapter 3: Arithmetic for computers Chapter 4: The Processor: Datapath and Control Chapter 5: Interfacing CPU, Memory and peripherals Lab: SoPC and Nios IDE programming environment Assembly language programming Instruction Set Debug and Memory I/O communication Device control
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30 - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheat sheet per student is allowed in the exam room: 20

	- Final exam (90 minutes): Written exam, a four-page cheat sheet per student is allowed in the exam room: 50
Media employed	Text Books
Reading list	1. David A. Patterson, John L. Hennessy, Peter J. Ashenden, James R. Larus, Daniel J. Sorin, Computer Organization and Design: The Hardware/Software Interface, Morgan Kaufmann, Third Edition 2008. References 2. S. Brown and Z. Vranesic, Fundamentals of Digital Logic with Verilog Design, Mc Graw Hill, Second Edition, 2008 3. Document of CPU NIOS II at www.altera.com 4. Document of DE2 board computer Organization Labs at http://www.altera.com/education/univ/materials/boards/unv-de2-board.html

Written/updated by
Huynh Huu Thuan

13. Computer Structure Lab-ETC10006

Module name	Computer Structure Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10006
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Bachelor Dang Tan Phat
Lecturer	MSc. Tran Tuan Kiet
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2 hours/week Office hours: Based on the timetable made by the teacher for the students.
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Digital Electronics, Programming Techniques.
Module objectives/intended learning outcomes	- Understand and clarify the architecture and the operation of CPU, specific for NIOS II CPU. - Understand the instruction set of CPU NIOS II and differentiate many kind of instruction and the usage of them. - Investigate the structure and syntax of assembly programming language and use it to program CPU NIOS II. - Explain the operations of co-microprocessors based on the specification and explore how to control them through the CPU. - Distinguish stack definitions from queues, subfunctions, and exceptions. - Design software by assembly language to control the peripheral devices and submodules of the DE2-115 Computer system.
Content	Chater 1: CPU NIOS II and DE2-115 COMPUTER system - Get an overview of the NIOS II CPU. - Get an overview of the DE2-115 COMPUTER system. - Get acquainted with the programming tool and run the assembly program of the NIOS II CPU on the DE2-115 COMPUTER system. Chapter 2: CPU NIOS II instruction set

	- Investigate the instructions and the process of reading and decoding instructions of the NIOS II CPU. - Familiarize with arithmetic and logic calculation instructions, comparison, and branching statements. Chapter 3: Guidance for use of simulation software - How to use the nios2simulator simulation support tool. - Steps to find and fix the error of a simple program using the assembly instruction set of the Nios II CPU. Chapter 4: Communication with peripheral device - Help students understand how to design assembly programs interface between the NIOS II CPU and peripheral components including LEDs, SWITCHs, BUTTONs, seven-segment LEDs. Chapter 5: Stack and sub-function and investigate of JTAG_UART - Stack architecture and NIOS II CPU operation every time a subfunction is called. - How the JTAG_UART module works. Chapter 6: Exceptions - The workflow of NIOS II CPU whenever exceptions occur. - Investigate the operation of Interval_Timer and Parallel ports. - How to configure and handle interrupt of Interval_Timer and Parallel ports.
Study and examination requirements and forms of examination	- Diligence score: Students attend more than 80% of course: 20 - Assignments: 2 exercises per a lesson, total of 14 exercises during the course: 40 - Final exam (60 minutes): Write two programs by assembly programming to control NIOS II CPU for certain requirements: 40
Media employed	Text Books
Reading list	1. Computer Organization and Design, 2nd edition, David Patterson & John Hennessy. 2. Computer Architecture, John Hennessy & David Patterson. http://www.altera.com/education/univ/materials/boards/unv-de2-board.html

Written/updated by
Tran Tuan Kiet

14. Basic Computer Networks- ETC10007

Module name	Basic Computer Networks
Module level, if applicable	Undergraduate
Code, if applicable	ETC10007

Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Nguyen Viet Ha, Ph.D.
Lecturer	Nguyen Viet Ha, Ph.D.
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	<i>None</i>
Module objectives/intended learning outcomes	-Understand the network definitions (layers, packet, link parameters, datagram forwarding, topology, congestion, etc). -Differentiate the Ethernet standards and switch types. Analyze the packet matters such as delay, size, error detection, etc. -Generalize the IPv4 and IPv6 and their related protocols. -Understand the UDP and TCP transportation. -Understand TCP Reno and congestion management algorithm.
Content	Chapter: 1: Overview of Network. Introduce the computer network definitions. Study the network parameters that affect the transmission performance (data rate, throughput, bandwidth, delay, and congestion). Study the Datagram Forwarding and Topology. Introduce the LAN and Ethernet. Introduce the network protocol suites (OSI, TCP/IP) Chapter 2: Ethernet. Differentiate the Ethernet standards. Study the media collision detection/avoidance algorithm. Study the switching operation. Differentiate the Ethernet switch types. Chapter 3: Packet. Study how to calculate the packet delay and RTT. Discuss how the packet size affect the transport performance. Study the packet error detection algorithm. Chapter 4: IP version 4. Introduce the IPv4 header, fragmentation.

	Introduce the IPv4 addresses. Study IPv4 subnetting. Introduce the IPv4 Address Resolution Protocol, Dynamic Host Configuration Protocol, Internet Control Message Protocol. Chapter 5: IP version 6. Introduce the IPv6 standard header and extension headers. Introduce the IPv6 addresses. Introduce the IPv6 neighbor discovery and host address assignment. Study IPv6 subnetting. Introduce the IPv6 Dynamic Host Configuration Protocol, Internet Control Message Protocol. Chapter 6: UDP transport. Introduce the UDP related protocols. Introduce the UDP application examples. Discuss the UDP fundamental transport issues. Chapter 7: TCP transport. Introduce the TCP header. Study the TCP connection establishment, timeout, retransmission. Study the path MTU discovery function. Study the TCP flow control. Chapter 8: TCP Reno and Congestion Management. Study the slow start, fast retransmission, fast recovery algorithms. Study the TCP NewReno and its partial ACK function. Study the TCP Selective Acknowledgments.
Study and examination requirements and forms of examination	In-class assignments: Quiz, problem-solving, discussion: 20 Attendance: attendance all the lectures: 10 Presentation: groups of students study the new network definition and do the presentation: 20 Final exam: single choice quiz: 50
Media employed	Text Books
Reading list	Peter L Dordal, An Introduction to Computer Networks, open source and available on http://intronetworks.cs.luc.edu/ , Sep. 2021.

Written/updated by

Nguyen Viet Ha, Ph.D.

15. Basic Computer Network Lab- ETC10008

Module name	Basic Computer Networks Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10008
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	MSc Ngo Thanh Hai
Lecturer	MSc Ngo Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	None
Module objectives/intended learning outcomes	- Understand the network definitions (layers, packet, link parameters, datagram forwarding, topology, congestion, etc). - Differentiate the Ethernet standards and switch types. - Analyze the packet matters such as delay, size, error detection, etc. - Generalize the IPv4 and IPv6 and their related protocols. - Understand the UDP and TCP transportation. - Understand TCP Reno and congestion management algorithm.
Content	Chapter 1 : IP version 4 Subnetting IPv4 to establish a logical network model Chapter 2 : Basic configurations of Cisco network devices + Modeling network models in Cisco packet tracer software consists of Router, Switch, PC, and Server + Configuring some basic commands Chapter 3 : Study some basic services of computer networks + Modeling network models in Cisco packet tracer software consists of Router, Switch, PC, and Server

	+ Configuring and using services such as WEB, DNS, MAIL, and FTP - Chapter 4 : IP version 6 + Modeling network models in Cisco packet tracer software consists of Router, Switch, PC, and Server + Adopting IPv6 to network devices + Configuring and using services such as WEB, DNS, MAIL, and FTP with IPv6 - Chapter 5 : Wireless router configurations + Setting up a network model in Cisco packet tracer software with the wireless router + Configuring wireless router - Chapter 6 : Static routing + Setting up a network model of small office + Configuring static routing - Chapter 7 : DHCP and NAT services + Setting up a network model of small office + Configuring static routing + DHCP and NAT configuration - Chapter 8 : Summary lab + Modeling a network model of a small office with network devices on IPv4 and IPv6 + Static Routing + Implement some network services such as DHCP, NAT, Web, DNS, Mail,... + Troubleshooting problems regarding network configurations
Study and examination requirements and forms of examination	- Attendance: 15 - Lab reports: 35 - Final lab exam: 50
Media employed	Text Books
Reading list	Peter L Dordal, An Introduction to Computer Networks, opensource and available on http://intronetworks.cs.luc.edu/ , Sep. 2021.

Written/updated by

Ngo Thanh Hai

16. Microcontroller-ETC10009

Module name	Microcontroller
Module level, if applicable	Undergraduate
Code, if applicable	ETC10009
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	
Lecturer	MSc Nguyen Quoc Khoa
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hours/week Office hours: Thursday 08:30AM – 11:30AM Saturday 08:30AM – 11:30AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Students are encouraged to have basic knowledge in Computer Architecture; Assembly & C Programming; Analog & Digital Processing.
Module objectives/intended learning outcomes	- Understanding of basic concepts of Microprocessor/ Microcontroller - Reading, understanding of English technical documents that used in this course - Know how to program microcontroller using Assembly/ C Language - Know how to communicate with I/O Devices - Know how to design a control system using Microcontroller
Content	1: Introduction to the PIC 18 Microcontroller Reading: Chapter 1 (except section 1.11) 2: PIC18 Assembly Language Programming and Tools Reading: Chapters 2 and 3 (except sections 3.4, 3.7, 3.8) 3: PIC18 Assembly Language Programming and Tools (cont') Reading: Chapters 2 and 3 (except sections 3.4, 3.7, 3.8) 4: C Programming for the PIC18 Reading: Chapter 5 (except sections 5.17, 5.19) 5: Parallel Ports

	Reading: Chapter 7 (except sections 7.7, 7.10, 7.11) 6: Interrupts and Resets Reading: Chapter 6 (except section 6.7) 7: Timers and CCP Modules Reading: Chapter 8 (except sections 8.7, 8.8) 8: Timers and CCP Modules (cont') Reading: Chapter 8 (except sections 8.7, 8.8) 9: Serial Interface Reading: Chapter 9 10: Analog-to-Digital Conversion Reading: Chapter 12 (except sections 12.15 □ 12.18) 11: Final Exam
Study and examination requirements and forms of examination	- Assignments: 5 mini assignments during the course: 50 - Final exam (60 minutes): written exam, all hard and soft materials are allowed in the exam: 50
Media employed	Text Books
Reading list	PIC Microcontroller: An Introduction to Software & Hardware Interfacing, Han-Way, Thompson, 2005

Written/updated by

Nguyen Quoc Khoa

17. Microcontroller Lab-ETC10010

Module name	Microcontroller Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10010
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	4th semester
Person responsible for the module	Bachelor Dang Tan Phat
Lecturer	MSc. Tran Tuan Kiet
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2 hours/week Office hours: Based on the timetable made by the teacher for the students.
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Digital Electronics, Programming Techniques.
Module objectives/intended learning outcomes	- Understand and explain the datasheet of Microcontroller families. - Distinguish the definition and works of firmware/soft and hardware in a microcontroller system. - Program a firmware and use it to control the microcontroller's hardware for a certain purpose. - Verify the realistic operations of the microcontroller based on the EDA tools and support equipment. - Analyze and characterize the suitable hardware features to meet requirements. - Design a digital system to use the microcontroller to handle real problems.
Content	Chapter 1: Introduction - Introduce the microcontroller which used in the course. - Use EDA tools and simulation tools to simulate microcontrollers. Chapter 2: I/O Ports - Learn how to use the development board. - Learn the way to read and understand the datasheet. - Analyze and explain the function of the I/O port.

	- Differentiate I/O port modes and use firmware to control them. Chapter 3: Interrupt - Describe the definition of interrupt and the workflow of interrupt handling in microcontrollers. - Figure out the way to configure the interrupt controller in the microcontroller. - Program I/O ports with interrupt feature. Chapter 4: Timer - Analyze and explain the function of Timer types from the datasheet. - Use the firmware to control the Timer module to work in the various modes it supports. - Program I/O ports with interrupt feature. Chapter 5: ADC - Short describes the definition of Analog to Digital converter. - Determine the type of ADC used in the microcontroller. - Analyze and figure out the operation of ADC and the way to control it. - Use the firmware to control the Timer ADC to read the temperature sensor and convert it to a digital number from an analog signal. Chapter 6 UART - Describe the protocol of UART and the way to control it from the datasheet. - Program the firmware to transmit and receive data by UART protocol. - Verify the operation of the UART core in the microcontroller by Oscilloscope. Chapter 7 SPI - Describe the protocol of SPI and the way to control it from the datasheet. - Program the firmware to transmit and receive data by SPI protocol. - Verify the operation of SPI core in microcontroller by Oscilloscope. - Compare UART and SPI protocols on advantages and disadvantages.
Study and examination requirements and forms of examination	- Diligence score: Students attend more than 80% of course: 20 - Assignments: 2 exercises per a lesson, total of 14 exercises during the course: 40 - Project: Students make groups that include 5 people and choose an item from the teacher project list. The product of a group includes a document that describes their system and a digital system use microcontroller to process a certain problem: 40

Media employed	Text Books
Reading list	PIC M-icrocontroller: An Introduction to Software & Hardware Interfacing, Han-Way, Thompson, 2005

Written/updated by
Tran Tuan Kiet

18. Numerical Analysis and Matlab- ETC10011

Module name	Numerical Analysis and Matlab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10011
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	MSc. Huynh Quoc Thinh
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hours/week Office hours: Monday 07:30 AM – 9:30 AM Thursday 07:30 AM – 9:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 145 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Linear algebra, Differential integral.
Module objectives/intended learning outcomes	- Use the Matlab command line, graph, and GUI proficiently. - Demonstrate understanding of common numerical algorithms and application. - Apply numerical methods to obtain approximate solutions to mathematical problems. - Implement numerical methods in Matlab. - Analyze and evaluate the accuracy of common numerical methods. - Apply numerical methods to solve practice problems.
Content	Chapter 1: Matlab Basic - Variables - Array, vector, matrix - Operators, expressions - Complex numbers -Language structure, condition, loop, ... -Matlab function - Graph 2D, 3D -File input/output -Symbolics, GUI, Simulink Chapter 2: Approximations and Round-Off Errors - Accuracy and precision - Absolute and relative error - Truncation Errors and the Taylor Series

	Chapter 3: Approximate real roots of equations -Bracketing methods: Graphical methods, bisection -Open methods: Fixed-point iteration, Newton- Chapter 4: Raphson, Secant Method Chapter 5: Linear algebraic equations - Gauss Elimination - LU Decomposition and Matrix Inversion - Cholesky method - Jacobi -Gauss-Seidel Chapter 6: Curve fitting - Least-Squares Regression - Linear regression - Polynomial regression - Nonlinear regression Chapter 7: Interpolation - Newton polynomial - Lagrange polynomial -Spline Chapter 8: Numerical differentiation and integration - Taylor Series -High-accuracy formulas - Richardson extrapolation - Newton-Cotes Integration Formulas - Trapezoidal rule - Simpson's rules - Gauss quadrature Chapter 9: Ordinary differential equations - Euler's method - Heun and midpoint methods - Runge-Kutta method
Study and examination requirements and forms of examination	- Assignments: three problems during the course: 30 - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheat sheet per student is allowed in the exam room: 30 - Final exam (90 minutes): Written exam, a four-page cheat sheet per student is allowed in the exam room: 40
Media employed	Text Books
Reading list	1. <i>Phương pháp tính</i>, Tạ Văn Đình, NXB Giáo dục, 2011 2. <i>Numerical methods for engineers</i>, Steven C. Chapra, Raymond P. Canale, McGraw-Hill, 2010.

Written/updated by

Huynh Quoc Thinh

19. Numerical Analysis and MATLAB Lab- ETC10012

Module name	Numerical Analysis and Matlab Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10012
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	B.S. Thai Hong Hai
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Labs: 8 hrs/week (2 classes divided into 4 shifts)
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Linear Algebra, Calculus, Numerical Analysis and MATLAB
Module objectives/intended learning outcomes	- Understand and use MATLAB at basic level - Read and apply instructions in MATLAB Helps and other English documentations - Write MATLAB programs to deploy numerical analysis methods - Analyze and choose optimal solutions for calculation problems - Create graphics user interface (GUI) with MATLAB
Content	Chapter 1: MATLAB's basic commands and applying in circuit analysis. Chapter 2: Calculate approximate root of equations, polynomial's interpolation and approximation. Chapter 3: Calculate derivatives and integrals approximately with MATLAB. Chapter 4: Solving differential equations approximately with MATLAB. Chapter 5: MATLAB GUIDE: Designing numerical analysis application with user interface.
Study and examination requirements and forms of examination	- Assignments: 5 reports for 5 chapters: 50 - Final exam (30 minutes): Writing a program following requirements listed in exam paper: 50
Media employed	Text Books
Reading list	1. Essential MATLAB for Engineers and Scientists, Brian D.Hahn, Daniel T.Valentine, Elsevier, 2007.

	2. Numerical methods for engineers, Steven C. Chapra, Raymond P. Canale, McGraw-Hill, 2010.
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Written/updated by
Thai Hong Hai

20. Digital Signal Processing- ETC10013

Module name	Digital Signal Processing
Module level, if applicable	Undergraduate
Code, if applicable	ETC10013
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	MSc. Tran Thi Huynh Van
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 12:30 PM – 14:30 PM Friday 12:30 PM – 14:30 PM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Analog Electronics
Module objectives/intended learning outcomes	- Understand and explain discrete – time signals and systems - Analyze signal processing and systems in time and frequency domain - Analyze and calculate Discrete Fourier Transform, Fast Fourier Transform and Z transform - Demonstrate and generalize FIR and IIR filters - Describe and apply filter implementation
Content	Chapter 1: Discrete – time signals and systems Chapter 2: Analysis in time domain Chapter 3: Analysis in frequency domain Z-transform Chapter 4: Design of FIR filters Chapter 5: Design of IIR filters Chapter 6: Filter implementation and finite word-length effect Chapter 7: Discrete Fourier Transform and Fast Fourier Transform.
Study and examination requirements and forms of examination	- Assignments: exercises of each chapter: 30 - Mid-term exam (60 minutes): Multiple choice exam: 20 - Final exam (90 minutes): Multiple choice and written exam: 50
Media employed	Text Books

Reading list	1. <i>“Digital Signal Processing”</i>, Nguyen Huu Phuong, 2012 2. <i>“Lọc số FIR và IIR”</i>, Nguyen Huu Phuong, Vietnam National University Ho Chi Minh City Publishing House, 2014 3. <i>“Biến đổi z trong xử lý tín hiệu số”</i>, Nguyen Huu Phuong, Vietnam National University Ho Chi Minh City Publishing House, 2014 4. <i>“Xử lý tín hiệu số”</i>, Nguyen Huu Phuong, Statistical publisher, 2003
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Written/updated by

Tran Thi Huynh Van

21. Digital Signal Processing Lab-ETC10014

Module name	Digital Signal Processing Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10014
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	Nguyen Thai Cong Nghia Nguyen Thi Xuan Uyen
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 9:00 AM – 11:30 AM Thursday 9:00 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Digital Signal Processing
Module objectives/intended learning outcomes	- Understand and explain discrete – time signals and systems - Analyze signal processing and systems in time and frequency domain - Analyze and calculate Discrete Fourier Transform, Fast Fourier Transform and Z transform - Demonstrate and generalize FIR and IIR filters - Describe and apply filter implementation
Content	Chapter 1: Discrete-time signals in time domain - Discrete signals (unit sample, unit step, unit ramp, sin, cos, complex...) - Nyquist Sampling Theorem and signal reconstruction Chapter 2: Discrete-time systems in time domain - Systems classification - Linear time-invariant systems Chapter 3: Discrete-time systems in time domain (cont) - Impulse response - Convolution Chapter 4: Analysis in frequency domain - Discrete-time Fourier transform - Signal analysis in frequency domain - Discrete Fourier transform

	Chapter 5: Analysis in frequency domain (cont) - Fast Fourier transform - Circular convolution - System analysis in frequency domain - Frequency response (amplitude response and phase response) Chapter 6: Design of FIR filters - Window method (triangle, Blackman, Hamming, Hann, Kaiser...) - Filter frequency response (lowpass, highpass, bandpass, bandstop, multi-band...) Chapter 7: Design of IIR filters - Filter types (Butterworth, Chebyshev, elliptic, Bessel...) - Filter frequency response (lowpass, highpass, bandpass, bandstop, multi-band...) - Comparison of FIR filters and IIR filters Chapter 8: Filter implementation - Audio signal analysis - Audio signal filtering
Study and examination requirements and forms of examination	- Attendance: 20 - Lab reports: 30 - Final lab exam: 50
Media employed	Text Books
Reading list	1. Nguyen Huu Phuong, Digital Signal Processing, 2012 2. Nguyen Huu Phuong, Lọc số FIR và IIR, Vietnam National University Ho Chi Minh City Publishing House, 2014 3. Nguyen Huu Phuong, Biến đổi z trong xử lý tín hiệu số, Vietnam National University Ho Chi Minh City Publishing House, 2014 4. Nguyen Huu Phuong, Xử lý tín hiệu số, Statistical publisher, 2003

Written/updated by
 Nguyen Thai Cong Nghia

22. Communication Systems-ETC10015

Module name	Communication Systems
Module level, if applicable	Undergraduate
Code, if applicable	ETC10015
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa MSc. Truong Tan Quang
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Signals, Systems and Circuit Analysis
Module objectives/intended learning outcomes	- Understand the functionalities of the main parts of communication systems - Analyze each signal in time and frequency domains in analog modulation systems - Calculate bit rate, baud rate, bandwidth of PCM signal and digital communication system - Calculate encoding and decoding for Shannon-Fano coding, Huffman coding, linear code, cyclic code, and convolutional codes - Build technical report on modulation and coding for preset communication systems.
Content	Chapter 1: Introduction Basic Block Diagram, Typical Communication systems Channels, Spectrum Allocation Chapter 2: Amplitude Modulations and Demodulations Reasons of modulation Different AMs: DSB-SC, DSB-FC, SSB, VSB, Demodulation Chapter 3: Angle Modulation and Demodulation Angle Modulation Characteristics FM and PM, Narrow Band and Wide Band FM, Angle Demodulation

	Chapter 4: Pulse Code Modulation and Multiplexing Digital transmission, multiplexing Signaling, TDM Hierarchy Chapter 5: Basic Digital Modulation The raised cosine filter ASK, OOK, MASK, FSK, MFSK, BPSK, DBPSK, MPSK, MQAM, Bandwidth Chapter 6: Introduction to Information Theory Entropy and Information rate, Shannon-Fano Coding, Huffman Coding Chapter 7: Error Correcting Codes Linear Code, Cyclic code, Convolutional Code, Viterbi algorithm
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 15 - Mid-term exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30 - Seminar: write a research report on the selected topic and give an oral presentation: 15 - Final exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 40
Media employed	Text Books
Reading list	1. B.P. Lathi, Zhi Ding, <i>Modern Digital and Analog Communication Systems, 4th edition</i>, Oxford University Press, 2010 2. John G. Proakis and Masoud Salehi, <i>Communication Systems Engineering, Second Edition</i>, Prentice Hall, 2002.

Written/updated by

Dang Le Khoa

23. Communication Systems Lab-ETC10016

Module name	Communication Systems Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10016
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa MSc. Ngo Thanh Hai Mr. Nguyen Thai Cong Nghia Mr. Ngo Minh Nghia Ms. Nguyen Thi Xuan Uyen
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Communication Systems
Module objectives/intended learning outcomes	- Understand the functionalities of the main parts of communication systems - Analyze each signal in time and frequency domains in analoge in analog modulation systems - Calculate bit rate, baud rate, bandwidth of PCM signal and digital communication system - Calculate encoding and decoding for Shannon-Fano coding, Huffman coding, linear code, cyclic code, and convolutional codes - Build technical report on modulation and coding for preset communication systems.
Content	Chapter 1: Introduction to Communication Systems Simulation Chapter 2: Amplitude Modulations and Demodulations Chapter 3: Angle Modulation and Demodulation Chapter 4: Pulse Code Modulation and Multiplexing Chapter 5: Basic Digital Modulation Chapter 6: Introduction to Information Theory Chapter 7: Error Correcting Codes

Study and examination requirements and forms of examination	- Lab report: 50 - Final exam: computer simulation and oral interview: 50
Media employed	Text Books
Reading list	1. B.P. Lathi, Zhi Ding, <i>Modern Digital and Analog Communication Systems, 4th edition</i>, Oxford University Press, 2010 2. John G. Proakis and Masoud Salehi, <i>Communication Systems Engineering, Second Edition</i>, Prentice Hall, 2002.

Written/updated by

Dang Le Khoa

24. Programmable Logic Design-ETC10017

Module name	Programmable Logic Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10017
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Le Duc Hung
Lecturer	PhD. Le Duc Hung – MSc. Cao Tran Bao Thuong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Wednesday 07:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Digital Electronics, Programming Techniques
Module objectives/intended learning outcomes	- Ability to design complete digital circuits - Ability to implement hardware based on HDL - Ability to have skills in forming ideas to design circuit boards based on - on digital ICs - Ability to plan and implement
Content	Chapter 1: Overview of Digital Design Chapter 2: Number base systems Chapter 3: Binary operations Chapter 4: Boolean Logic Minimization Chapter 5: Combinational Logic Design Chapter 6: Sequential Logic Design Chapter 7: Combinational Logic Design with Verilog HDL
Study and examination requirements and forms of examination	- Assignments: Problems and short tests during the course: 20 - Mid-term Exam (90 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 30 - Final exam (90 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 50
Media employed	Text Books
Reading list	1. Digital Logic Design with Verilog HDL – Le Duc Hung, Cao Tran Bao Thuong – Vietnam National University Ho Chi Minh City Press, 2020 (First Edition).

Written/updated by
Le Duc Hung,
Cao Tran Bao Thuong

25. Programmable Logic Design Lab-ETC10018

Module name	Programmable Logic Design Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10018
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Le Duc Hung
Lecturer	PhD. Le Duc Hung – MSc. Cao Tran Bao Thuong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Wednesday 07:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Digital Electronics, Programming Techniques
Module objectives/intended learning outcomes	- Ability to design complete digital circuits - Ability to implement hardware based on HDL - Ability to have skills in forming ideas to design circuit boards based on digital ICs - Ability to plan and implement
Content	Chapter 1: Switches, LEDs, and Buttons Chapter 2: Multiplexers and Seven-segment display decoders Chapter 3: Arithmetic Circuits Chapter 4: Comparators Chapter 5: Latches and Flip-Flops Chapter 6: Counters Chapter 7: Shift Registers Chapter 8: ALU Chapter 9: CPU
Study and examination requirements and forms of examination	- Assignments: Problems and short tests during the course: 20 - Mid-term Exam (90 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 30 - Final exam (90 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 50
Media employed	Text Books
Reading list	1. Thiết kế logic mạch số với Verilog HDL – Lê Đức Hùng, Cao Trần Bảo Thương – NXB ĐHQG TP.HCM, 2020 (Tái bản lần 1).

Written/updated by
Le Duc Hung,
Cao Tran Bao Thuong

26. Introduction to Electronics and Telecommunications-ETC10019

Module name	Introduction to Electronics and Telecommunications
Module level, if applicable	Undergraduate
Code, if applicable	ETC10019
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	5th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 15 hrs Assignments and independent learning: 30 hrs Total Working hours: 45 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Introduction to Engineering
Module objectives/intended learning outcomes	- Explain the learning outcomes of electronics and telecommunications program - Explain the process of product design - Introduction to the electronic and telecommunication industry
Content	Chapter 1: Introduction to Electronics and Telecommunications Firm Chapter 2: The Engineering Design Process Chapter 3: Conceptual design Chapter 4: Introduction to the electronic and telecommunication industry
Study and examination requirements and forms of examination	- Group Project: Design a product/service and presentation: 50 - Final exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 50
Media employed	Text Books
Reading list	Nhập môn Kỹ thuật – Phạm Ngọc Tuấn, Hồ Thị Thu Nga, Đỗ Thị Ngọc Khánh, Trần Đại Nguyên, Nguyễn Văn Tường, Nguyễn Minh Hà – NXB ĐHQG TP.HCM, 2014

Written/updated by
PhD. Dang Le Khoa

27. Electronic Circuit Techniques - ETC10101

Module name	Electronic Circuit Techniques
Module level, if applicable	Undergraduate
Code, if applicable	ETC10101
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Nguyen Thi Thien Trang
Lecturer	MSc. Nguyen Thi Thien Trang
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, digital electronics
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the concepts of electronic circuit techniques and its analysis to specific circuit design - Calculate the time behavior, power dissipation, frequency response and analysis and design specific circuits - Design, build circuits and take measurements of circuit variables using LTSPICE tools - Assemble related documents, imitate and calibrate electronic circuit designs, combine theory and tools - Demonstrate the systematization of knowledge
Content	Chapter 1: Review – Fundamental of Electric Circuit Basic law, method of analysis, circuit theorem Chapter 2: Feedback circuits and system Chapter 3: Multistage amplifiers Chapter 4: Power amplifiers Chapter 5: Amplifier frequency response and filters Chapter 6: Oscillators Chapter 7: Amplitude and frequency modulation Lab 1: Feedback circuit and multi-stage amplifier Lab 2: Differential amplifiers Lab 3: Power amplifier Lab 4: Analog filter circuits Lab 5: Oscillators

Study and examination requirements and forms of examination	- Daily quizzes: 6 quizzes during the course: 10% - Assignments: 3 problems during the course: 20% - Labs: 6 reports: 30% - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Electronic Circuit Analysis and Design, 3rd edition, Donald A. Neamen, McGraw-Hill, 2007. 2. Fundamentals of Microelectronics, Behzad Razavi, 2nd edition, 2003.

Written/updated by
 Nguyen Thi Thien Trang

28. VLSI Circuits and Technology - ETC10102

Module name	VLSI Circuits and Technology
Module level, if applicable	Undergraduate
Code, if applicable	ETC10102
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6 semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	PhD. Bui Trong Tu
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 5 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 60 hrs Assignments and independent learning: 120 hrs Total Working hours: 180 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, digital electronics
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand VLSI technology - Understand CMOS digital circuit design flow - Understand the operation of CMOS digital circuits - Develop the VLSI implementation of combinatorial circuits - Develop the VLSI implementation of sequential circuits - Understand major VLSI design tradeoffs - Use VLSI circuit simulators
Content	Chapter 1: Introduction MOS Transistors, CMOS Logic, CMOS Fabrication and Layout, Logic Design, Circuit Design, Physical Design Chapter 2: Delay Transient Response, Delay Model, Linear Delay Model, Logical Effort of Paths Chapter 3: Power Introduction, Dynamic Power, Static Power, Low Power Architectures Chapter 4: Interconnect Engineering Interconnect Modeling, Interconnect Impact, Interconnect Engineering, Logical Effort with Wires Chapter 5: Combinational Circuit Design Circuit Families, Circuit Pitfalls, Subthreshold Circuit Design Chapter 6: Sequential Circuit Design Sequencing Static Circuits, Circuit Design of Latches and Flip-Flops

	Chapter 7: Basic Principles of IC Fabrication Basic steps of CMOS IC fabrication process, corners
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30% - Final exam: Written exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. CMOS VLSI Design - A Circuits and Systems Perspective, Neil H. E. Weste, David M. Harris, Addison-Wesley. 2. Introduction to VLSI Circuits and Systems, J. Uyemura, Wiley, 2001

Written/updated by
PhD. Bui Trong Tu

29. Advanced Programming Techniques - ETC10103

Module name	Advanced Programming Techniques
Module level, if applicable	Undergraduate
Code, if applicable	ETC10103
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Huynh Quoc Thinh
Lecturer	MSc. Huynh Quoc Thinh
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hours/week Office hours: Monday 09:30 AM – 11:30 AM Wednesday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Basic Programming, Numerical Analysis and Matlab.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts related to object-oriented programming - Use the principles of object-oriented programming to write programs - Organize code and testing - Implement small to medium software programs - Improve programming skills and style - Practice collaborative programming skills in groups
Content	Chapter 1: Microsoft C# and Visual Studio - Introduce basic of C# - Working with variables, operators, and expressions - Writing method, and applying scope - Using decision and iteration statement Chapter 2: Understanding the C# Object Model - Creating and managing classes and object - Understanding value and reference type - Enumeration and structure Chapter 3: Oriented-object Programming: - Inheritance and polymorphism - Encapsulation and Properties - Interface and abstract class

	Chapter 4: -Managing errors and exceptions - Event in C# Chapter 5: Building Universal Window Platform Application with C# Chapter 6: Principles of object-oriented programming Design patterns: Template, Strategy, và Singleton. Unit Test
Study and examination requirements and forms of examination	- Assignments: two problems during the course: 20% - Mid-term exam (60 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40% - Final project: Design UWP Application (Text report and presentation): 40%
Media employed	Text book
Reading list	C# reference Microsoft Docs (https://docs.microsoft.com/)

Written/updated by
Huynh Quoc Thinh

30. Fundamentals of Biomedical Electronics - ETC10104

Module name	Fundamentals of Biomedical Electronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC10104
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Le Duc Hung
Lecturer	PhD. Le Duc Hung
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Tuesday 12:30 AM – 15:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Digital Electronics, Analog Electronics.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Know the basics of biomedical potential - Know the origin and classification of biomedical signals - Know the basic operating principles of biomedical sensors - Know the operating principles of basic biomedical electronic circuits, basic biomedical electronic devices. - Ability to apply knowledge related to biomedical electronics to implement a project - The ability to think, analyze and interpret data in biomedical electronics. - Ability to understand and interpret English literature on biomedical electronics
Content	Chapter 1: Overview of Biomedical Electronics Chapter 2: Basis of cell physiology Chapter 3: Biology Potential Chapter 4: Electrodes and Sensors Chapter 5: Biomedical Signals Chapter 6: Biomedical Electronic Circuits Chapter 7: Biomedical Equipment
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Seminar: Teamwork presentation, project-based learning: 30% - Final exam (90 minutes): Written exam, 4-page cheatsheets are allowed in the exam: 50%
Media employed	Text book

Reading list	1. Myer Kutz, Standard Handbook of Biomedical Engineering & Design, Volume 1, Mc Graw Hill, 2003 2. John Enderle J., Blanchard S., Bronzino J., Introduction to Biomedical Engineering, Academic Press, Inc., San Diego, California, 2000
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Written/updated by
Le Duc Hung

31. Fundamentals of Automatic Control - ETC10105

Module name	Fundamentals of Automatic Control
Module level, if applicable	Undergraduate
Code, if applicable	ETC10105
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Tran Xuan Tan
Lecturer	MSc. Tran Xuan Tan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Tuesday 12:30 – 15:00
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Signals, Systems, and Circuit Analysis, Matlab
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Express basic concepts of automatic control systems. - Express technical documentations in English that used in this course. - Apply related knowledges to calculate values in automatic control systems. - Analyze automatic control systems. - Appraise automatic control systems: stable of system, response of system in time domain and frequency domain.
Content	Chapter 1: Mathematical formulas in continuous automatic control systems - Concepts of automatic control systems. - Mathematical formulas - Block diagram - Transfer function - Signal graph Chapter 2: Properties of continuous automatic control systems - Properties in time domain - Properties in frequency domain - Stable of system, stability criterions Chapter 3: Synthesis of continuous automatic control systems - Methods increase the quality of system - Proportional-Integration-Derivative (PID) control Chapter 4: Mathematical formulas in discrete automatic control systems

	- Block diagram - Discrete-time signal - Mathematical formulas. - Transfer function Chapter 5: Analysis and synthesis for discrete automatic control systems - Stable of system. - Quality of system - System synthesis Chapter 6: Application of MatLab in automatic control systems - Simulink - GUIDE - Interfacing Simulink with a Matlab GUIDE
Study and examination requirements and forms of examination	- Reinforcement exercises in each lesson: 30% - Attendance: 30% - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Nguyễn Hữu Phương, Biến đổi Z trong xử lý tín hiệu số, Nhà xuất bản Đại học Quốc gia TP HCM, 2014

Written/updated by
Tran Xuan Tan

32. Advanced Computer Networks – ETC10201

Module name	Advanced Computer Networks
Module level, if applicable	Undergraduate
Code, if applicable	ETC10201
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Nguyen Quang Anh
Lecturer	MSc. Nguyen Quang Anh
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 09:30 AM – 11:30 AM Friday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand, analysis and applicate networking services - Read, understand terminology in English course - Ability to operate and maintain the network system - Describe how routing protocols work, understand and applicate Networking services - Understand about wireless networking - Ability to configure network devices in layer 2 & 3
Content	Chapter 1: Overview-Introduction Course + The basic network concepts + TCP/IP Advanced + ARP/ MAC address Chapter 2: The routing Protocols + Static Route + RIP + Single OSPF area + OSPF multi-area + EIGRP + BGP Chapter 3: LAN/WAN Learning about VLAN + VTP + STP + Router on a stick Chapter 4: Networking Services

	+ advanced DNS/NetBioSName + DHCP/DHCP Relay Agent + Email (POP3/SMTP) + Telnet, + FTP,TFTP + HTTP, WAP + SNMP Chapter 5: Domain Controller System + Single Domain + Multi Domain (Child Domain, Tree Domain) Chapter 6: Wireless Networking + Wireless (A,B,G,N,AC) + Bluetooth + NFC
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (30 minutes): Multiple choice exam, 4-page cheatsheets are allowed in the exam room: 20% - Final exam (60 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 50%
Media employed	Text book
Reading list	Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose and Keith W.Ross.

Written/updated by
 Nguyen Quang Anh

33. Data Structures and Algorithms - ETC10202

Module name	Data Structures and Algorithms
Module level, if applicable	Undergraduate
Code, if applicable	ETC10202
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Cao Tran Bao Thuong
Lecturer	MSc. Cao Tran Bao Thuong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Programming Technique of Electronics – Telecommunication Branch
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to program and analyze algorithms - Ability to propose solutions to complex computer problems - Ability to implement software
Content	Chapter 1: Recursion Chapter 2: Searching Algorithms Chapter 3: Basic Sorting Algorithms Chapter 4: Advanced Sorting Algorithms Chapter 5: Linked Lists Chapter 6: Binary Search Trees
Study and examination requirements and forms of examination	- Assignments: 1 problems during the course: 20% - Mid-term exam (45 minutes): Multiple choice exam or Written exam: 30% - Final exam (60 minutes): Written exam: 50%
Media employed	Text book
Reading list	- Kỹ thuật lập trình, Trần Đan Thư, Nguyễn Thanh Phương, Đinh Bá Tiến, Trần Minh Triết, Đặng Bình Phương, nxb KH và KT, 2014.

	- Data Structures and Algorithms in C++, Michael T. Goodrich, Roberto Tamassia, David M. Mount, John Wiley & Sons, Inc, 2011.
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Written/updated by

Huynh Huu Thuan

34. Data Structures and Algorithms Lab - ETC10203

Module name	Data Structures and Algorithms Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10203
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Cao Tran Bao Thuong
Lecturer	MSc. Cao Tran Bao Thuong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Programming Technique of Electronics – Telecommunication Branch
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to program and analyze algorithms - Ability to propose solutions to complex computer problems - Ability to implement software
Content	Chapter 1: Recursion Chapter 2: Searching Algorithms Chapter 3: Basic Sorting Algorithms Chapter 4: Advanced Sorting Algorithms Chapter 5: Linked Lists Chapter 6: Binary Search Trees
Study and examination requirements and forms of examination	- Final exam (30 minutes): 100%
Media employed	Text book
Reading list	- Kỹ thuật lập trình, Trần Đan Thư, Nguyễn Thanh Phương, Đinh Bá Tiến, Trần Minh Triết, Đặng Bình Phương, nxb KH và KT, 2014. - Data Structures and Algorithms in C++, Michael T. Goodrich, Roberto Tamassia, David M. Mount, John Wiley & Sons, Inc, 2011.

Written/updated by Huynh Huu Thuan

35. VLSI Design - ETC10204, ETC10205

Module name	VLSI Design VLSI Design Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10204 ETC10205
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Do Quoc Minh Dang
Lecturer	MSc. Do Quoc Minh Dang
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lab: 4 hrs/week Office hours: Monday 07:30 AM – 11:30 AM
Workload	VLSI Design: • Contact hours: 45 hrs • Assignments and independent learning: 90 hrs • Total Working hours: 135 hrs VLSI Design Lab: • Contact hours: 30 hrs • Assignments and independent learning: 30 hrs • Total Working hours: 60 hrs
Credit points	VLSI Design: 3 VLSI Design Lab: 1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Digital Circuit Design, Programmable Logic Design, Physics of Semiconductor Device
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the internal structure of an integrated circuit - Understand how to design digital integrated circuits by using CMOS transistors - Understand CMOS circuits delay and power analysis - Logical and physical design digital integrated circuits - Optimize an integrated circuit design in terms of performance, power consumption and area
Content	Chapter 1: CMOS logical design Schematic-level design of combinational and sequential digital circuits based on CMOS transistors Chapter 2: Semiconductor fabrication and physical design Introduce to semiconductor fabrication process and layout design of a CMOS circuit Chapter 3: CMOS transistor theory

	Describe the operation theory of a MOS transistor, establish its current/voltage relationship and analyze its parasitic capacitors Chapter 4: CMOS delay and power analysis Describe the models used to estimate a CMOS circuit's delay and analyze its power consumption Chapter 5: CMOS circuit family Introduce to a variety of CMOS circuit family and compare their advantages and disadvantages Lab 1: Schematic & Symbol design for CMOS logic gates: Inverter, NAND, NOR Lab 2: CMOS gates DC and Transient simulation & analysis Lab 3: Simulation using script and gate netlist in bash mode Lab 4: Layout design for CMOS logic gates Lab 5: Extensive lab for CMOS compound gates
Study and examination requirements and forms of examination	VLSI Design: - Classwork assignments: doing exercises in class: 15% - Homework assignments: doing exercises at home: 15% - Midterm exam: Written exam: 20% - Final exam: Written exam: 50% VLSI Design Lab: - Final exam: schematic, layout design and simulation of a CMOS compound gate: 100%
Media employed	Text book
Reading list	1. CMOS VLSI Design: A Circuits and Systems Perspective 4 th edition, Neil Weste, David Harris, Pearson, 2010

Written/updated by

Do Quoc Minh Dang

36. Object Oriented Programming - ETC10206

Module name	Object-Oriented Programming
Module level, if applicable	Undergraduate
Code, if applicable	ETC10206
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Le Duc Tri
Lecturer	MSc. Le Duc Tri
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 120 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Electronics and telecommunications programming techniques
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of object-oriented programming - Knowing the applicability and advantages of the method of object-oriented programming - Understand and analyze the problem of object-oriented programming - Programmable object-oriented software programs - Demonstrate the systematization of knowledge
Content	Chapter 1: Object-oriented programming Fundamentals + The basics + The advantages of OOP + The application OOP + The language OOP + The programming language C++ Chapter 2: Features of the C++ language + Input/output on the C ++ + Utilities for programmers + Inline function + Reference + Override function + Default parameter in the function call + Operator dynamic memory management Chapter 3: Objects and classes + Object

	+ Class + Class declare + Ability of function Chapter 4: Operator on the class definition + Introduction class operators + Operator function is the composition function + Operator function is the friend function + Limitations and abilities of the operator override + Strategy for using operator functions + Type conversion Chapter 5: Inheritance + Introducing inheritance + Single inheritance + Virtual functions and polymorphism + Multiple inheritance Chapter 6: Template + Function template + Using function template + Class template + Using class template
Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 30% - Mid-term exam (45 minutes): Written exam, open topic: 20% - Final exam (60 minutes): Multiple choice exam, open topic: 50% 50% cheat sheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. C++ and Object Oriented Programming, Pham Van At, Science and Technics Publishing House. 2. Beginning Programming with C++ For Dummies, Stephen R. Davis, John Wiley and Sons, 2015.

Written/updated by

Le Duc Tri

37. Object Oriented Programming Lab - ETC10207

Module name	Object Oriented Programming Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10207
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Le Duc Tri
Lecturer	MSc. Le Duc Tri
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Electronics and telecommunications programming techniques
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand and analyze the problem of object-oriented programming - Programmable object-oriented software programs
Content	Lab 1: Object-oriented programming C++ Lab 1 Lab 2: Object-oriented programming C++ Lab 2 Lab 3: Object-oriented programming C++ Lab 3 Lab 4: Object-oriented programming C++ Lab 4 Lab 5: Object-oriented programming C++ Lab 5 Lab 6: Object-oriented programming C++ Lab 6 Lab 7: Object-oriented programming C++ Lab 7 Lab 8: Object-oriented programming C++ Lab 8 Lab 9: Object-oriented programming C++ Lab 9 Lab 10: Object-oriented programming C++ Lab 10

Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 50% - Final exam (45 minutes): Practicing exam, open topic: 50%
Media employed	Text book
Reading list	1. C++ and Object Oriented Programming, Pham Van At, Science and Technics Publishing House. 2. Beginning Programming with C++ For Dummies, Stephen R. Davis, John Wiley and Sons, 2015.

Written/updated by

Le Duc Tri

38. Bus Architecture - ETC10216

Module name	Bus Architecture
Module level, if applicable	Undergraduate
Code, if applicable	ETC10216
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 08:00 AM – 11:00 AM Friday 08:00 AM – 11:00 AM
Workload	School time: 30 hours Self-learning time: 60 hours Total time: 90 hours
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital electronics, Computer Structure.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to analyze communication standards with peripheral devices - Understanding computer organization - Ability to design communication between CPU and peripheral devices - Understand technical terms - Ability to analyze communication standards with peripheral devices
Content	Chapter 1: Data transfer in Microprocessor systems Chapter 2: Bus cycle Chapter 3: Address decoding for memory and I/O Chapter 4: Theory and Operation of Cache Chapter 5: Direct Memory Access (DMA) Chapter 6: Interrupt Management Chapter 7: Real Time Clock (RTC) and Configurable Ram Chapter 8: Introduction to PCI and PCIe Architecture
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheat sheet per student is allowed in the exam room: 20% - Final exam (90 minutes): Written exam, a four-page cheat sheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. IP Compiler for PCI Express, (Intel) Altera company 2014.

Written/updated by

Huynh Huu Thuan

39. Database - ETC10217

Module name	Database
Module level, if applicable	Undergraduate
Code, if applicable	ETC10217
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6
Person responsible for the module	MSc. Le Duc Tri
Lecturer	MSc. Le Duc Tri
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Electronics and telecommunications programming techniques
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of database - Understand and design database - Understand and solve the relational algebra problem - Programmable SQL language - Demonstrate the systematization of knowledge
Content	Chapter 1: Object-oriented programming Fundamentals +The basics + Data Systems + The type of database + The database management system Chapter 2: Entity relationship model + The components of the entity relationship diagram (ERD) + Entity + Single-attributes + Multi-valued attributes + Complex attributes + Key Chapter 3: Data normalization + Depending function + Characteristics of functional dependencies + Closure of attributes + Algorithm for finding the key + Data Normalization

	Chapter 4: Relational algebra + Union + Intersect + Subtraction + Descart + Division Chapter 5: SQL language + Introduction to SQL language + The features of the SQL language + Select clause + The arithmetic operators in SQL + Order by clause
Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 30% - Mid-term exam (45 minutes): Written exam, open topic: 20% - Final exam (60 minutes): Written exam, open topic: 50%
Media employed	Text book
Reading list	- Database system, Dương Tuấn Anh - Nguyễn Trung Trực, Ho Chi Minh City National University Publishing House.

Written/updated by

Le Duc Tri

40. Introduction to cryptography - ETC10218

Module name	Introduction to cryptography
Module level, if applicable	Undergraduate
Code, if applicable	ETC10218
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Programming Technique of Electronics – Telecommunication Branch.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basics of encryption - cryptography - Ability to use the encryption - cryptography algorithms - Ability to implement the encryption - cryptography algorithms
Content	Chapter 1: The basic concepts Chapter 2: Secret key cryptosystems Chapter 3: Public key cryptosystems Chapter 4: Digital signatures Chapter 5: Key management and distribution Chapter 6: Modern encryption
Study and examination requirements and forms of examination	- Assignments: 1 problem during the course: 20% - Mid-term exam (45 minutes): Multiple choice exam or Written exam: 30% - Final exam (60 minutes): Written exam: 50%
Media employed	Text book
Reading list	- Mã hoá và ứng dụng, Dương Anh Đức, Trần Minh Triết, nxb Đại học Quốc gia thành phố Hồ Chí Minh, 2008.

Written/updated by

Huynh Huu Thuan

41. Network Technology- ETC10307

Module name	Network Technology
Module level, if applicable	Undergraduate
Code, if applicable	ETC10307
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Tran Thi Huynh Van
Lecturer	MSc. Tran Thi Huynh Van
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2.5 hrs/week Office hours: Monday 12:30 PM – 15:00 PM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network, Communication Systems
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Generalize fundamental, operation of Active Directory, domain resources on Microsoft servers - Relate the network/system administration based on Windows Servers services: DHCP, DNS, VPN - Design the small/big company's network/system administration model based on related background of studied Windows Servers services - Synthesize systematization of knowledge - Set up systems/model to experience how to manage domain users, groups, file sharing - Organize to rewrite research documents and effectively present related knowledge of network administration: RAID, Virtual Machines control, etc.
Content	Chapter 1: Active Directory Chapter 2: User and Group management Chapter 3: File Sharing Chapter 4: Group Policy and Security Policy Chapter 5: Dynamic Host Configuration Protocol Service Chapter 6: Network Address Translation Chapter 7: Domain Name System Chapter 8: Wide Area Networks Chapter 9: Virtual Private Network Service Chapter 10: Frame Relay

Study and examination requirements and forms of examination	- Assignments: 4 problems during the course: 25% - Course small projects: Work in group of 4, will be assessed by presentation, course project summary report: 25% - Final exam (60 minutes): Multiple choice exam: 50%
Media employed	Text book
Reading list	1. “Bài giảng Công nghệ mạng”, Nguyen Viet Ha, University of Sciences Ho Chi Minh City, 2012

Written/updated by

Tran Thi Huynh Van

42. Digital Communication-ETC10301

Module name:	Digital Communication
Module level, if applicable	Undergraduate
Code, if applicable	ETC10301
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	Tran Thi Thao Nguyen, Ph.D. Truong Tan Quang, MSc.
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Friday 08:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Digital Signal Processing and Communications
Module objectives/intended learning outcomes	- Understand the features of a digital communication system. - Understand the structure to design a digital communication system. - Analysis of each block in the digital communication system. - Design each block in the transmitter and receiver. - Master in holding and managing the presentation.
Content	Chapter 1: Introduction to Digital Communication. Introduce the important features of a digital communication system (DCS). Review some basic concepts/definitions related to communication system. Chapter 2: Formatting and Baseband Modulation. Introduce the first important step in any DCS to transforming the information source to a form compatible with a digital system. Study the Sampling and Quantization. Study the baseband modulation (Binary PAM, M-PAM). Chapter 3: Receiver structure. Review the Formatting and Encoding. Study Receiver structure (Demodulation, Detection)

	Study first step for designing the receiver (Matched filter receiver) Chapter 4: Signal detection. Review the Receiver structure. Introduce the Signal detection in AWGN channels. Study SNR and Eb/No. Chapter 5: ISI, Nyquist theorem. Review Signal detection in AWGN channels. Introduce Another source of error: Inter-symbol interference (ISI). Study the techniques to reduce ISI with Pulse shaping and Equalization. Chapter 6: Modulation schemes. Review ISI and the techniques to reduce ISI. Introduce Bandpass modulation schemes (M-PAM, M-PSK, M-FSK, M-QAM). Study how to detect the transmitted information at the receiver with Coherent detection. Chapter 7: Channel coding. Review last chapter. Study two types of channel coding. Study Linear block codes. Study Convolutional codes. Study how to apply the channel coding to a DCS.
Study and examination requirements and forms of examination	- In-class assignments: Quiz, problem-solving, discussion: 20 - Attendance: attendance all the lectures: 10 - Presentation: groups of students study other important features of digital communication system and do the presentation: 20 - Final exam: single choice quiz: 50
Media employed	Text Books
Reading list	Bernard Sklar, Digital communications: Fundamentals and Applications, Prentice Hall, 2001, ISBN: 0-13-084788-7

by

Written/updated

Tran Thi Thao Nguyen, Ph.D.

43. Telecommunication Networks-ETC10302

Module name:	Telecommunication Networks
Module level, if applicable	Undergraduate
Code, if applicable	ETC10302
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	MSc. Tran Thi Huynh Van
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2.5 hrs/week Office hours: Monday 12:30 PM – 15:00 PM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%
Recommended prerequisites	Fundamental of Electronics, Communication Systems
Module objectives/intended learning outcomes	- Generalize fundamental of data communication and networking - Demonstrate data encoding, data link layer control; multiplexing/demultiplexing techniques - Classify packet versus circuit switching and routing algorithms in switching - Assemble related documents to have outline course project report and presentation - Demonstrate the systematization of knowledge.
Content	Chapter 1: Fundamental of data communication and networking Chapter 2: Data communication Chapter 3: Data encoding Chapter 4: Data Link Layer Control Chapter 5: Multiplexing/Demultiplexing Chapter 6: Packet and Circuit Switching Chapter 7: Asynchronous Transfer Mode Chapter 8: Routing in switched network Chapter 9: Congestion control in switched networks
Study and examination requirements and forms of examination	- Assignments: 4 problems during the course: 25 - Course small projects: Work in group of 4, will be assessed by presentation, course project summary report: 25 - Final exam (60 minutes): Multiple choice exam: 50
Media employed	Text Books

Reading list	1. “Fundamentals of Telecommunications”, Roger L. Freeman, John Wiley & Sons, 1999 2. “<i>Truyền Số liệu</i>”, Tong Van On and Le Tien Thuong, Vietnam National University Ho Chi Minh City Publishing House, 2000 3. “Kỹ thuật truyền số liệu”, Nguyen Hong Son, ICPublisher, 2004.
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Written/updated by

Tran Thi Huynh Van

44. Digital Communications and Telecommunication Network Lab- ETC10303

Module name	Digital Communications and Telecommunication Network Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10303
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Nguyen Thai Cong Nghia
Lecturer	Nguyen Thi Xuan Uyen Nguyen Thai Cong Nghia
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 9:00 AM – 11:30 AM Thursday 9:00 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Communications, Telecommunication Network
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the features and the structure to design a digital communication system - Analysis of each block in the digital communication system - Design each block in the transmitter and receiver - Generalize fundamental of data communication and networking - Demonstrate data encoding, data link layer control; multiplexing and demultiplexing techniques - Demonstrate the systematization of knowledge
Content	Lab 1: Analog and Digital signal - Signal generating and filtering - Time-domain and Frequency-domain analysis - Transmission channel and noise Lab 2: Line coding - Digital-digital conversion - Line coding types (NRZ, RZ, Bipolar-AMI, Manchester...) - Power spectral density analysis and comparison Lab 3: Pulse code modulation (PCM) - ADC DAC - Quantization, μ law and A law (companding algorithm) - PCM-TDM Lab 4: Error detection and correction

	- Cyclic redundancy check code - Linear block code and Hamming code - Convolutional code and Viterbi decoder Lab 5: Flow and Error Control - Frame mode - Stop-and-wait Automatic Repeat Request Protocol Lab 6: Baseband Modulation - Binary PAM, M-PAM - Bit error rate calculation, bertool, comparison between theory and script/model Lab 7: Pulse Shaping and Receiver Optimization - Pulse shaping (raised-cosine pulse) - Bandwidth comparison - Matched filter and Correlation receiver Lab 8: Bandpass Modulation - M-PAM, M-PSK, M-FSK, M-QAM - Coherent detection
Study and examination requirements and forms of examination	- Attendance: 20% - Lab reports: 30% - Final lab exam: 50%
Media employed	Text book
Reading list	1. John Proakis, Digital Communication, Fourth Edition, McGraw Hill, 2000 2. Bernard Sklar, Digital Communication: Fundamentals and Applications, Second Edition, Prentice Hall PTR, 2001, ISBN: 0-13-084788-7 3. Roger L. Freeman, John Wiley & Sons, Fundamentals of Telecommunications, 1999 4. Tong Van On and Le Tien Thuong, Truyền Số liệu, Vietnam National University Ho Chi Minh City Publishing House, 2000 5. Nguyen Hong Son, Kỹ thuật truyền số liệu, ICPublisher, 2004.

Written/updated

by

Nguyen Thi Xuan Uyen
Nguyen Thai Cong Nghia

45. High Frequency Electronics-ETC10304

Module name	High Frequency Electronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC10304
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Nguyen Thi Hong Ha
Lecturer	MSc. Nguyen Thi Hong Ha
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 08:30 AM – 10:30 AM Wednesday 08:30 AM – 10:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Basic Electronics, Analog Electronics.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Explaining and calculating high frequency circuits - Describe high frequency circuits on Smith Chart and understand impedance matching. - Understand and explain the characteristics of electronic components in high frequency. - Describe the working of Active filters, Oscillators and Mixers in high frequency. - Assemble related documents, imitate and calibrate high frequency circuit designs, combine theory and tools - Demonstrate the systematization of knowledge
Content	Chapter 1: Equivalent circuits of components electronics at HF 1.1 Wire 1.2 R, L, C 1.3 Diode 1.4 Transistor (Miller Effect) 1.5 Fet – MosFet Chapter 2: Smith Chart 2.1 Impedance transformation (Z Smith Chart) (Find Γ , Z_{in} , SWR, RL, special points: open circuit, short circuit) 2.2 Admittance transformation (Y Smith Chart) 2.3 Parallel and series elements (combined ZY Smith Chart) Chapter 3: Impedance Matching 3.1 Impedance matching using discrete components (type T and Pi..)

	Impedance matching using microstrip line (transferring from discrete components to microstrip line, impedance matching using single-stub, double-stub) Chapter 4: Single and Multiport Networks 4.1 Definition 4.2 Interconnecting Networks (series, parallel, cascade networks) 4.3 Scattering parameters 4.4 Meaning of S-parameter 4.5 Signal flow chart modeling Chapter 5: Resonant circuits and RF Filter 5.1 Resonant circuits 5.2 Filter types and basic parameters 5.3 Insertion Loss 5.4 Special filter (Butterworth filter, Chebyshev filter) 5.5 Design RF filter Chapter 6: RF amplifiers 6.1 Characteristics of amplifiers 6.2 Amplifier power 6.3 Conditions of amplifier operation (stability, gain, noise, VSWR...) Broadband, high-power and multistage amplifiers. Chapter 7: RF oscillators 7.1 Basic oscillator model 7.2 High frequency oscillator configuration 7.3 Fixed frequency oscillator 7.4 Dielectric resonator oscillator VCO Chapter 8: Mixers 8.1 Basic characteristics of mixers 8.2 Frequency domain of mixer 8.3 Single-ended mixer design 8.4 Single-balanced mixer Double-balanced mixer
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30% - Final exam (60 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 50%
Media employed	Text and Reference Books
Reading list	Text - RF Circuit and Design: Theory and Application, Reinhold Ludwig – Paven Bretchko, Prentice Hall, 2000.

Written/updated by
Nguyen Thi Hong Ha

46. Antennas and Wave Propagation- ETC10305

Module name	Antennas and Wave Propagation
Module level, if applicable	Undergraduate
Code, if applicable	ETC10305
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa MSc. Ngo Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday: 09:30 AM – 11:30 AM Wednesday: 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamentals of Calculus, Electromagnetic Field Theory
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the fundamentals of parameters of antennas - Demonstrate the radiation analysis procedures by the vector potentials function. - Apply the radiation analysis procedures to analyze specific antennas - Simulate numerically the fundamental parameters of antennas on Matlab software and construct a radiation model in HFSS software. - Based on the simulation, appraise the radiation performance of antennas by varying physical parameters of antennas. - Identify characteristics of radio wave propagation
Content	Chapter 1: Consolidate mathematical preliminaries and antennas introduction Chapter 2: Fundamental parameters of antennas Radiation pattern, Radiation power density, Radiation Intensity, Beamwidth, Directivity, Antenna efficiency, Gain, Polarization, Thevenin equivalent model of antennas, Friss Model in free space. Chapter 3: Radiation integrals and auxiliary potential functions Construct a vector potential function to analyze radiation problems Analyze the infinitesimal dipole

	Chapter 4: Linear wire antennas Use vector potential functions to analyze the radiation pattern of linear wire antennas as small dipoles, finite length dipoles, half-wavelength dipoles, and loop antennas Chapter 5: Antennas array Consider a linear antennas array to analyze the arrays factor and the radiation of the reference antenna. Adjust the physical parameters of an array to obtain the desired radiation pattern as Broadside and Endfire. Chapter 6: Seminar Analyze the radiation pattern of practical antennas such as Horn antennas, Microstrip antennas, Helix antennas, Yagi-Uda Chapter 7: Radio-wave Propagation Propagation in Free Space, Tropospheric Propagation, Ionospheric Propagation, Surface Wave, Low Frequency Propagation and Very Low Frequency Propagation, Extremely Low Frequency Propagation
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Seminars: Multiple topics in terms of antennas and wave propagation: 30% - Final exam (90 minutes): Written exam, 4 questions of increasing difficulty are allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Balanis, C. A. (2005). Antenna Theory: Analysis and Design, John Wiley & Sons. 2. Lê Tiến Thường, Trần Văn Sư. (2016). Truyền sóng và Anten, NXB. Đại học Quốc gia TP. Hồ Chí Minh.

Written/updated by

Dang Le Khoa & Ngo Thanh Hai

47. Antennas and Wave Propagation Lab- ETC10306

Module name	Antennas and Wave Propagation Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10306
Subtitle, if applicable	None
Courses, if applicable	None

Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Ngo Thanh Hai
Lecturer	MSc. Ngo Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday: 09:30 AM – 11:30 AM Wednesday: 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamentals of Calculus, Electromagnetic Field Theory
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the fundamentals of parameters of antennas - Demonstrate the radiation analysis procedures by the vector potentials function. - Apply the radiation analysis procedures to analyze specific antennas - Simulate numerically the fundamental parameters of antennas on Matlab software and construct a radiation model in HFSS software. - Based on the simulation, appraise the radiation performance of antennas by varying physical parameters of antennas. - Identify characteristics of radio wave propagation
Content	Lab 1: HFSS simulation of a transmission line + Investigating characteristic impedance of the line by theoretical calculation and simulation results + Finding the frequency at which the line could be used as a resonator. Lab 2: Smith chart + Considering a transmission line connected load impedance, using the Smith chart to find parameters: voltage reflection coefficient, voltage standing wave ratio, impedance at any point in a transmission line. + Using the Smith chart to implement impedance matching in a transmission line. + Verifying the above results in the Smith chart with theoretical calculations. Lab 3: Half-wavelength dipoles + Simulating half-wavelength dipoles in HFSS software + Investigating the fundamental parameters of half-wavelength dipoles: radiation pattern, radiation intensity, beamwidth, the total radiated power, radiation resistance, and directivity.

	+ Verifying the analysis results by HFSS with theoretical calculations. Lab 4: Broadside antenna array + Investigating radiation pattern of broadside antenna array Lab 5: Endfire antenna array + Investigating radiation pattern of endfire antenna array
Study and examination requirements and forms of examination	- Attendance: 15% - Lab reports: 35% - Final lab exam: 50%
Media employed	Text book
Reading list	1. Balanis, C. A. (2005). Antenna Theory: Analysis and Design, John Wiley & Sons. 2. Lê Tiến Thường, Trần Văn Sư. (2016). Truyền sóng và Anten, NXB. Đại học Quốc gia TP. Hồ Chí Minh.

Written/updated by

Ngo Thanh Hai

48. Routing Protocols-ETC10314

Module name	Routing Protocols
Module level, if applicable	Undergraduate
Code, if applicable	ETC10314
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Nguyen Dinh Lam
Lecturer	MSc. Nguyen Dinh Lam
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs / week Office hours: Monday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Networking.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of routing network device - Understand and explain concepts of static route, type of static route, how to implement static route on Cisco devices - Understand and explain concepts of dynamic routing, type of dynamic routing, how to implement static route on Cisco devices - Describe the working and how to implement Single Area OSPF Routing Protocol - Optimize and troubleshooting Single Area OSPF Routing Protocol - Describe the working and how to implement EIGRP Routing Protocol - Describe the working and how to implement Network Access Control List - Describe the working and how to implement Network Address Translation
Content	
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 20% - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	CCNA guide to Cisco Networking, Kelly Caudle, Kelly Cannon, Massachusetts : Thomson Course Technology, 2004

Written/updated by
Nguyen Dinh Lam

49. LAN Switching and Wireless- ETC10315

Module name	LAN Switching and Wireless
Module level, if applicable	Undergraduate
Code, if applicable	ETC10315
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	MSc. Ngo Thanh Hai
Lecturer	MSc. Ngo Thanh Hai
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Basic Computer Networks
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Describe a switched network in the small-sized to medium-sized business - Understand principles of Switch in switched networks such as VLAN, Inter-VLAN Routing, and Spanning Tree Protocol (STP) - Describe the components of a wireless LAN - Construct a medium-sized network model and configure network devices to achieve network convergence. - Troubleshoot common configuration issues.
Content	Chapter 1: Introduction to switched networks Describe convergence of data, voice, and video in the context of switched networks Explain the process of frame forwarding in a switched network Compare a collision domain to a broadcast domain Chapter 2: Basic switching concepts and configuration Configure initial settings on a Cisco switch Configure switch ports to meet network requirements and the management switch virtual interface Describe basic security attacks and security best practices. Configure the port security feature to restrict network access Chapter 3: VLANs

	Explain the purpose of VLANs in a switched network Analyze how a switch forwards frame based on VLAN configuration Configure a switch port to be assigned to a VLAN based on requirements Configure a trunk port on a LAN switch and Dynamic Trunking Protocol (DTP) Troubleshoot VLAN and trunk configuration in a switched network Chapter 4: Inter-VLAN routing Apply 3 routing methods to routing in a multiple switched environment Using multi-layer Switch in inter-VLAN routing Chapter 5: LAN Redundancy Spanning Tree Protocol (STP) concepts Varieties of Spanning Tree Protocols Spanning Tree configuration First-Hop Redundancy protocols Chapter 6: Link Aggregation Describe link aggregation Describe EtherChannel technology Configure link aggregation with EtherChannel Troubleshoot link aggregation with EtherChannel Chapter 7: Wireless LANs Describe wireless LAN technology and standards, the components, topologies, and the 802.11 frame structure Describe the media access method used by wireless technology and channel management in a WLAN Describe threats and security mechanisms in wireless LANs Configure a wireless router and wireless client in wireless LANs Troubleshoot common wireless configuration issues Chapter 8: IOS Images and Licensing Explain the IOS image naming conventions implemented by Cisco Manage Cisco IOS system image files to support network requirements in small-sized to medium-sized business networks Explain the licensing process for Cisco IOS software Configure a router to install a Cisco IOS software image license
Study and examination requirements and forms of examination	- Assignments: 3 labs during the course: 30% - Mid-term exam (45 minutes): An exam with about 30 multiple-choice questions is allowed in the exam room: 30%

	- Final exam (60 minutes): An exam with about 40 multiple-choice questions is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. https://www.netacad.com/ 2. Mai Van Cuong, Tran Trung Dung, Tran Hong Ngoc, Le Ngoc Son, Le Giang Thanh, Truong Thi My Trang, Dao Anh Tuan, ‘Giao trinh mang may tinh’ 2015, Nha xuat ban khoa hoc va ky thuat

Written/updated by

Ngo Thanh Hai

50. Cisco Networking Lab- ETC10316

Module name	Cisco Networking Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10316
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Ngo Minh Nghia
Lecturer	Ngo Minh Nghia
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 2 hrs/week Office hours: - Thursday, 07:30 – 09:30 - Thursday, 09:30 – 11:30
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Networking
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Describe a switched network in the small-sized to medium-sized business - Understand principles of Switch in switched networks such as VLAN, Inter-VLAN Routing, and Spanning Tree Protocol (STP) - Describe the components of a wireless LAN - Construct a medium-sized network model and configure network devices to achieve network convergence. - Troubleshoot common configuration issues. - Understand the basic concepts of routing network device - Understand and explain concepts of static route, type of static route, how to implement static route on Cisco devices - Understand and explain concepts of dynamic routing, type of dynamic routing, how to implement static route on Cisco devices - Describe the working and how to implement Single Area OSPF Routing Protocol - Optimize and troubleshooting Single Area OSPF Routing Protocol - Describe the working and how to implement EIGRP Routing Protocol - Describe the working and how to implement Network Access Control List

	- Describe the working and how to implement Network Address Translation
Content	Lab 1: Configure Static Route and CDP - Understand how to implement Static Route and Cisco Discovery Protocol in Cisco Packet Tracer. Lab 2: Configure Dynamic Routing Protocol – RIPv2 - Understand how to implement RIPv2 in Cisco Packet Tracer. Lab 3: Configure Dynamic Routing Protocol – EIGRP - Understand how to implement EIGRP in Cisco Packet Tracer. Lab 4: Configure Dynamic Routing Protocol – OSPF - Understand how to implement OSPF in Cisco Packet Tracer. Lab 5: Configure STP, VTP, Port Security - Understand how to implement Spanning Tree Protocol (STP), Vlan Trunking Protocol (VTP), and Port Security in Cisco Packet Tracer. Lab 6: Configure Inter-VLAN Routing - Understand how to implement Inter-VLAN Routing in Cisco Packet Tracer. Lab 7: Configure ACL, DHCP, NAT - Understand how to implement Network Access Control List, DHCP and Network Address Translation in Cisco Packet Tracer.
Study and examination requirements and forms of examination	- Midterm Lab Exam: 30% - Final Lab Exam: 70%
Media employed	Text book
Reading list	- https://www.netacad.com/ - Mai Van Cuong, Tran Trung Dung, Tran Hong Ngoc, Le Ngoc Son, Le Giang Thanh, Truong Thi My Trang, Dao Anh Tuan, ‘Giao trinh mang may tinh’ 2015, Nha xuất bản khoa hoc va ky thuat

Written/updated by

Ngo Minh Nghia

51. Fundamentals of Telecommunication Programming-ETC10317

Module name	Fundamentals of Telecommunication Programming
Module level, if applicable	Undergraduate
Code, if applicable	ETC10317
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Nguyen Minh Tri
Lecturer	PhD. Nguyen Minh Tri
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 2.5 hrs/week
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	C/C++ programming
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand basic algorithms in searching and sorting - Evaluate the efficiency of an algorithm - Understand the working of popular data structures - Choose the right data structure for the project - Optimize the code to make a better system.
Content	Chapter 1: Introduction Definition of algorithm, data structure and its types. Asymptotic Analysis: Big-O Notation Chapter 2: Searching Algorithms Linear Search Binary Search Chapter 3: Sorting Algorithms Selection Sort Insertion Sort Interchange Sort Bubble Sort Quick Sort Chapter 4: Linear Data Structure Array Structure Stack Queue Chapter 5: Linear Data Structure (cont.) Linked List Hash Table Chapter 6: Tree Data Structure Binary Tree

	Binary Search Tree Chapter 7: Tree Data Structure (cont.) B Tree: Insertion, Deletion B+ Tree: Insertion, Deletion Chapter 8: Graph Data Structure Spanning Tree Deep First Search Algorithm Dijkstra's Algorithm
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Mid-term exam: Essay exam, 30 minutes, without reference: 30% - Final exam: Essay exam, 60 minutes, without reference: 50%
Media employed	Text book
Reading list	1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, 3rd Edition 2009, The MIT Press. 2. Robert Sedgewick and Kevin Wayne, Algorithms, 4th Edition.

Written/updated by

Nguyen Minh Tri

52. Telecommunication Programming Lab- ETC10318

Module name	Telecommunication Programming Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10318
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	Mr. Nguyen Thai Cong Nghia
Lecturer	Mr. Nguyen Thai Cong Nghia
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3hrs/week Office hours: Wednesday 9:00 AM – 11:30 AM Friday 9:00 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamentals of Telecommunication Programming, Mobile Operating Systems
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand and diagram the structure of common mobile operating systems - Compare and evaluate the common mobile operating system - Distinguish between procedural programming and object-oriented programming - Understand and practice programming with Java - Assemble related documents, combine theory and tools - Apply Java programming knowledge and adapt to practical projects.
Content	Lab 1: Searching and Sorting Algorithms with Array - Linear Search and Binary Search - Interchange Sort, Bubble Sort, Selection Sort, Insertion Sort, Quick Sort Lab 2: Basic Java programming - Data and expressions - Conditionals and loops Lab 3: Recursive Algorithms and Non-recursive Algorithms - Fibonacci sequence - The Euclidean Algorithm - Binary Search, Quick Sort... Lab 4: Using and Writing Classes in Java - Objects and Classes, Encapsulation - Basic Classes in Java

	- Writing Classes Lab 5: Data Structures - Stack and Queue - Linked List Lab 6: Object-oriented concept with Java - Class relationships - Inheritance and Visibility - Abstract classes - Polymorphism Lab 7: Tree Data Structure - Binary Tree - Binary Search Tree Lab 8: Java Graphic User Interface and IO Stream - Window, Layout, Common Control, Event Listener, Dialog Box... - Input and output File Streams
Study and examination requirements and forms of examination	- Attendance: 20% - Lab reports: 30% - Final lab exam: 50%
Media employed	Text book
Reading list	1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, 3rd Edition 2009, The MIT Press. 2. Robert Sedgewick and Kevin Wayne, Algorithms, 4th Edition. 3. John Lewis, William Loftus, Java Software Solutions Foundations of Program Design, Eighth Edition, 2007.

Written/updated by

Nguyen Thai Cong Nghia

53. Mobile Operating Systems- ETC10319

Module name	Mobile Operating Systems
Module level, if applicable	Undergraduate
Code, if applicable	ETC10319
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	6th semester
Person responsible for the module	PhD. Dang Le Khoa Mr. Nguyen Thai Cong Nghia
Lecturer	Mr. Nguyen Thai Cong Nghia
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3hrs/week Office hours: Tuesday 9:00 – 11:30 AM Wednesday 9:00 – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand and diagram the structure of common mobile operating systems - Compare and evaluate the common mobile operating system - Distinguish between procedural programming and object-oriented programming - Understand and practice programming with Java - Assemble related documents, combine theory and tools - Apply Java programming knowledge and adapt to practical projects.
Content	Chapter 1: Overview of mobile devices and mobile operating systems: History, market, structures & application development of common mobile OS. Comparison of common mobile OS. (group seminar and debate) Chapter 2: Introduction of Java Programming Data and expressions, Conditionals and Loops Chapter 3: Object-oriented concept with Java Programming Objects, Classes, Arrays, Class definitions and Instance data, Encapsulation and Java modifiers, Method declaration and parameter passing, Constructors Chapter 4: Object-oriented design, Inheritance and Polymorphism

	Static variables and methods, Class relationships, Interfaces, Inheritance and Visibility, Subclass, Class Hierarchies, Overriding and Overloading, Abstract classes, Polymorphism via Inheritance and via Interfaces, I/O Stream Chapter 5: Java Graphic User Interface Window, Layout, Common Control, Event Listener, Dialog Box...
Study and examination requirements and forms of examination	- Seminar about Mobile OSs: in group of 04 students: 20% - Assignments: 4 problems during the course per students: 20% - Mid-term project: Java classes, in group of 02 students: 20% - Final project: practical project with GUI, in group of 04 student, including presentation, demonstration and Q&A section: 40%
Media employed	Text book
Reading list	1. Mitsunori Ogihara, Fundamentals of Java Programming, First Edition, Springer, 2018 2. John Lewis, William Loftus, Java Software Solutions Foundations of Program Design, Eighth Edition, 2015

Written/updated by

Nguyen Thai Cong Nghia

54. Sensors and Advanced Measuring Instruments - ETC10106

Module name	Sensors and Advanced Measuring Instruments
Module level, if applicable	Undergraduate
Code, if applicable	ETC10106
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Trung-Khanh Le
Lecturer	MSc. Trung-Khanh Le
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM ; Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Measurement and Instruments, Basic Computer Networks, Microcontroller, Digital Signal Processing, Communication Systems
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand concepts of noise on electronic devices - Understand and analyze noise cancellation techniques - Understand principle of operation of some popular sensors - Understand principle of speed and light measurements - Design and program IoT devices and applications for sensor and measurement
Content	Interface Electronic Circuits Touch sensors Occupancy, Motion Detectors and Speed Light measurements IoT systems HTTP and MQTT protocol RF systems for IoT
Study and examination requirements and forms of examination	Assignments: 40% Final exam: 60%
Media employed	Text book
Reading list	- Handbook of Modern Sensors: Physics, Designs, and Applications 4th edition, Jacob Fraden, Springer 2010.

Written/updated by Trung-Khanh

Le

55. Digital Integrated Circuit and IP-core Design - ETC10107

Module name	Digital Integrated Circuit and IP-core Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10107
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Bui An Dong
Lecturer	MSc. Bui An Dong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 5 hrs/week (3 theory hrs + 2 lab hrs) Office hours: Saturday 06:40 AM – 9:40 AM
Workload	Contact hours: 50 hrs Assignments and independent learning: 50 hrs Total Working hours: 100 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Advanced Programming Techniques.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Knowledge of the interface standards on mobile devices. - Knowledge of the operating systems on mobile devices. - Applying Java language and XML. - Design a simple application on Android. - Design applications communicating between smart mobile devices and peripherals on Android.
Content	Chapter 1: Overview of the architecture of mobile devices + History of mobile devices + Architecture of the mobile device and especially, smart mobile device Chapter 2: Overview of the mobile OS + History of mobile OS + Overview of mobile OS + Architecture of mobile OS + Setting environment for application on Android Chapter 3: Java Programming Language + History of Java + The basic Java + object-oriented, classes and inheritance, interface and packaging + Debugging the common mistakes when writing program Chapter 4: Design of an application on Android OS + Overview of the applications environment + Flow of Studio design + Flow of application testing

	+ The basic components of the application on Android + Activity and storing Activity Chapter 5: Design of application interface on Android OS + Using intuitive tool, XML, Java + Knowledge of Bluetooth or Network standard + Design of interface between mobile and peripheral devices uses Bluetooth or network Lab 1: Java Lab 2: Design of basic application on Android + Basic application Lab 3: Design of advance application on Android + Advance application + Interface of application Lab 4: Design of interface between mobile and peripheral devices + Design of a peripheral devices + Design of a interface
Study and examination requirements and forms of examination	- Assignments: 3 or 4 problems during the course: 30% - Mid-term exam (45 minutes): Projects: 20% - Final exam (60 minutes): Projects: 50%
Media employed	Text book
Reading list	1. Thiết kế vi mạch CMOS VLSI, Tống Văn On, NXB Phương Đông, 2007. 2. ASIC Design and Synthesis: RTL Design Using Verilog, Vaibbhav Taraate, Springer. 2021 (VNULIB)

Written/updated by
Bui An Dong

56. Digital Integrated Circuit and IP-core Design Lab - ETC10108

Module name	Digital Integrated Circuit and IP-core Design Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10108
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Bui An Dong
Lecturer	MSc. Bui An Dong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 5 hrs/week (3 theory hrs + 2 lab hrs) Office hours: Saturday 06:40 AM – 9:40 AM
Workload	Contact hours: 50 hrs Assignments and independent learning: 50 hrs Total Working hours: 100 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Electronic, Programmable Logic Design, Languages for VLSI Design, VLSI Circuits and Technology.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Applying Verilog language to design digital integrated circuit. - Applying Verilog language to create testbench for digital integrated circuit design. - Simulate and test the operating of design. - Knowledge of the IC digital library. - Use software to optimize, synthesize, layout and test timing of the design. - Having ability to comprehend English documents on digital integrated circuit design.
Content	Chapter 1: Lab 1: Verilog Compiler Simulator + Verilog code + Testbench + Simulation Chapter 2: Lab 2: Synthesis and Optimize Design + Constraints + Report + Optimize Chapter 3: Lab 3: Static Timing Analysis + Debugging timing violations + Optimize Chapter 4: Lab 4: Layout + Preparing Library data + Layout flow

	+ DRC and LVS
Study and examination requirements and forms of examination	- Assignments: 3 or 4 problems during the course: 30% - Mid-term exam (45 minutes): Written exam: 20% - Final exam (60 minutes): Multi-choice and Written exam: 50%
Media employed	Text book
Reading list	1. Thiết kế vi mạch CMOS VLSI, Tổng Văn On, NXB Phương Đông, 2007. 2. ASIC Design and Synthesis: RTL Design Using Verilog, Vaibbhav Taraate, Springer. 2021 (VNULIB)

Written/updated by
Bui An Dong

57. Computer Vision - ETC10109

Module name	Computer Vision
Module level, if applicable	Undergraduate
Code, if applicable	ETC10109
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Huynh Quoc Thinh
Lecturer	MSc. Huynh Quoc Thinh
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hours/week Office hours: Tuesday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Basic Programming, Probability and statistics, Numerical Analysis and Matlab.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basics of digital image and computer vision - Apply basic image processing algorithms (filters, convert color, image transformations, ...) - Apply algorithms to extract features, search, recognition... - Build a basic computer vision system using OpenCV - Select algorithms to solve the problem - Improve programming skills
Content	Chapter 1: Introduce Computer vision and application Introduce OpenCV library Image information and image acquisition Handling files, camera Chapter 2: Basic image processing: - Converting images between difference color mode - Histogram equalization - Filter methods - Transformations. Chapter 3: Detect image features: -Find edges and corners. -Find and describe image features Chapter 4: Image segmentation Retrieving images and searching using image descriptors: -Harris corners, SIFT, SUFF, or ORB -KNN

	Chapter 5: Detecting and Recognizing Faces - Using Haar cascades to detect faces in still images and videos - Gathering images to train and evaluate a face recognizer - Face recognition algorithms: Eigenfaces, Fisherfaces, and Local Binary Pattern Histograms (LBPHs). Chapter 6: Object detector - Histogram of oriented gradients (HOG) descriptor - Understanding non-maximum suppression, also called non-maxima suppression (NMS) - Understanding of support vector machines (SVMs) Chapter 7: Tracking Object - Detect moving objects using the Mixture of Gaussians (MOG) and k-nearest neighbors (KNN) background subtractors - MeanShift and CamShift algorithm
Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 20% - Mid-term exam (60 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40% - Final project: Design of computer vision application (Text report and presentation): 40%
Media employed	Text book
Reading list	1. OpenCV (https://opencv.org) 2. Computer and robot vision, Robert M. Haralick, Massachusetts : Addison-Wesley, 1993

Written/updated by
Huynh Quoc Thinh

58. Biomedical Electronic Circuits - ETC10110

Module name	Biomedical Electronic Circuits
Module level, if applicable	Undergraduate
Code, if applicable	ETC10110
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Nguyen Duy Manh Thi
Lecturer	MSc. Nguyen Duy Manh Thi
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 07:30 AM – 09:30 AM Thursday 07:30 AM – 09:30 AM
Workload	Contact hours: 60 hrs Assignments and independent learning: 90 hrs Total Working hours: 150 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Biomedical Electronics
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts related to biomedical electronics, bioelectric potential - Understand operation of analog filter circuits, Op-Amp based circuits, and analog to digital converters in biomedical instrumentations. - Understand and classify of biomedical electrodes, sensors, transducers. Understand of biomedical instrumentation circuits and designs, and medical safety. - Describe the operations of active filters, AD/DA converters in biomedical circuits. - Apply biomedical analog circuit designs with theories and tools to build up biomedical instrumentation systems. - Demonstrate the systematization of knowledge in biomedical systems.
Content	Chapter 1: Review Op-Amp Fundamentals, Equivalent circuits, Op-Amp basics, Op-Amp specification. Differential Amplifier circuits, Comparators and applications. Chapter 2: Introduction to Biomedical Instrumentation Systems Bio-potentials signals, Biomedical system diagrams, Biomedical Safety and regulations

	Chapter 3: Biomedical Electronics: Analog. Types of Active Filters design and applications Biomedical Electronics: Digital. Biomedical signal processing, AD/DA converter, FIR/IIR filters Chapter 4: Biomedical Electrodes, Sensors and Transducers Definition and Classification of Electrodes, Sensors and Transducers Model of Biomedical Electrode. Types of Electrodes. Performance Characteristics of Electrodes, Sensors and Transducers Pressure, Flow, Temperature, Optical ... Transducers Chapter 5: Electromyography (EMG) Instrumentation EMG Electrodes and Circuits EMG Circuits Design of an Instrumentation Noise and Signal Processing in EMG Systems Chapter 6: Instrumentation in Diagnostic Cardiology (ECG) ECG Signal ECG Leads Noisy or poor quality ECG Traces Block diagram of an ECG Machine ECG Machines and Monitors ECG Circuits Design Chapter 7: Electroencephalography (EEG) Instrumentation The organization of the brain and Neural Communication Bioelectric Potentials and brain Waves EEG Leads, Electrodes and Accessories and EEG Circuits Design EEG Machines and Procedures Chapter 8: SPO2 (Saturation of peripheral Oxygen) Measurement Principles and Concepts of SPO2 measurement SPO2 circuits and devices Chapter 9: Measurement Procedures and Circuits of Biomedical Instrumentations Instrumentation in Diagnostic Ultrasound Blood Pressure Measurement Lung Volume Measurement Glucosamine Measurement The X-Ray Machine, CT Scanner MRI Machines
Study and examination requirements and forms of examination	- Experiment Assignments: 5 problems during the course: 20%

	- Project-based Lab: Group activity on experiment, building analog biomedical circuits for specific biomedical instrumentation systems: 30% - Final exam (90 minutes): Written exam: 50%
Media employed	Text book
Reading list	1. Mạch xử lý tín hiệu y sinh, GS.TS. Nguyễn Đức Thuận, PGS. TS. Vũ Duy Hải, TS. Nguyễn Phan Kiên, NXB Bách Khoa Hà Nội, 2016. 2. Giáo trình Thực tập Mạch điện tử Y sinh, Nguyễn Thanh Hải, Nguyễn Thanh Nghĩa, NXB ĐH Quốc gia TPHCM, 2019.

Written/updated by
 Nguyen Duy Manh Thi

59. Internship - ETC10111

Module name	Internship
Module level, if applicable	Undergraduate
Code, if applicable	ETC10111
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	A supervisor in industrial companies and a faculty member
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: NA Office hours: NA
Workload	Assignments and independent learning: 300 Total Working hours: 300
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Know contemporary engineering practice - Use engineering skills, and modern engineering tools necessary for engineering practice - Self-orient in the desired professional field - Understand the impact of engineering solutions in a global, economic, environmental, and societal context - Conduct work responsibly in cooperation with others - Write and present a report in an adequate format
Content	Attendance: Students should attend 100%. Attendance will be regulated and checked Individual Assignments: Tasks are assigned by the industrial company. Internship: At an industrial company in the field corresponding to the registered major
Study and examination requirements and forms of examination	Internship certificate issued by the company: 50% Written report and presentation graded by the faculty supervisor: 50%
Media employed	None
Reading list	None

Trong Tu

Written/updated by PhD. Bui

60. Analog Integrated Circuit Design - ETC10112

Module name	Analog Integrated Circuit Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10112
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	PhD. Bui Trong Tu
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 08:00 AM – 09:30 AM Thursday 08:00 AM – 09:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	VLSI Circuits and Technology
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the large and small signals and use MOS transistor models. - Identify three basic amplifier configurations: common source, common gate and common drain - Design current sources and mirrors for integrated circuits - Understand the gain-bandwidth concept and frequency response of basic amplifiers - Estimate dominant pole frequencies using pole splitting Miller effect - Use Bode plots to estimate the phase margin of operational amplifiers in feedback. Understand operational amplifier frequency compensation. - Design a basic operational amplifier.
Content	Chapter 1: CMOS device physic review Chapter 2: Spice models and layout Chapter 3: Analog Subcircuits Switches, Diodes/active resistors, Current mirrors, Current sources/current sinks Current/voltage references Chapter 4: Amplifiers Simple Amplifiers, Cascode Amplifiers, Current Amplifiers, Output Amplifiers Differential Amplifiers Chapter 5: Frequency Response of Amplifiers

	Chapter 6: CMOS Operational Amplifiers Chapter 7: Stability and compensation
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30% - Final exam: Written exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Analog Integrated Circuit Design, David A. Johns and Ken Martin, John Wiley and Sons. 2. Analysis and Design of Analog Integrated Circuits, Paul R. Gray, Paul J. Hurst, Stephen H. Lewis , Robert G. Meyer, John Wiley and Sons. 3. CMOS Analog IC Design: Fundamentals, Erik Bruun, bookboon.com

Written/updated by
PhD. Bui Trong Tu

61. Analog Integrated Circuit Design Lab - ETC10113

Module name	Analog Integrated Circuit Design Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10113
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Trung-Khanh Le
Lecturer	MSc. Trung-Khanh Le
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Basic Electronics Lab, Analog and Digital Electronics Lab, Physics of Electronic Devices, Signals, Systems and Circuit Analysis, Analog Integrated Circuit Design.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Manipulate of CAD tool for Analog CMOS design - Analyze operation and characteristics of Analog CMOS circuit with SPICE - Design schematic and layout of Analog CMOS circuit.
Content	Chapter 1: CAD Tools and basic usage Chapter 2: Netlist and simulation Chapter 3: Analyze CMOS circuit in DC and transient mode Chapter 4: Analyze CMOS circuit in AC mode Chapter 5: Analyze effect of PVT and noise Chapter 6: Design layout of analog CMOS circuit Chapter 7: LVS and LPE Chapter 8: ESD and IO PAD design Chapter 9: Design resistor, capacitor and inductor in CMOS circuit Chapter 10: Export to GDS II and Review
Study and examination requirements and forms of examination	Assignments: 30% Final exam: 70%
Media employed	Text book
Reading list	- Thực hành Thiết Kế Vi Mạch Tương Tự, Lê Trung Khanh

Written/updated by Trung-Khanh
Le

62. Languages for VLSI Design - ETC10114

Module name	Languages for VLSI Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10114
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Nguyen Duy Manh Thi
Lecturer	MSc. Nguyen Duy Manh Thi
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 07:30 AM – 09:30 AM Thursday 07:30 AM – 09:30 AM
Workload	Contact hours: 60 hrs Assignments and independent learning: 90 hrs Total Working hours: 150 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Programming Techniques in Electronics.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the usage of scripting languages in VLSI IC design - Realize scripting languages in EDA tools during IC design - Distinguish the differences between scripting languages and software programming - Use of scripting language during IC design phase. - Describe and analyze application programs and working environment with EDA tools - Program and debug scripting program during IC design phases
Content	Chapter 1: Introduction to Linux Operation System and Environment Linux Terminal and Commands Chapter 2: Linux Programming Shell programming (C-Shell, Bash Shell) VI/EMACS Editors Chapter 3: Awk scripting language programming Increasing productivity with command-line tools Find, Grep, Sed utilities Chapter 4: Tcl/Tk scripting programming Language The Basics of Tcl

	Tcl: Interactive and Batch mode Tcl: Arithmetic Tcl: List Tcl: Array Tcl: Regular Expression Tcl: String Tcl: Loop-If-While-Case Tcl: Procedure Chapter 5: Perl programming language Perl Introduction and Environment Perl Syntax, Data Types, Variables, Scalars, Arrays, Hashes Perl If..else, Loops Perl File I/O Perl Regular Expression Chapter 6: Python Programming Language Python: Indentation Python: Print Python: String Python: List, Getting elements, Add elements Python List: Push & Pop Python List: Processing Python List: Copy vs. Mirror Python: Dictionary Chapter 7: Supporting programming Tools: Makefile, Concurrent Versions System (CVS) Program building tool Makefile Rule introduction and writing rules How make works. Writing Makefiles Makefiles Conventions CVS Introduction and Environment Creating CVS, Import and Export CVS Commands
Study and examination requirements and forms of examination	- Assignments: 10 programming problems during the course: 40% - Experiment Lab: 1 practical project Lab: 10% - Final exam (90 minutes): Written exam: 50%
Media employed	Text book
Reading list	1. CAD Scripting Language, Quan Nguyen, RAMACAD Publisher, 2009. 2. The CAD Connection, Quan Nguyen, RAMACAD Publisher, 2007. 3. Lập trình Linux 1, Nguyễn Phương Lan, Hoàng Đức Hải, NXB Lao động Xã hội, 2007.

Written/updated by

Nguyen Duy Manh Thi

63. PCB Design - ETC10115

Module name	PCB Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10115
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Nguyen Thi Thien Trang
Lecturer	MSc. Nguyen Thi Thien Trang
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Digital Electronics, Analog Electronics, Electronic Circuit Techniques.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of PCB designing, packaging - Apply advance techniques, skills and modern tools for designing and fabrication of PCBs - Apply the knowledge and techniqueto fabricate Multilayer, SMT and HDI PCB - Assemble related documents, imitate and calibrate analog circuit designs, combine theory and tools - Demonstrate the systematization of knowledge
Content	Chapter 1: Introduction to Printed circuit board Layout planning, general rules and parameters, ground conductor considerations, thermal issues, check and inspection of artwork Chapter 2: Design rules & Electronic Design Automation (EDA) tools for PCB designing Chapter 3: Printed circuit board production techniques Photo printing, filmmaster production, reprographic camera, basic process for double sided PCBs photo resists, Screen printing process, plating, relative performance and quality control, Etching machines, Solders alloys, fluxes, soldering techniques, Mechanical operations Chapter 4: PCB Technology Trends

	Multilayer PCBs. Multiwire PCB, Flexible PCBs, Surface mount PCBs, Reflow soldering, Introduction to High-Density Interconnection (HDI) Technology Chapter 5: PCB design for EMI/EMC Subsystem/PCB Placement in an enclosure, Filtering circuit placement, decoupling and bypassing, Electronic discharge protection; Printed circuit boards Recycling techniques, Introduction to Integrated Circuit Packaging and footprints, NEMA and IPC standards
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 30% - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. Clyde F. Coombs, Printed Circuit Handbook, Mc Graw Hill, 1997. 2. Nguyễn Việt Hùng, Giáo trình điện tử mô phỏng và thiết kế mạch in với sự trợ giúp của máy tính tự học Pspice 7.1 bằng hình ảnh, Thống kê, 2003. 3. Nguyễn Việt Hùng, Phạm Quang Huy, Nguyễn Phương Quang, Vẽ và thiết kế mạch in với Orcad, Lao động Xã hội, 2006.

Written/updated by
 Nguyen Thi Thien Trang

64. SoC Design - ETC10208

Module name	SoC Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10208
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 08:00 AM – 11:00 AM Friday 08:00 AM – 11:00 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital electronics, Programming Techniques, Computer Structure.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understanding the FPGA-based embedded system design process - Understanding bus architecture on embedded systems - Capable of designing master and slave components - Understanding software (drivers) for hardware design - Ability to develop applications on embedded systems
Content	Chapter 1: Overview of embedded systems Chapter 2: Hardware in embedded systems Chapter 3: Software in embedded systems. Chapter 4: Embedded systems based on microcontrollers and FPGA Chapter 5: Design master and slave components on FPGA
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (45 minutes): Multiple choice exam, a four-page cheat sheet per student is allowed in the exam room: 20% - Final exam (60 minutes): Written exam, a four-page cheat sheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. S. Brown and Z. Vranesic, Fundamental of Digital Logic with Verilog Design, Mc Graw Hill, Second Edition, 2008 2. SOPC Builder Support and Nios® II Processors Support (www.intel.com)

Written/updated by
Huynh Huu Thuan

65. SoC Design Lab - ETC10209

Module name	SoC Design Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10209
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Bachelor. Dang Tan Phat
Lecturer	Bachelor. Dang Tan Phat
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2 hours/week Office hours: Based on the timetable made by the teacher for the students.
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Electronics, Programming Techniques.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the signals of Avalon bus and clarify the usage of them. - Describe and explain the design flow of SoC system using RTL. - Characterize IP cores from Intel library and design custom IP core. - Use design and simulation tools to create and verify individual IPs and entire systems. - Analyze and characterize the suitable hardware features to meet requirements. - Design a SoC system to meet real-world problems.
Content	Lab 1: Introduction - The SoC system design flow includes both hardware and software. - How to use Quartus Prime and Platform Designer tools to create computer hardware, synthesize, and download to FPGA development board. - How to use the NIOS II tool to compile and download software to the FPGA SoC system. Lab 2: Communication with IO peripheral devices use PIO module - Intel's PIO (Parallel Input/Output) structure (registers, functions...).

	- How to build hardware systems with PIO using Platform Designer to perform simple I/O functions. - How to access PIO hardware block designed on C program through Nios II - Eclipse. Lab 3: Custom design parallel IO port - How to add a custom module to a library in Platform Designer. - How to build and connect IO interfaces (built-in Verilog language) to an existing SoC system. - How to access IO interfaces from the Nios II engine. Lab 4: Functional module design and survey with signal tab - Design a simple module as certain required by using Verilog HDL, and access this module through Nios II. - How to analyze signal logic and debug using the Signal Tab tool. Lab 5: System design with simple memory - Design a simple memory using Verilog code. - How to add custom components to the SoC system. - How to verify this module with the Signal Tab tool. Lab 6: Modelsim and Testbench - How to build testbench to initialize input data and check the output of hardware blocks. - How to use Modelsim tool to simulate hardware blocks. Lab 7: Communication interval timer and interrupt handling - Intel's Interval Timer architecture (registers, functions, ...). - How to build a hardware system using Platform Designer to communicate with the Interval Timer. - How to build C program on Nios II – EDS tool to communicate with Interval Timer, and use Timer interrupt.
Study and examination requirements and forms of examination	- Diligence score: Students attend more than 80% of course: 20% - Assignments: 2 exercises per a lesson, total 14 exercises during the course: 20% - Final exam (60 minutes): Write two programs by assembly programming to control NIOS II CPU for certain requirements: 60%
Media employed	Text book
Reading list	1. Intel's Avalon Interface Specifications. 2. Intel's Embedded Peripherals IP User Guide.

Written/updated by
Tran Tuan Kiet

66. Advanced Computer Networks Lab - ETC10210

Module name	Advanced Computer Networks Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10210
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Nguyen Quang Anh
Lecturer	MSc. Nguyen Quang Anh
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Practice building a network through virtual machines - Practice building a domain network system through building internal DNS - Analyze and practice building a basic multidomain system - Practice to create one system including DHCP, NAT, WEB services from the internal network are accessible from the external WAN - Practice to create VPN client to site, site to site system - Analyze and evaluate the quality of an actual network model through measurement tools
Content	Chapter 1: DNS-DNS Backup Server Chapter 2: Domain Controller + Additional Domain + Child Domain + Tree Domain Chapter 3: DHCP -: DHCP -: DHCP Relay Agent Chapter 4: Networking Services + Web + NAT (inbound/outbound) Chapter 5: Virtual Private Network + Client to site + Site to site

	Chapter6: Routing Protocol + RIP v2 + OSPF + EIGRP + Static Route
Study and examination requirements and forms of examination	- Assignments: 9 exercises during the course: 30% - Midterm exam: do test in the computer: 20% - Final exam (60 minutes): Practice in the computer: 50%
Media employed	Text book
Reading list	Computer Networking: A Top-Down Approach Featuring the Internet, James F.Kurose and Keith

Written/updated by
Nguyen Quang Anh

67. Internship - ETC10211

Module name	Internship
Module level, if applicable	Undergraduate
Code, if applicable	ETC10211
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	A supervisor in industrial companies and a faculty member
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: NA Office hours: NA
Workload	Assignments and independent learning: 300 Total Working hours: 300
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Know contemporary engineering practice - Use engineering skills, and modern engineering tools necessary for engineering practice - Self-orient in the desired professional field - Understand the impact of engineering solutions in a global, economic, environmental, and societal context - Conduct work responsibly in cooperation with others - Write and present a report in an adequate format
Content	Attendance: Students should attend 100%. Attendance will be regulated and checked Individual Assignments: Tasks are assigned by the industrial company. Internship: At an industrial company in the field corresponding to the registered major
Study and examination requirements and forms of examination	Internship certificate issued by the company: 50% Written report and presentation graded by the faculty supervisor: 50%
Media employed	None
Reading list	None

Written/updated by PhD. Huynh Huu

Thuan

68. Connectivity and DAQ for IoT - ETC10219

Module name	Connectivity and DAQ for IoT
Module level, if applicable	Undergraduate
Code, if applicable	ETC10219
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Nguyen Quoc Khoa
Lecturer	Nguyen Quoc Khoa
Language	English/Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hours/week Office hours: Thursday 08:30 AM – 11:30 AM Saturday 08:30 AM – 11:30 AM
Workload	Contact hours: 30 hours Assignment and independent learning: 60 hours Total working hours: 90 hours
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Students are encouraged to have intermediate knowledge in Computer Architecture; C & Python Programming; Analog & Digital Processing; Computer Network.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understanding of basic concepts of Communicating & Data acquisition in IOT - Reading, understanding of English technical documents that used in this course - Know how to configure IOT devices and Sensors Technologies - Know how data acquisition and processing using machine learning - Know how to design and control an IOT system using Wireless Protocols - Understanding of basic concepts of Big data
Content	Chapter 1: IOT Introduction Chapter 2: From Internet to IOT & Applications Chapter 3: Short-range Wireless Protocols: ZigBee; Bluetooth LE Chapter 4: Long-range Wireless Protocols: LoRa Chapter 5: Arduino/ ESP8266 Chapter 6: MQTT Chapter 7: Machine Learning/AI Chapter 8: Machine Learning/AI Chapter 9: Big Data Chapter 10: IOT Devices/ Sensors Technology

Study and examination requirements and forms of examination	- Assignments: 5 mini assignments during the course: 50% - Final exam (60 minutes): written exam, all hard and soft materials are allowed in the exam: 50%
Media employed	Text book and References
Reading list	References - ZigBee.org - Bluetooth.com - ESP8266.com - lora-alliance.org - Lora-developers.semtech.com - MQTT.org - Tensorflow.org - Coursera.org

Written/updated by
 Nguyen Quoc Khoa

69. Connectivity and DAQ for IoT Lab - ETC10220

Module name	Connectivity and DAQ for IoT Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10220
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Bachelor. Dang Tan Phat
Lecturer	Bachelor. Dang Tan Phat
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 2 hours/week. Office hours: According to the timetable made a teacher for the students.
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Electronics, Programming Techniques.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts related to communication protocol in IoT - Explain the working of IoT devices through protocol, struture of mesage. - Understand and cover modules from libraries provided by vendors. - Use libraries flexibly and apply them in many contexts. - Analyze the purpose of the design and the strengths and weaknesses of the communication protocols. - Select communication protocols and design the system to meet actual requirements.
Content	Chapter 1: Introduction, UART protocol and peripherals. - Get familiar with the EFR32 Blue Gecko Bluetooth Starter Kit (SLWSTK6020B). - How to create projects from Simplicity Studio software. - UART communication. - Interface peripheral devices such as button, LED, LCD on EFR32. - Using the library. Chapter 2: I2C protocol - I2C communication process between EFR32 board devices. - How to build a simple I2C communication program through the lab provided by Silabs. Chapter 3: Bluetooth Low Energy

	- Basic knowledge of Bluetooth Low Energy. - Use the GATT configurator interface to build a local database. - Run the basic Bluetooth Low Energy program to communicate with the Smart Phone. Chapter 4: Bluetooth Low Energy (continue) - Understand how to communicate between GATT server and GATT client. - Build GATT client to connect to GATT server and get data from there. - Using temperature and humidity sensor (Si7021). Chapter 5: Project Practice - Presenting problems, analyzing problems and offering solutions based on learned knowledge, applying knowledge skills to analyze the advantages and disadvantages of the system, thereby selecting solutions: communication protocols, devices,...
Study and examination requirements and forms of examination	- Diligence score: Students attend more than 80% of course: 15% - Assignments: 2 exercises per a lesson, total 8 exercises during the course: 15% - Final exam: Presentation and report on the project: 70%
Media employed	Text book
Reading list	1. UG279: EFR32BG13 Blue Gecko Bluetooth Starter Kit User's Guide. 2. AN0822: Simplicity Studio™ User's Guide. 3. EFR32xG13 Wireless Gecko Reference Manual. 4. EFR32BG13 Blue Gecko Bluetooth® Low Energy SoC Family Data Sheet. 5. QSG139: Getting Started with Bluetooth® Software Development. 6. UG103.14: Bluetooth® LE Fundamentals. 7. UG118: Blue Gecko Bluetooth® Profile Toolkit Developer's Guide. 8. Bluetooth software API Reference Manual. 9. UG118: Blue Gecko Bluetooth® Profile Toolkit Developer's Guide.

Written/updated by
Dang Tan Phat

70. Operating System - ETC10221, ETC10222

Module name	Operating System Operating System Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10221 ETC10222
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Do Quoc Minh Dang
Lecturer	MSc. Do Quoc Minh Dang
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 07:30 AM – 11:30 AM Lab: 4 hrs/week Office hours: Monday 07:30 AM – 11:30 AM
Workload	- Operating System: Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs - Operating System Lab: Contact hours: 30 hrs Lab preparation: 30 hrs Total Working hours: 60 hrs
Credit points	- Operating System: 2 - Operating System Lab: 1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Computer Architecture, C Programming Language
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand design idea and implementation of an application OS - Understand how OS manages and runs application processes - Understand how OS abstracts and shares hardware among processes in the computer - Use the OS system calls as a foundation for system-level programming - Extend the OS to support new sophisticated features - Develop device drivers in the OS to interface with new hardware
Content	Chapter 1: Operating System Interface Describe the services that xv6 provides and how user applications make use of them

	Chapter 2: Virtual Memory and Page Table Explain how a private virtual address space is created for each user process and how xv6 manages physical memory in the computer Chapter 3: Interrupts and Traps Describe the procedure that RISC-V take when an interrupt or exception occurs. Describe how xv6 handles these events and interfaces with hardware in the computer Chapter 4: Locking and Scheduling Describe the mechanisms that xv6 uses to synchronize concurrent access of a sharing resource and how xv6 schedules and multiplexes user processes on each processor core Chapter 5: File System Describe the data structure used by xv6 to organize data on disk to provide a high-level abstraction of file and folder to users Lab 1: Linux environment and utility tools, bash shell and its features Lab 2: C program, C library and OS system calls Lab 3: OS system calls extension Lab 4: Page Table content of a process and process's virtual memory address space Lab 5: Multithread application, race condition and program performance
Study and examination requirements and forms of examination	- Operating System: • Assignments: homework exercises for each lesson: 50% • Reading preparation: reading topics in textbook in advance before class: 10% • Final exam: Oral exam: 40% - Operating System Lab: • Assignments: do lab and write lab report: 100%
Media employed	Text book
Reading list	1. xv6: a simple, Unix-like teaching operating system, Russ Cox, Frans Kaashoek, Robert Morris, MIT, 2021 https://pdos.csail.mit.edu/6.828/2021/xv6/book-riscv-rev2.pdf 2. Operating Systems: Three Easy Pieces, Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau, University of Wisconsin-Madison, 2018 https://pages.cs.wisc.edu/~remzi/OSTEP/

Written/updated by
Do Quoc Minh Dang

71. Introduction to Image and Video Processing - ETC10223, ETC10224

Module name	Introduction to Image and Video Processing Introduction to Image and Video Processing Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10223 ETC10224
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 08:00 AM – 11:00 AM Friday 08:00 AM – 11:00 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Numerical Analysis and Matlab Lab, Digital Signal Processing.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to design and implement filters using for image processing - Ability to implement DCT and Wavelet transforms and their applications in image and video processing. - Ability to implement image and video compression
Content	Chapter 1: Course Introduction Chapter 2: Digital filters, Fourier Transform, Wavelet Transform and Sampling Chapter 3: Image Enhancement Chapter 4: Image Recovery Chapter 5: Image Compression; Jpeg and Jpeg2000 Chapter 6: Motion Estimation and Video Compression Lab 1: 2-D filter and edge detection Lab 2: DCT transform Lab 3: Wavelet transform Lab 4: Jpeg compression Lab 5: Motion Estimation

Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (45 minutes): Multiple choice exam, a four-page cheat sheet per student is allowed in the exam room: 20% - Final exam (60 minutes): Written exam, a four-page cheat sheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. Y. Wang, J. Ostermann, and Y.Q.Zhang, Video Processing and Communications. Prentice Hall, 2002. 2. R. C. Gonzalez and R. E. Woods, Digital Image Processing, Prentice Hall, (3rd Edition) 2008.

Written/updated by
Huynh Huu Thuan

72. Java Programming - ETC10225

Module name	Java Programming
Module level, if applicable	Undergraduate
Code, if applicable	ETC10225
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 2 hrs/week Office hours: Wednesday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Programming Technique of Electronics – Telecommunication Branch.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to design, code, test and debug Java application program - Ability to self learn advanced Java language - Ability to implement software
Content	Chapter 1: Getting Started with Java Chapter 2: Console Input and Output Chapter 3: Flow of Control Chapter 4: Defining Classes Chapter 5: Array Chapter 6: Inheritance Chapter 7: Polymorphism and Abstract Classes Chapter 8: Exception Handling Chapter 9: File I/O handling Chapter 10: Java Applications: Applets, HTML and Swing
Study and examination requirements and forms of examination	- Final exam (60 minutes): Written exam: 100%
Media employed	Text book
Reading list	1. DiMarzio Jerome F., Android : a programmer's guide, McGraw-Hill, 2008

	2. Trương Thị Ngọc Phượng, Lập trình Android, Thời Đại, 2012
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Written/updated by Huynh Huu Thuan

73. Java Programming Lab - ETC10226

Module name	Java Programming Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10226
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 1.5 hrs/week Office hours: Saturday 01:30 PM – 3:00 PM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Programming Technique of Electronics – Telecommunication Branch.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to design, code, test and debug Java application program - Ability to self learn advanced Java language - Ability to implement software
Content	Chapter 1: Getting Started with Java Chapter 2: Console Input and Output Chapter 3: Flow of Control Chapter 4: Defining Classes Chapter 5: Array Chapter 6: Inheritance Chapter 7: Polymorphism and Abstract Classes Chapter 8: Exception Handling Chapter 9: File I/O handling Chapter 10: Java Applications: Applets, HTML and Swing
Study and examination requirements and forms of examination	- Final exam (30 minutes): 100%
Media employed	Text book

Reading list	1. DiMarzio Jerome F., Android : a programmer's guide, McGraw-Hill, 2008 2. Trương Thị Ngọc Phượng, Lập trình Android, Thời Đại, 2012
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Written/updated by Huynh Huu Thuan

74. Network Security - ETC10227

Module name	Network Security
Module level, if applicable	Undergraduate
Code, if applicable	ETC10227
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Quang Anh
Lecturer	MSc. Nguyen Quang Anh
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network, Advanced Computer Networking.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand and explain the concept of computer network security - Identify some of the factors driving the need for network security - Identify and classify particular examples of attacks - Understand various protocols for network security to protect against the threats in the networks. - Compare and contrast symmetric and asymmetric encryption systems and their vulnerability to attack, and explain the characteristics of hybrid systems. - Learn about how to maintain the Confidentiality, Integrity and Availability of a data.
Content	Chapter 1: Overview, introduces the subject

	• The theory of network security (process development, current trends, infrastructure policies) • Introduction to worms, viruses, trojans • Referral forms of attack: polls Attack, attack accounts for access, denial of service attacks Chapter 2: Protecting network devices • Methods of protecting border routers • Assign the access routers • Enhance security remote login • Configuring SSH • Assign user rights on routers • Oversee and manage routers in the network infrastructure Chapter 3: AAA model + Security settings, Firewall settings on the hard / soft + AAA on Cisco Router • ISA + Firewall 2006 & TMG 2010 Chapter 4: Implement firewall technologies • Access Control List (Access Control Lists) • Introduction of firewall techniques: Protecting the network using a firewall, the firewall type, model design firewall • Access control based on the scope (Context-Based Access Control) • Firewall management access partition Chapter 5: The type of encryption + Overview of PKI + Certificate + Kerberos + The digital signature + Https + Hash (Hash) • MD5 and SHA-1 • Authentication with HMAC • Key Management • Basics of confidentiality of data + The symmetric encryption algorithms: DES, 3DES, AES + The algorithm asymmetric encryption: RSA and PKI Chapter 6: The system safety solutions + Firewall + Snort + DHCP Snooping + DAI
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (30 minutes): Multiple choice exam, 4-page cheatsheets are allowed in the exam room: 20% - Final exam (60 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 50%
Media employed	Text book
Reading list	Network security and cryptography Robert Karamagi, 2021

Written/updated by
Nguyen Quang Anh

75. Network Security Lab - ETC10228

Module name	Security Networking Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10228
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Quang Anh
Lecturer	MSc. Nguyen Quang Anh
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Practice protect for Cisco Router - Practice cyber attack techniques to understand the risk of data loss - Practice to build a security web server using SSL - Practice to create mail server using Ipsec to encrypt - Practice to create VPN client to site, site to site system - Analyze and evaluate the quality of an actual network model through measurement tools
Content	Chapter 1: Configure to protect Cisco Router Chapter 2: Cyber attack techniques + scanning + password hacking + create a basic trojan + arp poisoning Chapter 3: HTTPS using SLL Chapter 4: Ip-security Chapter 5: Virtual Private Network using Ipsec and Certificate +Client to site +Site to site Chapter 6: Build AAA System

Study and examination requirements and forms of examination	- Assignments: 9 exercises during the course: 30% - Midterm exam: do test in the computer: 20% - Final exam (60 minutes): Practice in the computer: 50%
Media employed	Text book
Reading list	- Practice Course of advanced computer networks (internal circulation only).

Written/updated by
 Nguyen Quang Anh

76. Network Technology- ETC10307

Module name	Network Technology
Module level, if applicable	Undergraduate
Code, if applicable	ETC10307
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Tran Thi Huynh Van
Lecturer	MSc. Tran Thi Huynh Van
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2.5 hrs/week Office hours: Monday 12:30 PM – 15:00 PM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network, Communication Systems
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Generalize fundamental, operation of Active Directory, domain resources on Microsoft servers - Relate the network/system administration based on Windows Servers services: DHCP, DNS, VPN - Design the small/big company's network/system administration model based on related background of studied Windows Servers services - Synthesize systematization of knowledge - Set up systems/model to experience how to manage domain users, groups, file sharing - Organize to rewrite research documents and effectively present related knowledge of network administration: RAID, Virtual Machines control, etc.
Content	Chapter 1: Active Directory Chapter 2: User and Group management Chapter 3: File Sharing Chapter 4: Group Policy and Security Policy Chapter 5: Dynamic Host Configuration Protocol Service Chapter 6: Network Address Translation Chapter 7: Domain Name System Chapter 8: Wide Area Networks Chapter 9: Virtual Private Network Service Chapter 10: Frame Relay

Study and examination requirements and forms of examination	- Assignments: 4 problems during the course: 25% - Course small projects: Work in group of 4, will be assessed by presentation, course project summary report: 25% - Final exam (60 minutes): Multiple choice exam: 50%
Media employed	Text book
Reading list	1. “Bài giảng Công nghệ mạng”, Nguyen Viet Ha, University of Sciences Ho Chi Minh City, 2012

Written/updated by

Tran Thi Huynh Van

77. Network Technology Lab-ETC10308

Module name	Network Technology Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10308
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc. Ngo Thanh Hai
Lecturer	MSc. Ngo Thanh Hai
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 12:30 PM – 15:00 PM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Network, Communication Systems
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Generalize fundamental, operation of Active Directory, domain resources on Microsoft servers - Relate the network/system administration based on Windows Servers services: DHCP, DNS, VPN - Design the small/big company's network/system administration model based on related background of studied Windows Servers services - Synthesize systematization of knowledge - Set up systems/model to experience how to manage domain users, groups, file sharing - Organize to rewrite research documents and effectively present related knowledge of network administration: RAID, Virtual Machines control, etc.
Content	Chapter 1: Active Directory Chapter 2: Group Policy and Security Policy Chapter 3: File Sharing Chapter 4: Dynamic Host Configuration Protocol Service, Routing, Network Address Translation Chapter 5: Domain Name System Chapter 6: Mail Service Chapter 7: Virtual Private Network Service Chapter 8: IP security
Study and examination requirements and forms of examination	- Attendance: 15% - Lab reports: 35% - Final lab exam: 50%

Media employed	Text book
Reading list	1. “Bài giảng Công nghệ mạng”, Nguyen Viet Ha, University of Sciences Ho Chi Minh City, 2012

Written/updated by

Ngo Thanh Hai

78. Wireless Communications- ETC10309

Module name	Wireless Communications
Module level, if applicable	Undergraduate
Code, if applicable	ETC10309
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7 semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Communications
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Calculate coherence bandwidth and coherence time of wireless channels, channel modeling - Design of equalizer for wireless channels - Design of high-speed communication systems based on OFDM, spread spectrum, MIMO - Create, simulate, and present a part of wireless communication systems.
Content	Chapter 1: Overview of wireless communications Basic concepts and definitions Overview of wireless networks Chapter 2: Wireless channels Large-scale path loss Small-scale multipath propagation Chapter 3: Equalization Linear equalizers Nonlinear equalization Chapter 4: OFDM technique Principle of OFDM Use of FFT/IFFT to implement the OFDM, cyclic prefix Applications of OFDM: IEEE802.11, LTE Chapter 5: Spread spectrum Direct sequence spread spectrum (DSSS) Frequency-hopping spread spectrum (FHSS) Chapter 6: MIMO technique

	Diversity Space time block coding Space division multiplexing
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Group Project: Presentation and demonstration: 40% - Final exam: Multiple choice exam and Written exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Theodore S. Rappaport, Wireless Communications: Principles and Practice, 2nd Ed. Prentice Hall P T R, USA, 2001 2. Andrea Goldsmith, Wireless Communications, Cambridge University, UK, 2005

Written/updated by

Dang Le Khoa

79. Mobile Communications- ETC10310

Module name	Mobile Communications
Module level, if applicable	Undergraduate
Code, if applicable	ETC10310
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7 semester
Person responsible for the module	MSc. Truong Tan Quang
Lecturer	MSc. Truong Tan Quang
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Wednesday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Communications (ETC10301); Telecommunication Networks (ETC10302); Antenna and Wave Propagation (ETC10305);
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the concepts of mobile communication systems, analyze the operating principles of equipment components in a basic mobile communication network. - Evaluate the performance quality of some key components in existing mobile networks: GSM900, W-CDMA, LTE, etc. - Plan a basic digital cellular system with the following parameters: capacity, number of users, cell area, number of sites, frequency reuse factor, co-channel interference. - Synthesize, analyze and evaluate a basic mobile communication network. - Presents development goals and roadmap for future generations of mobile communication networks.
Content	Chapter 1: Overview - Introduction Course + Mobile communication history + The mobile communication systems Chapter 2: GSM Networks Structure + Overall structure + The structure of network components + The geographical network structure + Function layered network structure Chapter 3: Cellular mobile systems & planning network + Frequency reuse techniques

	+ Cell classification & coverage methods + Planning network Chapter 4: Channel Structure – Signal Processing – Signalling Procedure + Physical & Logic channel structure + Signal processing + Signalling Procedure Chapter 5: Spread Spectrum Techniques & CDMA + Pseudo noise sequences + Gold & Walsh-Hadamard sequences + Direct sequence- spreading spectrum & CDMA + RAKE receiver Chapter 6: Third Generation Mobile Network – 3G + Evolution of 2G to 3G + W-CDMA Radio interface + 3G network structure + The geographical network structure + Channel structure & spreading spectrum techniques + The network model and the release Chapter 7: 4G mobile networks & developments + After the 3G mobile systems + High-speed transmission of broadband mobile information + Techniques used in 4G mobile networks + Overview of 5G networks
Study and examination requirements and forms of examination	- Assignments: 4 problems during the course: 60% - Final exam (60 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Principles of Mobile Communication, Gordon L. Stuber; New York, Boston, Dordrecht, London, Moscow, 2002. 2. Modern Wireless Communications; Simon Haykin, Pearson Prentice Hall, 2005. 3. Giáo trình lộ trình phát triển thông tin di động 3G lên 4G, TS. Nguyễn Phạm Anh Dũng, Nhà xuất bản thông tin và truyền thông, 2010.

Written/updated by

Truong Tan Quang

80. Wireless and Mobile Communications Labs- ETC10311

Module name	Wireless and Mobile Communications Labs
Module level, if applicable	Undergraduate
Code, if applicable	ETC10311
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa Ms. Nguyen Thi Xuan Uyen
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Communications
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Calculate coherence bandwidth and coherence time of wireless channels, channel modeling - Design of equalizer for wireless channels - Design of high-speed communication systems based on OFDM, spread spectrum, MIMO - Create, simulate, and present a part of wireless communication systems.
Content	Chapter 1: BER performance of communication system Chapter 2: Channel fading Chapter 3: Equalization Chapter 4: OFDM technique Chapter 6: OFDM system over Rayleigh fading channels Chapter 7: Spread spectrum Chapter 8: MIMO technique
Study and examination requirements and forms of examination	- Lab report: 50% - Final exam: computer simulation and oral interview: 50%
Media employed	Text book
Reading list	1. Theodore S. Rappaport, Wireless Communications: Principles and Practice, 2nd Ed. Prentice Hall P T R, USA, 2001 2. Andrea Goldsmith, Wireless Communications, Cambridge University, UK, 2005

Written/updated by

Dang Le Khoa

81. Optical Communications- ETC10312

Module name	Optical Communications
Module level, if applicable	Undergraduate
Code, if applicable	ETC10312
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc.Ngo Dac Thuan
Lecturer	MSc.Ngo Dac Thuan
Language	English
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2hrs/week Office hours: Tuesday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Communication Systems, Digital Communications, Antenna and Wave Propagation.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the role of the main parts, principle activity of every parts in optical system; - To be able to analyze, evaluate some of essential parameter in transmission; - Can be estimate, inference of system quality in optical transmission channels; - Can be patent search and analyze entities which deal with optical communications; - Build technical report based on essential optical parameters in optical communications.
Content	Chapter 1: Introduction and Overview of Optical Communications - List of main parts in optical Communications, Block diagram,... - Advantage of optical transmission: bandwidth, data rate, reliability, applications,... - Optical fiber, transmission media Chapter 2: Characteristics of optical Transmission - Attenuation, dispersion, - Non-linear effectively - Type of new optical fiber and manufacturing technologies Chapter 3: Optical transmitter unit - Electronic-Optical convert principle and method - Components of Optical transmit as LED, LASER

	- Line coding in Optical Transmission Chapter 4: Optical receiver unit - Optical-Electronic convert principle and method - Components of Optical transmit as PIN, APD - Characteristics of optical receiver Chapter 5: Single-channel terminal transmission - Evaluation of data rate, distance for getting a single channel terminal transmission Chapter 6: WDM optical system - Principle of Wavelength Division Multiplexer - Effectively of multi-channel transmission Chapter 7: Review and Q&
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 15% - Mid-term exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room or essay exam: 30% - Seminar: Using patent database in the world to search & analyze some of patents dealing with optical communications to write a research report on the selected topic and give an oral presentation: 15% - Final exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room or essay exam: 40%
Media employed	Text book
Reading list	1. Digital optical communications, Le Nguyen Binh, CRC Press, 2009 2. Cơ sở lý thuyết truyền tin, Nguyễn Văn Hậu, Nguyễn Hiếu Minh, Khoa học và Kỹ thuật, 2008 3. Mạng thông tin quang thế hệ sau, Hoàng Văn Võ, nxb Thông tin và Truyền thông, 2010

Written/updated by

Ngo Dac Thuan.

82. Optical Communications Lab- ETC10313

Module name	Optical Communications Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10313
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	MSc.Ngo Dac Thuan
Lecturer	MSc.Ngo Dac Thuan Ngo Minh Nghia Nguyen Thi Xuan Uyen
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 2hrs/week Office hours: - Thursday, 07:30 – 09:30 - Thursday, 09:30 – 11:30 - Saturday, 13:00 – 17:00
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Communication Systems, Digital Communications, Antenna and Wave Propagation.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the role of the main parts, principle activity of every parts in optical system; - To be able to analyze, evaluate some of essential parameter in transmission; - Can be estimate, inference of system quality in optical transmission channels; - Can be patent search and analyze entities which deal with optical communications; - Build technical report based on essential optical parameters in optical communications.
Content	Chapter 1: Optical Fiber - Effect of Dispersion - Effect of Group Velocity Dispersion (GVD) - Effect of Four-Wave Mixing (FWM) Chapter 2: Optical Transmitters - LED Modulation Response - Semiconductor Laser Modulation Response - Transmitter – External Modulated Laser Chapter 3: Optical Receivers

	- Modulation Formats: RZ, NRZ - Receiver Noise – PIN - Receiver Noise - Shot Noise Enhancement with APD - Receiver Sensitivity - Bit Error Rate (BER) - Receiver sensitivity - Minimum input power Chapter 4: Doped Optical Fiber Amplifiers - Analysis of Gain and Noise in Erbium doped fiber. - Parameters affecting the amplifier: length of optical fiber, input power, power of Pump Laser. Chapter 5: WDM System - Comparison of RZ and NRZ Modulation Formats.
Study and examination requirements and forms of examination	- Attendance: 15% - Lab reports: 35% - Final lab exam: 50%
Media employed	Text book
Reading list	1. Digital optical communications, Le Nguyen Binh, CRC Press, 2009 2. Cơ sở lý thuyết truyền tin, Nguyễn Văn Hậu, Nguyễn Hiếu Minh, Khoa học và Kỹ thuật, 2008 3. Mạng thông tin quang thế hệ sau, Hoàng Văn Võ, nxb Thông tin và Truyền thông, 2010

Written/updated by

Ngo Minh Nghia

83. Internship- ETC10111/ETC10211/ETC10320

Module name	Internship
Module level, if applicable	Undergraduate
Code, if applicable	ETC10320
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	A supervisor in industrial companies and a faculty member
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: NA Office hours: NA
Workload	Assignments and independent learning: 300 Total Working hours: 300
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Know contemporary engineering practice - Use engineering skills, and modern engineering tools necessary for engineering practice - Self-orient in the desired professional field - Understand the impact of engineering solutions in a global, economic, environmental, and societal context - Conduct work responsibly in cooperation with others - Write and present a report in an adequate format
Content	Attendance: Students should attend 100%. Attendance will be regulated and checked Individual Assignments: Tasks are assigned by the industrial company. Internship: At an industrial company in the field corresponding to the registered major
Study and examination requirements and forms of examination	Internship certificate issued by the company: 50% Written report and presentation graded by the faculty supervisor: 50%
Media employed	None
Reading list	None

Written/updated by PhD. Dang Le Khoa

84. Outside Visit, Practical Work- ETC10321

Module name	Outside Visit, Practical Work
Module level, if applicable	Undergraduate
Code, if applicable	ETC10321
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	A supervisor in industrial companies and a faculty member
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: NA Office hours: NA
Workload	Contact hours: Assignments and independent learning: 100 hrs Total Working hours: 100 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Know contemporary engineering practice - Use engineering skills, and modern engineering tools necessary for engineering Practice - Self-orient in the desired professional field - Understand the impact of engineering solutions in a global, economic, environmental, and societal context - Write and present a report in an adequate format
Content	None
Study and examination requirements and forms of examination	- Internship certificate issued by the company: 50% - Written report and presentation graded by the faculty supervisor: 50%
Media employed	None
Reading list	None

Written/updated by

Dang Le Khoa

85. Mobile Programming- ETC10322

Module name	Mobile Programming
Module level, if applicable	Undergraduate
Code, if applicable	ETC10322
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	PhD. Dang Le Khoa Mr. Nguyen Thai Cong Nghia
Lecturer	Nguyen Thai Cong Nghia
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3hrs/week Office hours: Tuesday 9:00 – 11:30 AM Wednesday 9:00 – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	No
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Build applications include activities, fragments, services... combined - Build user interfaces with views, layouts, adapter views - Construct database for storage and access - Combine application, database and hardware to create practical system - Diagram the activities relationship, database arrangement and system structure - Assemble related documents, combine theory and tools
Content	Chapter 1: Overview of Android App Development and Android studio Android OS, Android Developer Documents, SDK, API, Android Studio IDE, Project Structure, Run Application on Android Virtual Device Chapter 2: User Interfaces UI Overview, Widgets, Layouts, AdapterView, ListView, RecyclerView, Event Listener, Resource Types, Alternative resources and Configuration, Resource Access Chapter 3: Activity, Fragment and more Application Lifecycle, Activity, Activity Lifecycle, Intent, Intent Filters, Navigation, Fragment, Fragment Lifecycle, Thread, Service, Permission, Camera, Google Maps... Chapter 4: Firebase

	Firebase Documents, Add firebase to an Android app, Firebase Realtime Database, JSON, Authentication, Storage...
Study and examination requirements and forms of examination	- Coding challenges: 04 challenges during class per students: 20% - Projects review: idea, solution and system structures presentation for final project, related projects review (at least 3 related projects), in group of 04 students: 20% - Assignments: 04 project during the course, in group of 02 students: 20% - Final project: practical project (application and database, hardware is optional), in group of 04 student, including presentation, demonstration and Q&A section: 40%
Media employed	Text book
Reading list	1. Android Developers' Documents (https://developer.android.com/docs) 2. Android: A programmer's guide, DiMarzio Jerome F., McGraw-Hill, 2008 3. Android OpenSource Project (http://source.android.com/)

Written/updated by
Dang Le Khoa & Nguyen Thai Cong Nghia

86. Mobile Programming Lab- ETC10323

Module name	Mobile Programming Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10323
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	7th semester
Person responsible for the module	Nguyen Thai Cong Nghia
Lecturer	Nguyen Thai Cong Nghia
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3hrs/week Office hours: Wednesday 9:00 AM – 11:30 AM Friday 9:00 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Mobile Programming
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Build applications include activities, fragments, services... combined - Build user interfaces with views, layouts, adapter views - Construct database for storage and access - Combine application, database and hardware to create practical system - Diagram the activities relationship, database arrangement and system structure - Assemble related documents, combine theory and tools
Content	Lab 1: User Interfaces (Widgets and Layouts) Lab 2: Activity and Event Listener, Intent and Intent Filters Lab 3: ListView and RecyclerView Lab 4: Notification and Service Lab 5: Fragment and Navigation Lab 6: Permission (Camera, Google Maps...) Lab 7: Firebase (Realtime Database, Cloud Firestore) Lab 8: Firebase (cont) (Authentication, Storage)
Study and examination requirements and forms of examination	- Attendance: 20% - Lab reports: 30% - Final project: practical project (database and hardware is optional), including presentation, demonstration and Q&A section: 50%
Media employed	Text book

Reading list	1. Android Developers' Documents (https://developer.android.com/docs) 2. Android: A programmer's guide, DiMarzio Jerome F., McGraw-Hill, 2008 3. Android OpenSource Project (http://source.android.com/)
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Written/updated by

Nguyen Thai Cong Nghia

87. Biomedical Signal Processing – ETC10116

Module name	Biomedical Signal Processing
Module level, if applicable	Undergraduate
Code, if applicable	ETC10116
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Le Duc Hung
Lecturer	PhD. Le Duc Hung
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Friday 07:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Lab hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 120 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Signal Processing, Fundamentals of Biomedical Electronics
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand concepts of biomedical signal processing - Implement biomedical signal processing algorithms - Have skills in solving problems in biomedical signal processing - Ideate, plan, and implement an application of Biomedical signal processing
Content	Chapter 1: Biomedical Signals (Cardiac electrophysiology, relation of electrocardiogram (ECG) components to cardiac events, clinical applications) Chapter 2: Biomedical Images (Survey of major modalities for medical imaging: ultrasound, X-ray, CT, MRI, PET, and SPECT; Physics and signal processing for magnetic resonance imaging) Chapter 3: Probability and Random Signals Chapter 4: Digital Filtering for Biomedical Signals Chapter 5: Blind source separation: Principal component analysis (PCA) and Independent component analysis (ICA) Chapter 6: Digital Image Processing 1 (Filtering and Fourier methods to 2-D signals and systems) Chapter 7: Digital Image Processing 2 (Interpolation, noise reduction methods, edge detection, homomorphic filtering) Chapter 8: Image Segmentation and Registration

Study and examination requirements and forms of examination	- Assignments: Problems and short tests during the course: 20% - Mid-term Exam (60 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 30% - Final exam (60 minutes): Written exam, 4-page cheatsheets are allowed in the exam room: 50%
Media employed	Text book
Reading list	1. A. V. Oppenheim, R. W. Schaffer, J. R. Buck, Discrete-Time Signal Processing, 2nd Edition, Upper Saddle River, NJ: Prentice-Hall, 1999. 2. A. Papoulis, Pillai S. U., Probability, Random Variables, and Stochastic Processes, New York, NY: McGraw Hill, 2002. 3. R. Gonzalez, and R. E. Woods, Digital Image Processing, 2nd Edition, Upper Saddle River, NJ: Prentice-Hall, 2002. 4. S. Webb, The Physics of Medical Imaging, New York, NY: Taylor & Francis, 1988.

Written/updated by
Le Duc Hung

88. System on Chip Design - ETC10117

Module name	System on Chip Design
Module level, if applicable	Undergraduate
Code, if applicable	ETC10117
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Duy Manh Thi
Lecturer	MSc. Nguyen Duy Manh Thi
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 07:30 AM – 09:30 AM Thursday 07:30 AM – 09:30 AM
Workload	Contact hours: 60 hrs Assignments and independent learning: 90 hrs Total Working hours: 150 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Electronics, Digital Circuits and IP Core Design.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the SoC design flow and each step in the flow - Realize necessary data and requirement during design process - Understand function and usage of digital, analog circuits in IC design - Use digital, analog circuits into IC design and implementation phase. - Implement IC design in script environment with practical EDA tools - Describe and verify IC design using EDA tools and scripting languages
Content	Chapter 1: Overview of System on Chip Design Technology Trends and Scaling Challenges in Modern technology Chapter 2: Generalized SoC design flow HW/SW Co-Design Bus Architecture Design Introduction to Front-End and Back-End Design flow Low power SoC Design Flow New Trends in SoC Design

	Chapter 3: Front-end Design flow at Block level RTL Coding Design Simulation and Verification Design Constraints and Synthesis Pre-Layout verification Formal Verification between RTL vs. Netlist Chapter 4: Design For Testability (DFT) Introduction to DFT DFT design flow and techniques Scan Insertion: Scan cell, Scan architecture, Test coverage BIST (Built-In Self Test) Chapter 5: Back-End design flow Floorplaning and power planning Cell Placement Clock Tree Synthesis Routing Post-Layout Verification Chapter 6: System on Chip (SoC) verification Verification Methodology Verification Strategy and Plans Physical Verification and Design Sign-Off
Study and examination requirements and forms of examination	- Experiment Lab: 5 practical Labs: 30% - Mid-term exam (60 minutes): Multiple choice exam: 20% - Final exam (90 minutes): Written exam: 50%
Media employed	Text book
Reading list	1. CMOS VLSI Design: A Circuits and Systems Perspective (4th Edition), Neil Weste, David Money Harris, Addison-Wesley, 2011.

Written/updated by
Nguyen Duy Manh Thi

89. MEMS, NEMS & Nanoelectronics - ETC10118

Module name	MEMS, NEMS & Nanoelectronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC10118
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Prof. PhD. Dinh Sy Hien

Lecturer	Prof. PhD. Dinh Sy Hien
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hours/week Office hours: Thursday: 7:30 AM – 11:30
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand material properties for MEMS & NEMS - Be familiar with the important concepts applicable to MEMS & NEMS, and their fabrication - Be fluent of design, analysis and testing of MEMS - Be fluent of technologies of NEMS - Apply the MEMS for different applications - Know about design, fabrication of capacitive accelerometer MEMS and pressure sensor MEMS
Content	Introduction History of MEMS development, Characteristics of MEMS, micro-electronics integration – mass production. Micro-fabrication – microelectronics fabrication process – Silicon based MEMS processes – new material and fabrication processing – points of consideration of processing. Chapter 1: Materials for MEMS 1.1 Introduction 1.2 Silicon monocrystalline 1.3 Polysilicon 1.4 Silicon dioxide 1.5 Silicon nitride 1.6 Ge based material 1.7 Metal 1.8 Silicon carbide 1.9 Diamond 1.10 Material of III-V groups 1.11 Piezoelectric material Chapter 2: MEMS Design 2.1 Introduction 2.2 Overview of MEMS design systems 2.3 Method of MEMS design 2.3.1 Design for manufacturability of MEMS devices 2.3.2 Organization structure of MEMS Chapter 3: MEMS fabrication 3.1 Introduction 3.2 Wet bulk micromachining 3.3 Isotropic wet etching and anisotropic wet etching

	3.4 Surface micromachining 3.5 Surface micromachining processes 3.6 Comparison between bulk micromachining and surface micromachining 3.7 Examples of surface micromachining Chapter 4: LIGA Method 4.1 Introduction 4.2 Basic information of LIGA 4.3 LIGA and process steps 4.4 LIGA applications Chapter 5: NEMS Fabrication Technologies 5.1 Introduction 5.2 NEMS compatible processing technologies 5.3 Nano-machine fabrication: interfacing with biology 5.4 NEMS – multidiscipline field Chapter 6: MEMS Packaging 6.1 Introduction 6.2 Requirement of Materials 6.3 Wafer level packaging 6.4 High temperature electric connection system 6.5 Ceramic packaging system 6.6 MEMS RF packaging Chapter 7: Capacitive Accelerometer 7.1 Accelerometers 7.2 Design of structure of accelerometer 7.3 Introduction to schematics of accelerometer 7.4 Introduction to ConvectWare software 7.5 Comparison between CA parameters and sample Chapter 8: Pressure Sensor MEMS 8.1 Introduction 8.2 Pressure micromachining sensors 8.3 Development of recent pressure sensors 8.4 Future prospect Chapter 9: MEMS and Their Applications 9.1 Introduction 9.2 MEMS – pioneer miniaturization technology 9.3 Some pictures of MEMS applications
Study and examination requirements and forms of examination	Midterm exam: 30% Final exam: 70%
Media employed	Text book
Reading list	1. Dinh Sy Hien, MEMS & NEMS: materials, design and fabrication, VNU HCMC publishing, HCM City, 2011. (in Vietnamese Language), recommended textbook. 2. Mohamed Gad_El_Hak, MEMS Design and Fabrication, Taylor & Francis Group, 2006

Written/updated by Dinh Sy Hien

90. Audio and Video Engineering – ETC10119

Module name	Audio and Video Engineering
Module level, if applicable	Undergraduate
Code, if applicable	ETC10119
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Bui An Dong
Lecturer	MSc. Bui An Dong
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week (3 theory hrs) Office hours: Saturday 06:40 AM – 9:40 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 30 hrs Total Working hours: 60 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Digital Signal Processing.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Digital Signal Processing - Digital Audio Compression - Digital Video Compression - Digital Television
Content	Chapter 1: Digital Signal Processing + Sampling and Quantization + Filtering + ADC and DAC Chapter 2: Digital Audio Compression + Introduction to Digital Audio Technology, + Introduction to editing and mixing audio to a stereo format, + WAV, MP3 format. Chapter 3: Digital Video Compression + Introduction to Digital Video Technology, + H.264 Chapter 4: Digital Television + Advance Television Systems, + 3D TV, LCD TV, HDTV
Study and examination requirements and forms of examination	- Assignments: 3 or 4 problems during the course: 40% - Final exam (60 minutes): Mini Seminar: 60%
Media employed	Text book
Reading list	www.101science.com/Radio.htm#Television

Written/updated by Bui An Dong

91. Architecture of mobile devices and programming – ETC10120

Module name	Architecture of mobile devices and programming
Module level, if applicable	Undergraduate
Code, if applicable	ETC10120
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Bui An Dong
Lecturer	MSc. Bui An Dong
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 5 hrs/week (3 theory hrs + 2 lab hrs) Office hours: Saturday 06:40 AM – 9:40 AM
Workload	Contact hours: 50 hrs Assignments and independent learning: 50 hrs Total Working hours: 100 hrs
Credit points	3 (2 Theories + 1 Lab)
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Advanced Programming Techniques.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Knowledge of the interface standards on mobile devices. - Knowledge of the operating systems on mobile devices. - Applying Java language and XML. - Design a simple application on Android. - Design applications communicating between smart mobile devices and peripherals on Android.
Content	Chapter 1: Overview of the architecture of mobile devices + History of mobile devices + Architecture of the mobile device and especially, smart mobile device Chapter 2: Overview of the mobile OS + History of mobile OS + Overview of mobile OS + Architecture of mobile OS + Setting environment for application on Android Chapter 3: Java Programming Language + History of Java + The basic Java + object-oriented, classes and inheritance, interface and packaging + Debugging the common mistakes when writing program Chapter 4: Design of an application on Android OS + Overview of the applications environment + Flow of Studio design + Flow of application testing

	+ The basic components of the application on Android + Activity and storing Activity Chapter 5: Design of application interface on Android OS + Using intuitive tool, XML, Java + Knowledge of Bluetooth or Network standard + Design of interface between mobile and peripheral devices uses Bluetooth or network Lab 1: Java Lab 2: Design of basic application on Android + Basic application Lab 3: Design of advance application on Android + Advance application + Interface of application Lab 4: Design of interface between mobile and peripheral devices + Design of a peripheral devices + Design of a interface
Study and examination requirements and forms of examination	- Assignments: 3 or 4 problems during the course: 30% - Mid-term exam (45 minutes): Projects: 20% - Final exam (60 minutes): Projects: 50%
Media employed	Text book
Reading list	1. Thiết kế vi mạch CMOS VLSI, Tống Văn On, NXB Phương Đông, 2007.

Written/updated by
Bui An Dong

92. Power Electronics - ETC10121

Module name	Power Electronics
Module level, if applicable	Undergraduate
Code, if applicable	ETC10121
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Thi Thien Trang
Lecturer	MSc. Nguyen Thi Thien Trang
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Digital Electronics.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the characteristics and working principles of power electronic devices - Understand and explain the working principle of converters AC-DC, DC-DC, DC-AC, AC-AC - Analyze and design power electronic circuits with specific description - Assemble related documents, imitate and calibrate power electronic circuit designs, combine theory and tools - Demonstrate the systematization of knowledge
Content	Chapter 1: Introduce to power electronic devices with diode, thyristor, GTO, IGBT and MOSFET Chapter 2: AC – DC converters Single phase half, full-wave rectifier, Single phase half, full-wave phase-controlled rectifier, Three phase half, full-wave rectifier, Three phase half, full-wave phase-controlled rectifier Chapter 3: DC – DC converters Boost (step-up) converter, Buck (step-down) converter Chapter 4: DC – AC converters Square wave voltage inverter, PMW wave voltage inverter Chapter 5: AC – AC converters Single phase cyclo converter, Single phase AC controller Lab 1: Single phase half, full-wave rectifier, Single phase half, full-wave phase-controlled rectifier

	Lab 2: Three phase half, full-wave rectifier, Three phase half, full-wave phase-controlled rectifier Lab 3: Boost (step-up) converter, Buck (step-down) converter Lab 4: Square wave voltage inverter, PMW wave voltage inverter Lab 5: Single phase cyclo converter, Single phase AC controller
Study and examination requirements and forms of examination	- Daily quizzes: 5 quizzes during the course: 10% - Assignments: 3 problems during the course: 20% - Mini Projects: teamwork, 2-3 students/project: 30% - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book
Reading list	1. Power Electronics: Converters, Applications, and Design, Mohan, Ned, Tore Undeland, and William Robbins, John Wiley & Sons, 2003. 2. Fundamentals of Power Electronics, Robert W. Erickson, Chapman & Hall, 2020.

Written/updated by
 Nguyen Thi Thien Trang

93. Graduation seminar project- ETC10190

Module name	Graduation seminar project
Module level, if applicable	Undergraduate
Code, if applicable	ETC10190
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	Various lecturers
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	None
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	4
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Successfully completed 124/134 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Point out the research problems - Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem - Use appropriate techniques for solutions - Demonstrate the scientific contribution and practical relevance of the research carried out - Provide a consistent, well-structured report - Plan and manage learning process - Present and defend results of the project at the Students' Seminar Conference
Content	Various topics
Study and examination requirements and forms of examination	Final defense: 100%
Media employed	None
Reading list	None

Written/updated by
PhD. Bui Trong Tu

94. Bachelor Thesis- ETC10195

Module name	Bachelor Thesis
Module level, if applicable	Undergraduate
Code, if applicable	ETC10195
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Bui Trong Tu
Lecturer	Various lecturers
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	None
Workload	Contact hours: 30 hrs Assignments and independent learning: 270 hrs Total Working hours: 300 hrs
Credit points	10
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Successfully completed 124/134 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Point out the research problems - Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem - Use appropriate techniques for solutions - Demonstrate the scientific contribution and practical relevance of the research carried out - Provide a consistent, well-structured Bachelor Thesis - Plan and manage learning process - Present and defend results of the thesis at the Students' Bachelor Thesis conference.
Content	Various topics
Study and examination requirements and forms of examination	Final defense: 100%
Media employed	None
Reading list	None

Written/updated by
PhD. Bui Trong Tu

95. Application Programming on Mobile Devices - ETC10212, ETC10213

Module name	Application Programming on Mobile Devices Application Programming on Mobile Devices Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10212 ETC10213
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	PhD. Huynh Huu Thuan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Tuesday 08:00 AM – 11:00 AM Friday 08:00 AM – 11:00 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 60 hrs Total Working hours: 90 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Object-Oriented Programming, Java Programming
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Knowledge of mobile OSs - Knowledge of mobile application development process - Knowledge of object-oriented programming in Android - Knowledge of SQL database in Android - Knowledge of network programming in Android
Content	Chapter 1: Overview – course introduction + Mobile platforms + Mobile development constraints Chapter 2: Introduction to Android + Features + Application components Chapter 3: Android application life cycle Chapter 4: Android UI components + Layouts + Basics UI Chapter 5: Android UI components + Fragments + Dialogs + Menus + Tabs Chapter 6: Restore application state + Preferences

	+ Files Chapter 7: SQL Database Chapter 8: Multi-tasking + Multi-Threading + AsyncTask Chapter 9: Services and Widgets Lab 1: Create a simple application containing simple UI controls like TextView, EditText, ... Lab 2: Create a simple note application containing Fragments, Dialogs and Tabs UI components Lab 3: Add app-state-restore write-to-external storage features to BTVN#3 using Preferences and Files. Lab 4: Add SQL database features to BTVN#4 Lab 5: Create an application which allows users to download images from a remote server and display download progress Lab 6: Create an alarm clock application using Services and Widgets.
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (45 minutes): Multiple choice exam, a four-page cheat sheet per student is allowed in the exam room: 20% - Final exam (60 minutes): Written exam, a four-page cheat sheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. Professional Android 4 Application Development, Reto Meier, 2010. 2. Eclipse Android Developer Tools.

Written/updated by
Huynh Huu Thuan

96. Embedded System - ETC10214

Module name	Embedded System
Module level, if applicable	Undergraduate
Code, if applicable	ETC10214
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Hoang Anh Tuan
Lecturer	MSc. Hoang Anh Tuan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hours/week in 10 weeks
Workload	School time: 30 hours Self-learning time: 30 hours Total time: 60 hours
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Digital Electronics.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Compare embedded system design models using different processor technologies (single-purpose, general-purpose, application specific processors). - Describe and compare the various types of peripherals used in embedded systems. - Analyze a given embedded system design and identify its performance critical points. - Use modern engineering tools necessary for integrating software and hardware components in embedded system designs. - Utilize a top-down modular design process to complete a medium complexity embedded system design project under instructor specified design constraints. - Ideation and build up an embedded system for general purpose application
Content	Chapter 1: Introduction of Embedded system design Chapter 2: Embedded system design flow Chapter 3: Concept of Embedded system Chapter 4: Specification Chapter 5: Hardware component Chapter 6: Software component Chapter 7: HW/SW portioning Chapter 8: Validation and Evaluation
Study and examination requirements and forms of examination	- Assignments: Embedded project seminar: 30% - Final exam (90 minutes): Written exam: 70%

Media employed	Text book
Reading list	- Embedded Systems Design, Peter Madiwel.

Written/updated by

MSc. Hoang Anh Tuan

97. Embedded System Lab - ETC10215

Module name	Embedded system laboratory
Module level, if applicable	Undergraduate
Code, if applicable	ETC10215
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Hoang Anh Tuan
Lecturer	MSc. Hoang Anh Tuan
Language	Vietnamese
Relation to curriculum	Compulsory
Type of teaching, contact hours	Lectures: 3 hours/week in 10 weeks
Workload	School time: 30 hours Self-learning time: 30 hours Total time: 60 hours
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Electronics, Digital Electronics.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Compare embedded system design models using different processor technologies (single-purpose, general-purpose, application specific processors). - Describe and compare the various types of peripherals used in embedded systems. - Analyze a given embedded system design and identify its performance critical points. - Use modern engineering tools necessary for integrating software and hardware components in embedded system designs. - Utilize a top-down modular design process to complete a medium complexity embedded system design project under instructor specified design constraints. - Ideation and build up an embedded system for general purpose application.
Content	Lab 1: Introduction of Embedded system design Lab 2: Embedded system design flow Lab 3: General purpose input/output with Galileo Embedded Development Kit Lab 4: ADC/DAC with Galileo Embedded Development Kit Lab 5: Timer & PWM with Galileo Embedded Development Kit Lab 6: Communication with Galileo Embedded Development Kit Lab 7: RTOS project 1 with Galileo Embedded Development Kit

	Lab 8: RTOS project 2 with Galileo Embedded Development Kit
Study and examination requirements and forms of examination	- Assignments: 5 lab projects: 50% - Final exam: Embedded project: 50%
Media employed	Text book
Reading list	- Embedded Systems Design, Peter Mavivel.

Written/updated by

MSc. Hoang Anh Tuan

98. Computer Memory - ETC10231

Module name:	Computer Memory
Module level, if applicable	Undergraduate
Code, if applicable	ETC10231
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Cao Tran Bao Thuong
Lecturer	MSc. Cao Tran Bao Thuong
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	None
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Ability to identify problems, analyze problems in computer organization and architecture (RAM and ROM). - Ability to design systems that communicate with computer memory.
Content	Chapter 1: Cache Chapter 2: SRAM Chapter 3: Overview of DRAMs Chapter 4: DRAM Device Organization Chapter 5: DRAM Memory System Organization Chapter 6: Overview of Disks Chapter 7: The Physical Layer Chapter 8: The Data Layer Chapter 9: Drive Interface
Study and examination requirements and forms of examination	- Assignments: 1 problems during the course: 20% - Mid-term exam (45 minutes): Multiple choice exam or Written exam: 30% - Final exam (60 minutes): Written exam: 50%
Media employed	Text book
Reading list	Memory Systems: Cache, DRAM, Disk , Bruce Jacob, Spencer W. Ng, David T. Wang, Elsevier Inc., 2008.

Written/updated by Huynh Huu Thuan

99. Network Programming - ETC10232

Module name:	Network Programming
Module level, if applicable	Undergraduate
Code, if applicable	ETC10232
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Le Duc Tri
Lecturer	MSc. Le Duc Tri
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Electronics and telecommunications programming techniques, Basic computer network.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of network programming - Understand and apply network programming algorithms - Understand and be able to network programming in C and Java - Demonstrate the systematization of knowledge
Content	Chapter 1: Network programming Fundamentals + Concepts in network programming + Introduction to Socket + Components in network applications + Languages for network programming Chapter 2: Network programming with Winsock library + Introduction to the winsock library + Main functions in the winsock . library + Connected and connectionless communication + Bloking and non-blocking mode Chapter 3: Multicast programming + Multicast concept + IP address for multicast + Reliable multicast + Program about multicast to many machines Chapter 4: Network Programming with Java + Introduction to network programming in java

	+ Class InetAddress + Client-side socket + Socket on the server side + Programming with UDP Chapter 5: Network applications + Network programming with Web system + Network programming with email system + Network programming with file transfer system
Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 30% - Mid-term exam (45 minutes): Written exam, open topic: 20% - Final exam (60 minutes): Written exam, open topic: 50%
Media employed	Text book
Reading list	1. Java lập trình mạng : Các kỹ xảo lập trình với Java, Nguyễn Phương Lan, Education publisher House. 2. TCP/IP Sockets in C : Practical Guide for Programmers, Michael J. Donahoo, Kenneth L. Calvert, Amsterdam : Morgan Kaufmann, 2009.

Written/updated by

Le Duc Tri

100. Network Programming Lab - ETC10233

Module name:	Network Programming Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10233
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Le Duc Tri
Lecturer	MSc. Le Duc Tri
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Electronics and telecommunications programming techniques, Basic computer network.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand and apply network programming algorithms. - Understand and be able to network programming in C and Java.
Content	Network programming lab 1 Network programming lab 2 Network programming lab 3 Network programming lab 4 Network programming lab 5 Network programming lab 6 Network programming lab 7 Network programming lab 8 Network programming lab 9 Network programming lab 10
Study and examination requirements and forms of examination	- Assignments: 2 problems during the course: 50% - Final exam (45 minutes): Practicing exam, open topic: 50%
Media employed	Text book
Reading list	1. Java lập trình mạng : Các kỹ xảo lập trình với Java, Nguyễn Phương Lan, Education publisher House. 2. TCP/IP Sockets in C : Practical Guide for Programmers, Michael J. Donahoo, Kenneth L. Calvert, Amsterdam : Morgan Kaufmann, 2009.

Written/updated by

Le Duc Tri

101. Graduation seminar project- ETC10290

Module name	Graduation seminar project
Module level, if applicable	Undergraduate
Code, if applicable	ETC10290
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	Various lecturers
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	None
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	4
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Successfully completed 124/134 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Point out the research problems - Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem - Use appropriate techniques for solutions - Demonstrate the scientific contribution and practical relevance of the research carried out - Provide a consistent, well-structured report - Plan and manage learning process - Present and defend results of the project at the Students' Seminar Conference
Content	Various topics
Study and examination requirements and forms of examination	Final defense: 100%
Media employed	None
Reading list	None

Written/updated by
PhD. Huynh Huu Thuan

102. Bachelor Thesis- ETC10295

Module name	Bachelor Thesis
Module level, if applicable	Undergraduate
Code, if applicable	ETC10295
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Huynh Huu Thuan
Lecturer	Various lecturers
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	None
Workload	Contact hours: 30 hrs Assignments and independent learning: 270 hrs Total Working hours: 300 hrs
Credit points	10
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Successfully completed 124/134 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Point out the research problems - Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem - Use appropriate techniques for solutions - Demonstrate the scientific contribution and practical relevance of the research carried out - Provide a consistent, well-structured Bachelor Thesis - Plan and manage learning process - Present and defend results of the thesis at the Students' Bachelor Thesis conference.
Content	Various topics
Study and examination requirements and forms of examination	Final defense: 100%
Media employed	None
Reading list	None

Written/updated by
PhD. Huynh Huu Thuan

103. Network Security-ETC10227

Module name	Network Security
Module level, if applicable	Undergraduate
Code, if applicable	ETC10227
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Dinh Lam
Lecturer	MSc. Nguyen Dinh Lam
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs / week Office hours: Monday 09:30 AM – 11:30 AM
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	2
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Networking, Routing Protocol.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of Modern Network Security Threats - Understand and explain concepts of Authentication, Authorization and Accounting - Understand and explain concepts of Cryptographic Systems, secure communication to ensure integrity, authenticity and confidentiality - Describe the working and how to implement Virtual Private Networks - Describe the working and how to implement Firewall Technologies - Describe the working and how to implement Intrusion Prevention Systems - Describe the working and how to implement Cisco ASA Firewall - Understand and explain concepts of network security and how to create a technical security policy
Content	
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 30% - Mid-term exam (60 minutes): Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 20% - Final exam (90 minutes): Written exam, a four-page cheatsheet per student is allowed in the exam room: 50%
Media employed	Text book

Reading list	1. Bảo mật trên mạng : bí quyết và giải pháp, nxb Thống kê, 2008. 2. Nguyễn Thanh Cường, Mạng máy tính và hệ thống bảo vệ, nxb Thống kê, 2004. 3. Pollino David, Hacker's challenge 3: 20 brand-new forensic scenarios & solutions, Wiley Publishing, Inc, 2006.
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Written/updated by

Nguyen Dinh Lam

104. Security Networking Lab-ETC10228

Module name	Security Networking Lab
Module level, if applicable	Undergraduate
Code, if applicable	ETC10228
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Nguyen Dinh Lam
Lecturer	MSc. Nguyen Dinh Lam
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 1 hrs / week Office hours: Monday 09:30 AM – 11:30 AM
Workload	Contact hours: 15 hrs Assignments and independent learning: 15 hrs Total Working hours: 30 hrs
Credit points	1
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Networking, Routing Protocol.
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the basic concepts of Modern Network Security Threats lab - Understand and how to configure to enhance protecting routers lab - Understand and how to configure AAA model on Cisco Router lab - Understand and how to configure Access Control List, CBAC, Firewall Zone Based on Cisco Router lab - Understand and how to configure Intrusion Prevention System on Cisco Router lab - Describe the working and how to implement Intrusion Prevention Systems lab - Understand and how to configure to enhance protecting switches in LAN network lab - Understand and how to configure Virtual private network lab
Content	
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 50% - Final exam (90 minutes): Lab exam, a four-page cheatsheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. Bảo mật trên mạng : bí quyết và giải pháp, nxb Thống kê, 2008. 2. Nguyễn Thanh Cường, Mạng máy tính và hệ thống bảo vệ, nxb Thống kê, 2004.

	3. Pollino David, Hacker's challenge 3: 20 brand-new forensic scenarios & solutions, Wiley Publishing, Inc, 2006. Written/updated by Nguyễn Đình Lãm
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105. Entrepreneurship- ETC10324

Module name	Entrepreneurship
Module level, if applicable	Undergraduate
Code, if applicable	ETC10324
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Introduction to Electronics and Telecommunications
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Developing successful business ideas - Moving from an idea to an entrepreneurial firm - Managing and growing an entrepreneurial firm - Creating a business plan presentation
Content	Chapter 1: Introduction to Entrepreneurship Chapter 2: Recognizing Opportunities and Generating Ideas Chapter 3: Feasibility Analysis Chapter 4: Industry and Competitor Analysis Chapter 5: Developing an Effective Business Model Chapter 6: Writing a business plan Chapter 7: Preparing a Proper Ethical and Legal Foundation Chapter 8: Assessing a New Venture's Financial Strength and Viability Chapter 9: Building a New-Venture Team Chapter 10: Getting Financing or Funding Chapter 11: Unique Marketing Issues
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Group Project: Writing a business plan and presentation: 40% - Final exam: Multiple choice exam, a four-page cheatsheet per student is allowed in the exam room: 40%
Media employed	Text book

Reading list	Kotler Philip, Jain Dipak C. , Maesincee Suvit, Bước chuyển marketing : cách tiếp cận mới để tìm kiếm lợi nhuận, phát triển và đổi mới, nxb Trẻ, 2011
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Written/updated by

Dang Le Khoa

106. Network Systems- ETC10325

Module name	Network Systems
Module level, if applicable	Undergraduate
Code, if applicable	ETC10325
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Nguyen Viet Ha, Ph.D.
Lecturer	Nguyen Viet Ha, Ph.D.
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Networks
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the difference between SDN network and traditional networks. - Understand the OpenFlow specification. - Differentiate the SDN types. - Analyze the Data Center requirement and the need for transition to SDN. - Generalize the Network Functions Virtualization. - Experiment with the SDN OpenFlow Applications. - Master in holding and managing the presentation.
Content	Chapter 1: Software Defined Networking introduction. Show the challenge of the traditional networks. Differentiate the Distributed networks and Centralized-control networks. Study the SDN and OpenFlow. Introduce the OpenFlow Switch and Controller Chapter 2: OpenFlow specification. Introduce the history of OpenFlow. Discuss the definitions of ports, port queues, Flow table, Packet matching, and Forwarding actions. Study the OpenFlow Messages. Chapter 3: Alternative definitions of SDN. Review the Open SDN.

	Introduce the SDN via API. Introduce the SDN via hypervisor-based overlays. Chapter 4: SDN-OpenFlow Applications. Introduce the RYU controller. Demonstrate the simple examples of RYU application. Study the RYU application API. Chapter 5: Data Center Requirements. Introduce the data center definition. Discuss the data center demands. Study the tunneling technologies for data center. Study the path technologies for data center. Study the ethernet fabrics in the data center. Chapter 6: SDN in the data center. Investigate the advantage and disadvantage of each SDN type in data center environment. Discuss the use case of SDN in data center. Introduce the real-world data center implementations. Chapter 7: Network Functions Virtualization (NFV). Introduce the NFV definitions. Study the Open platform for NFV. Comparison of the SDN and NFV. Investigate the in-line network functions.
Study and examination requirements and forms of examination	- In-class assignments: Quiz, problem-solving, discussion: 20% - Project: implement an SDN application on Mininet: 20% - Presentation: groups of students study the SDN in other environments and do the presentation: 20% - Final exam: single choice quiz: 40%
Media employed	Text book
Reading list	1. James F. Kurose, Keith W. Ross: Computer Networking: A Top-Down Approach, 6th edition. 2. Behrouz A. Forouzan, TCP/IP Protocol Suite, 3rd Edition, 2005, McGraw-Hill, ISBN 0-07- 296772-2.

by Nguyen Viet Ha, Ph.D.

Written/updated

107. Optical Network- ETC10326

Module name	Optical Network
Module level, if applicable	Undergraduate
Code, if applicable	ETC10326
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	Nguyen Viet Ha, Ph.D.
Lecturer	Nguyen Viet Ha, Ph.D.
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 3 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Fundamental of Computer Networks
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Understand the evolution of Optical Transport Networks. - Analyze the key performance characteristics of optical fibers in the communication systems. - Generalize the core network interworks with various types of access networks. - Experiment with the high-performance computing (HPC) systems. - Understand optical communication and networking technologies in the access network - Master in holding and managing the presentation.
Content	Chapter 1: The Evolution of Optical Transport Networks. Describe the role of optics in networks, their capabilities, and their scaling limitations. Discuss types of optical networks installed around the globe as well as their impact on society, market structure, and future perspectives. Chapter 2: Optical Subsystem for Transmission and Switching. Examine the evolution of optical fibers. Present the key performance characteristics of optical fibers in a communication systems context and discusses the outlook for fiber technologies.

	Cover optical fiber cabling technologies and deployment-related aspects, thus providing a comprehensive account of this key medium for information transmission. Chapter 3: Core Network. Describe Core (backbone) network architectures as deployed by telecommunication carriers. Discuss how the core network interworks with various types of access networks. Outlines the resilience strategies used by telecom carriers in their networks Chapter 4: Datacenter and Super-Computer Networking. Defines the interconnect types used in the global network of datacenters. Discuss the high-performance computing (HPC) systems and analysis of the topology and characteristics of HPC interconnects. Present the interconnection topologies and technologies for implementing intra-datacenter interconnections. Discuss the implications of hyperscale datacenter architectures and interconnection requirements for optical transceivers. Chapter 5: Optical Access and Wireless Networks. Covers optical communication and networking technologies in the access network segment and recent trends in optical wireless communication.
Study and examination requirements and forms of examination	- In-class assignments: Quiz, problem-solving, discussion: 20% - Presentation: groups of students study the Optical system in other environments and do the presentation: 20% - Mid-term exam: single choice quiz: 20% - Final exam: single choice quiz: 40%
Media employed	Text book
Reading list	1. Công nghệ truyền dẫn quang, nxb Khoa học và Kỹ thuật, 1997 2. Hoàng Văn Võ, Mạng thông tin quang thế hệ sau, nxb Thông tin và Truyền thông, 2010 3. Những công nghệ viễn thông hiện đại, nxb Thống kê, 2001

Written/updated

by Nguyen Viet Ha, Ph.D.

108. Satellite communications- ETC10327

Module name	Satellite communications
Module level, if applicable	Undergraduate

Code, if applicable	ETC10327
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	MSc. Tran Thi Huynh Van
Lecturer	MSc. Tran Thi Huynh Van
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 12:30 PM – 14:30 PM Friday 12:30 PM – 14:30 PM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Telecommunication systems, Wireless communications
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Generalize overview, fundamental, operation of satellites - Analyze and calculating characteristics of satellite's orbit, antennas, signal's attenuation, etc - Organize to rewrite research documents and effectively present related knowledge of satellite systems - Synthesize systematization of knowledge relating to satellites systems: constitution, operation frequency, antennas, access, services, interference. - Compare satellite access methods through various factors.
Content	Chapter 1: Overview of satellite systems Chapter 2: Orbits and launching methods Chapter 3: Geostationary Orbit Chapter 4: Radio Wave Propagation Chapter 5: Polarization Chapter 6: Satellite antennas Chapter 7: Space segment Chapter 8: Earth segment Chapter 9: Error control coding Chapter 10: Space link Chapter 11: Satellite access Chapter 12: Satellite services: Chapter 13: GPS technology
Study and examination requirements and forms of examination	- Assignments: 4 problems during the course: 25% - Course small projects: Work in group of 4, will be assessed by presentation, course project summary report: 25% - Final exam (60 minutes): Multiple choice exam: 50%
Media employed	Text book

Reading list	1. “Satellite Communications”, 4th Edition, Dennis Roddy, McGraw Hill, 2006.
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Written/updated by

Tran Thi Huynh Van

109. Digital Television- ETC10328

Module name	Digital television
Module level, if applicable	Undergraduate
Code, if applicable	ETC10328
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	PhD. Dang Le Khoa
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	Lectures: 4 hrs/week Office hours: Monday 09:30 AM – 11:30 AM Thursday 09:30 AM – 11:30 AM
Workload	Contact hours: 45 hrs Assignments and independent learning: 90 hrs Total Working hours: 135 hrs
Credit points	3
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Wireless communication
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Analysis of audio and video compression techniques for digital television. - Analyze MPEG transport frames. - Calculate advanced techniques in digital communication. - Analysis and design of transmission in digital television.
Content	Chapter 1: Introduction Chapter 2: Audio Compression Chapter 3: Video Compression Chapter 4: Modulation, Multiplexing and Multiple Access Chapter 5: MPEG Transport Stream (MPEG-TS) Chapter 6: Information Theory Chapter 7: Error Control Codin Chapter 8: Transmission Equipment & Link Budget Chapter 9: Link Budget (Terrestrial) Chapter 10: Link budget (Satellite)
Study and examination requirements and forms of examination	- Assignments: 3 problems during the course: 20% - Group Project: Presentation and demonstration: 30% - Final exam: Multiple choice exam and Written exam, a four-page cheatsheet per student is allowed in the exam room: 50%
Media employed	Text book
Reading list	1. Walter Fischer, “Digital Video And Audio Broadcasting Technology: A Practical Engineering Guide”, Springer, 2020

	2. John Arnold, Michael Frater, Mark Pickering, “Digital Television Technology and Standards”, John Wiley & Sons, 2007
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Written/updated by

Dang Le Khoa

110. Graduation seminar project- ETC10390

Module name	Graduation seminar project
Module level, if applicable	Undergraduate
Code, if applicable	ETC10290
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	Various lecturers
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	None
Workload	Contact hours: 30 hrs Assignments and independent learning: 90 hrs Total Working hours: 120 hrs
Credit points	4
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Successfully completed 124/134 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Point out the research problems - Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem - Use appropriate techniques for solutions - Demonstrate the scientific contribution and practical relevance of the research carried out - Provide a consistent, well-structured report - Plan and manage learning process - Present and defend results of the project at the Students' Seminar Conference
Content	Various topics
Study and examination requirements and forms of examination	Final defense: 100%
Media employed	
Reading list	

Written/updated by
PhD. Dang Le Khoa

111. Bachelor Thesis- ETC10395

Module name	Bachelor Thesis
Module level, if applicable	Undergraduate
Code, if applicable	ETC10395
Subtitle, if applicable	None
Courses, if applicable	None
Semester(s) in which the module is taught	8th semester
Person responsible for the module	PhD. Dang Le Khoa
Lecturer	Various lecturers
Language	Vietnamese
Relation to curriculum	Elective
Type of teaching, contact hours	None
Workload	Contact hours: 30 hrs Assignments and independent learning: 270 hrs Total Working hours: 300 hrs
Credit points	10
Requirements according to the examination regulations	Minimum attendance at lectures is 80%.
Recommended prerequisites	Successfully completed 124/134 Credits
Module objectives/intended learning outcomes	After completing the course, students will be able to: - Point out the research problems - Apply fundamental knowledge and theories to analyse the research problem; develop a research framework (model if applicable); and design a solution for the problem - Use appropriate techniques for solutions - Demonstrate the scientific contribution and practical relevance of the research carried out - Provide a consistent, well-structured Bachelor Thesis - Plan and manage learning process - Present and defend results of the thesis at the Students' Bachelor Thesis conference.
Content	Various topics
Study and examination requirements and forms of examination	Final defense: 100%
Media employed	None
Reading list	None

Written/updated by
PhD. Dang Le Khoa